

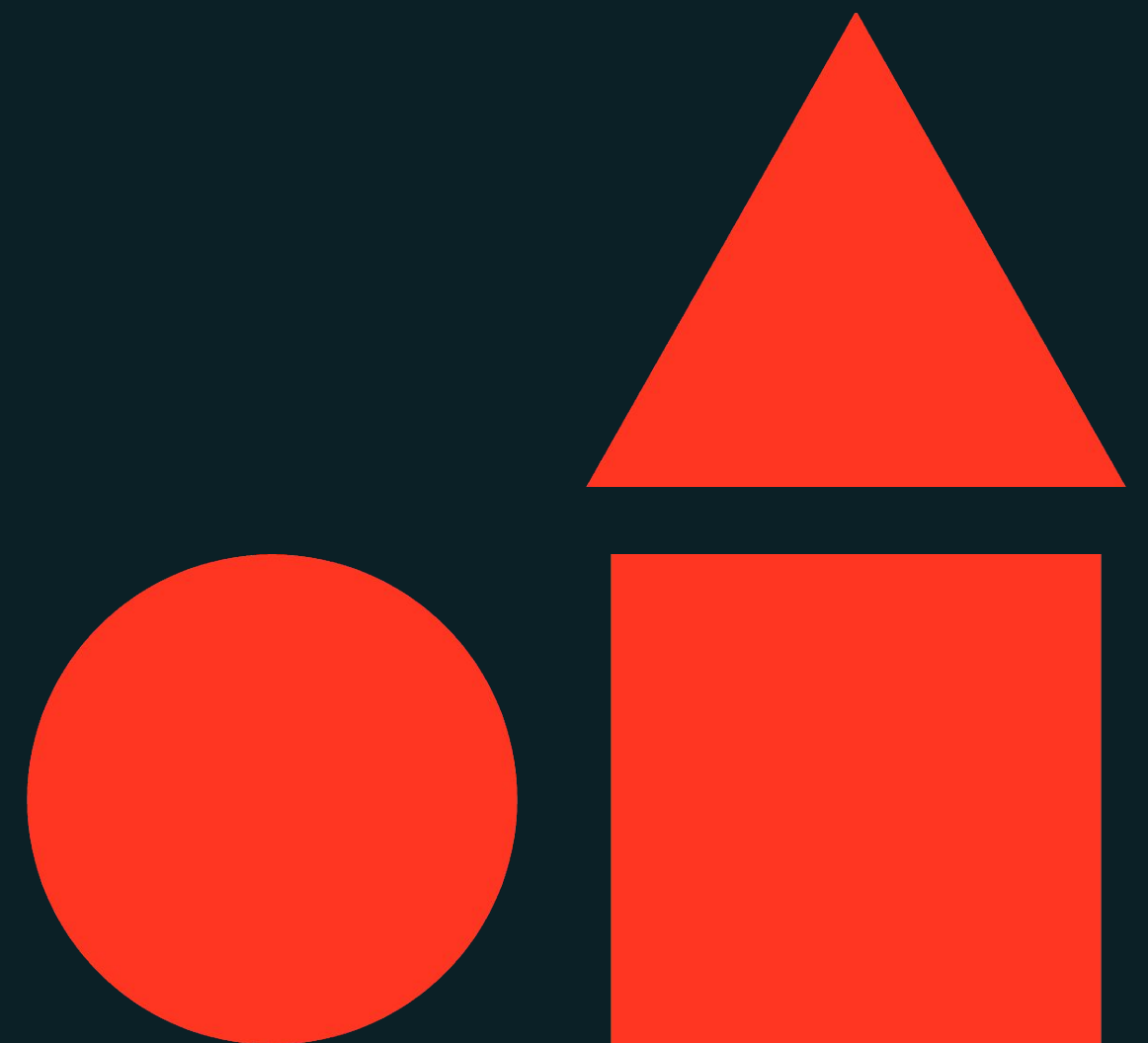


Data Warehousing with Databricks

Databricks Academy
November 2025



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Agenda

Databricks Overview	Time	Lecture	Demo	Lab
Data Warehousing with Databricks	15 mins	✓		
Unity Catalog Overview	10 mins	✓		
Databricks SQL Warehouses	5 mins	✓		
Preparing TPC-DI	20 mins		✓	



Agenda

Ingesting and Transforming Data	Time	Lecture	Demo	Lab
Data Ingestion Techniques	10 mins	✓		
Data Transformation	10 mins	✓		
Ingesting and Exploring Data	20 mins		✓	



Agenda

Ingesting and Transforming Data	Time	Lecture	Demo	Lab
Data Ingestion Techniques	10 mins	✓		
Data Transformation	10 mins	✓		
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Agenda

Orchestration	Time	Lecture	Demo	Lab
Introduction to Lakeflow Jobs	20 mins	✓		
Exploring Lakeflow Job Capabilities	10 mins		✓	



Agenda

Monitoring	Time	Lecture	Demo	Lab
Tagging	10 mins	✓		
System Tables	10 mins	✓		
Cost Optimization	10 mins	✓		



Agenda

Visualization	Time	Lecture	Demo	Lab
Introduction to AI/BI	5 mins	✓		
Creating a Dashboard in Databricks	25 mins		✓	✓



Agenda

Genie	Time	Lecture	Demo	Lab
Introduction to Genie	5 mins	✓		
Creating and Using Genie Spaces	25 mins		✓	✓



Agenda

Power BI Integration	Time	Lecture	Demo	Lab
Power BI Integration	5 mins	✓		
Star Schema Overview	5 mins	✓		



Course Learning Objectives

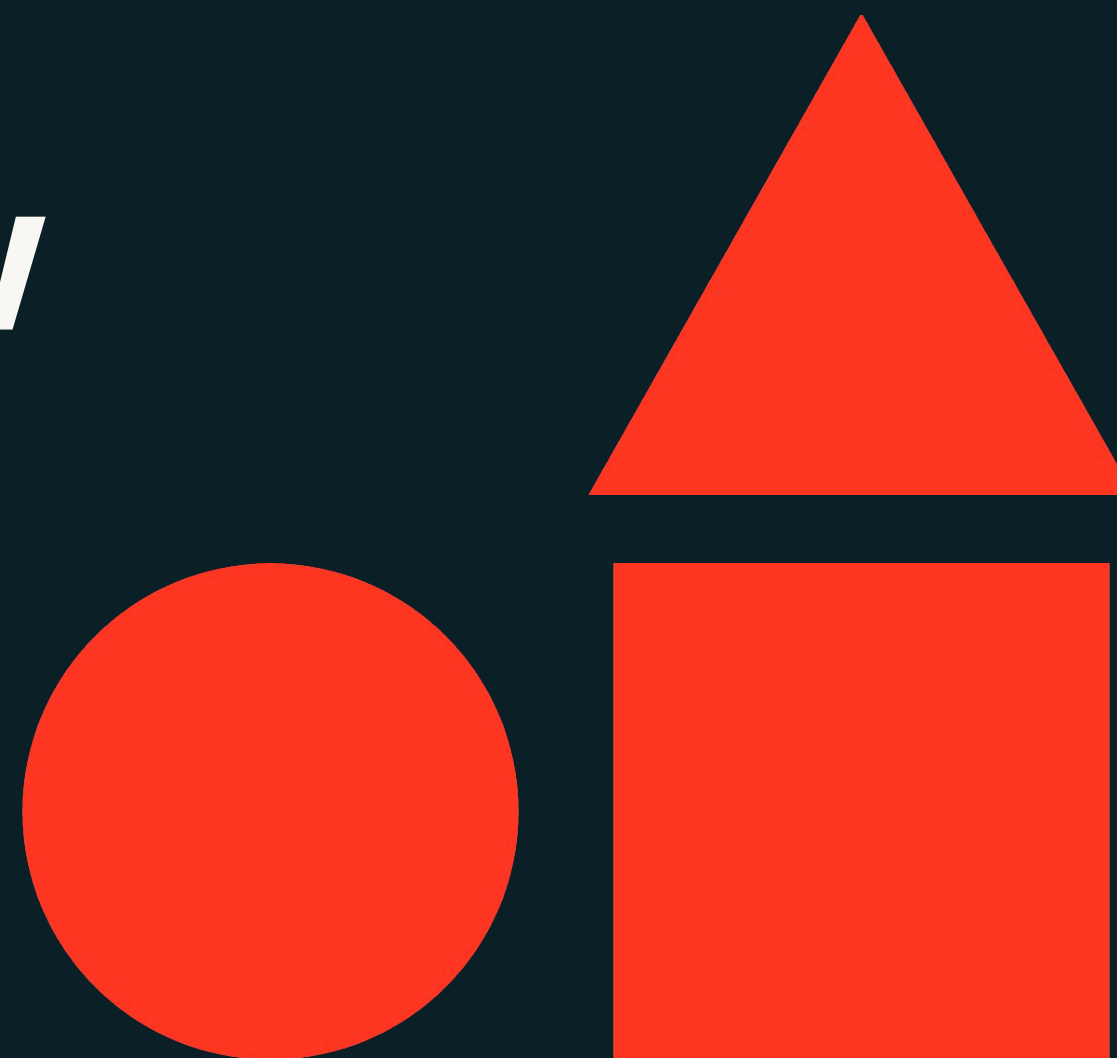
- Describe how data warehousing works on Databricks
- Describe the elements of Unity Catalog that support its role as the unified governance solution in Databricks
- Leverage Databricks capabilities to ingest, transform, and explore your data
- Use Lakeflow Jobs to automate data ingestion and transformation
- Apply best practices for controlling cost and optimizing your data warehouse
- Create meaningful visualizations of your data using AI/BI dashboards
- Create Genie Spaces that leverage natural language to garner insights from your data





Databricks Overview

Data Warehousing with Databricks



Objectives

- Describe challenges with classic data warehouses, and data lakes
- Describe how data warehousing works on Databricks
- Describe the elements of Unity Catalog that support its role as the unified governance solution in Databricks
- Become familiar with SQL warehouses and their configurable options





Databricks Overview

LECTURE

Data Warehousing with Databricks

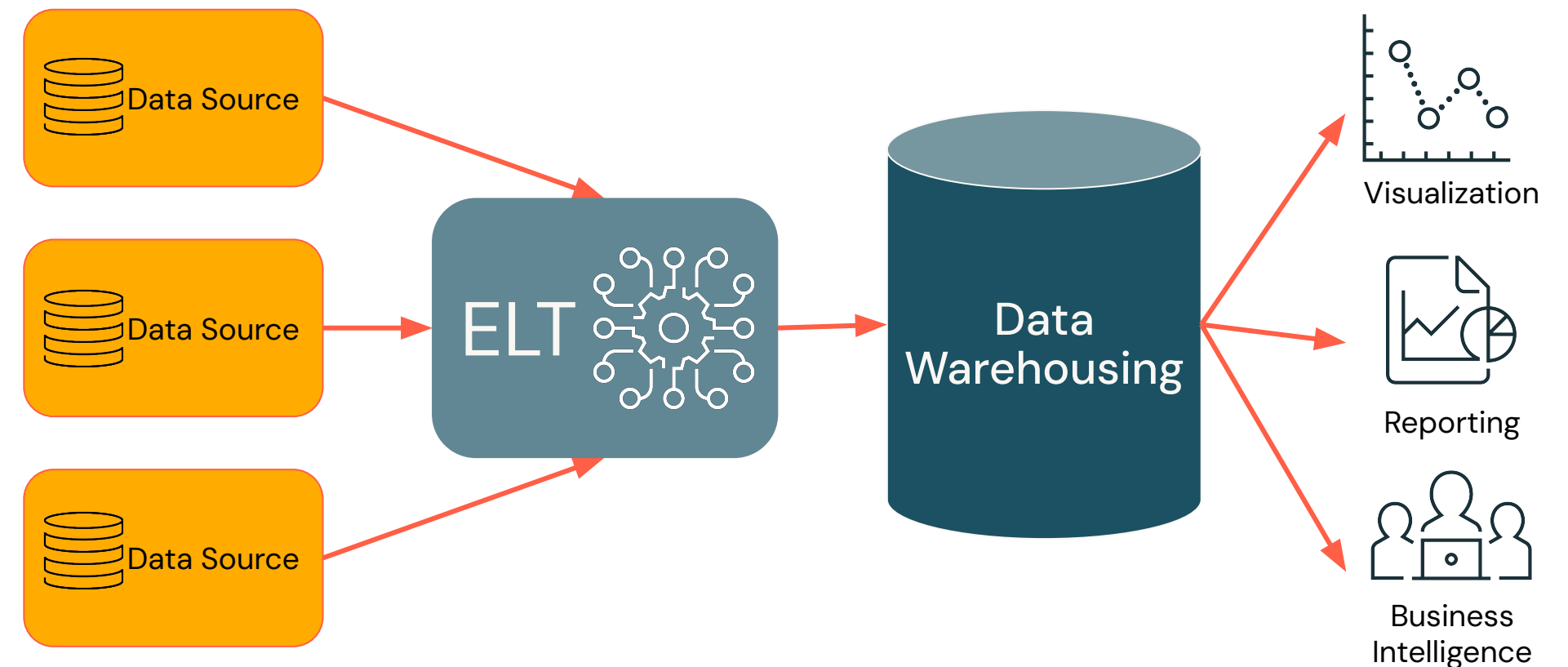


Data Warehousing

A Centralized Solution for Reporting, Analytics, and Insights

A data warehouse serves as the foundation for reporting, analytics, and BI, helping organizations make informed decisions.

- **Centralized Storage:** Integrates data from multiple sources (databases, logs, etc.).
- **ETL Process:** Data is extracted, cleaned, and organized for analysis.
- **Optimized for Insights:** Powers visualization, reporting, and BI tools.



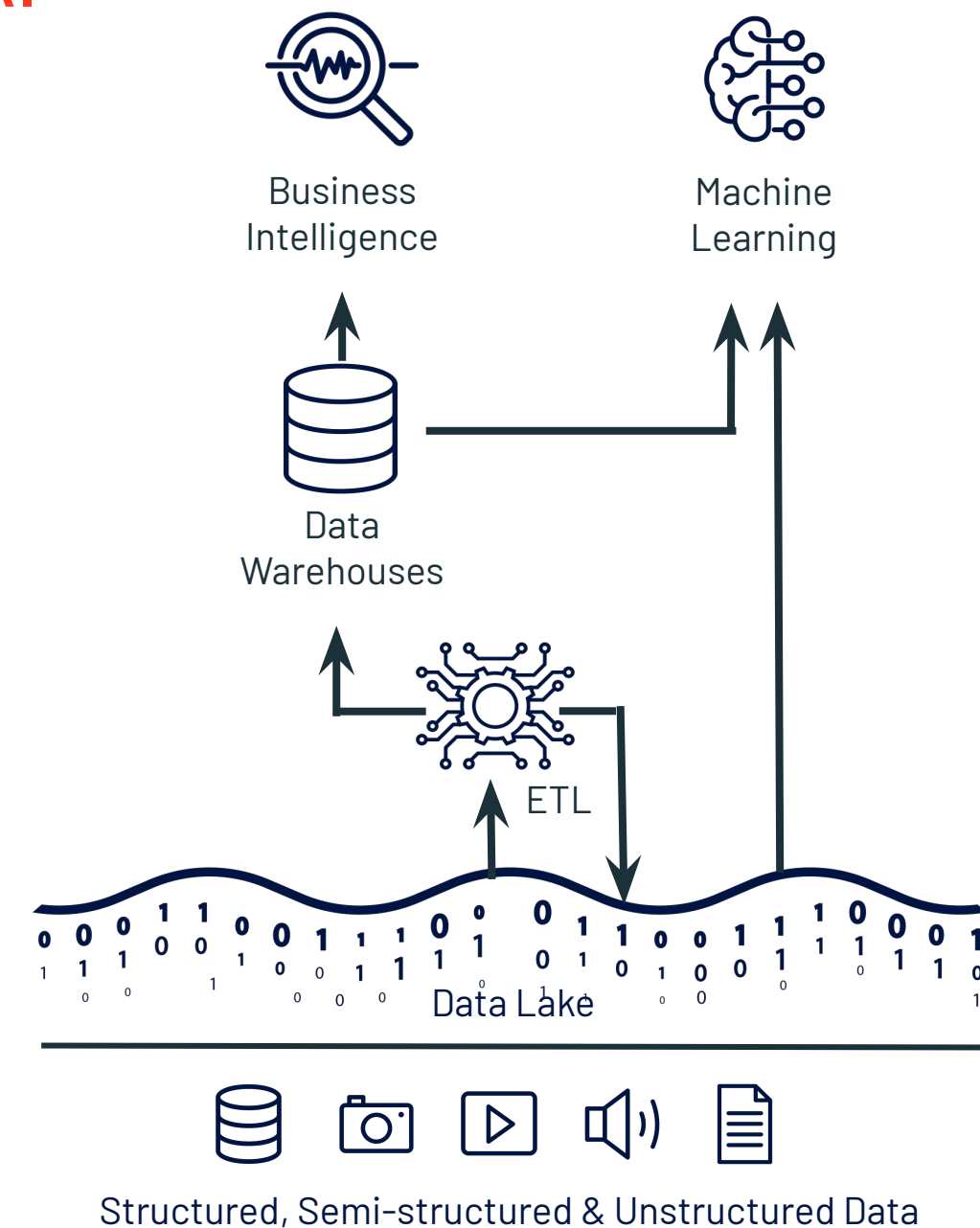
Data warehousing professionals ensure the ETL process runs smoothly and data is centralized, reliable, and optimized for reporting and insights.



Why Data Warehousing Matters

Unified data powers analytics, reporting, and AI

- Consolidates Data from Disparate Sources
- Improves Data Quality and Consistency
- Enables Reporting, Analytics, and BI
- Supports Better Decision-Making



Unified data enables faster insights and better decisions, driving organizational success.



Challenges in Data Warehousing

Data warehousing can be complex and expensive.



Users need specialized technical skills



Complex infrastructure



Data silos and fragmented systems



Issues with data governance and compliance



Slow performance at scale



Inconsistent data quality and access

- Users must have specialized knowledge to operate and manage traditional data warehouses.
- Talent is hard to obtain and retain as the industry is highly competitive.
- Expensive to train up these skills to have them lost to other companies.



Challenges in Data Warehousing

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Inconsistent data quality and access

- Redundant tooling within the data warehouse make it difficult to achieve consistency
- Systems and processes become inefficient over time and need to be frequently adjusted to maintain performance requirements
- Maintenance difficulties arise with easily outdated foundational components



Challenges in Data Warehousing

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Inconsistent data quality and access

- Data is spread across multiple systems in the ecosystem, fragmenting management
- Difficulty in consolidating insights and identifying accurate and clean data
- Delayed decision-making as processes are lengthy and heavily manual



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Inconsistent data quality and access

- Lack of centralized access controls in traditional platforms
- Challenges in ensuring compliance with regulations (GDPR, HIPAA)
- Limited capabilities to track data lineage and challenges with tackling quality issues



Challenges in Data Warehousing

Data warehousing can be complex and expensive.



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Complex infrastructure



Data silos and fragmented systems



Issues with data governance and compliance



Slow performance at scale



Inconsistent data quality and access

- Runaway increase in cost for cloud compute as the data warehouses scale
- Degraded performance as the amount of data increases from small initial testing
- Lack of access in traditional on-premise data warehouses to the latest technologies that impact performance capabilities



Challenges in Data Warehousing

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Issues with data governance and compliance



Slow performance at scale



Inconsistent data quality and access

- Inconsistent, incomplete, or outdated data persists within the environment
- Limited access for business users leads to manual overhead to process requests for access
- Complex processes requiring manual oversight to clean and validate data persist regardless of the volume of data in the warehouse

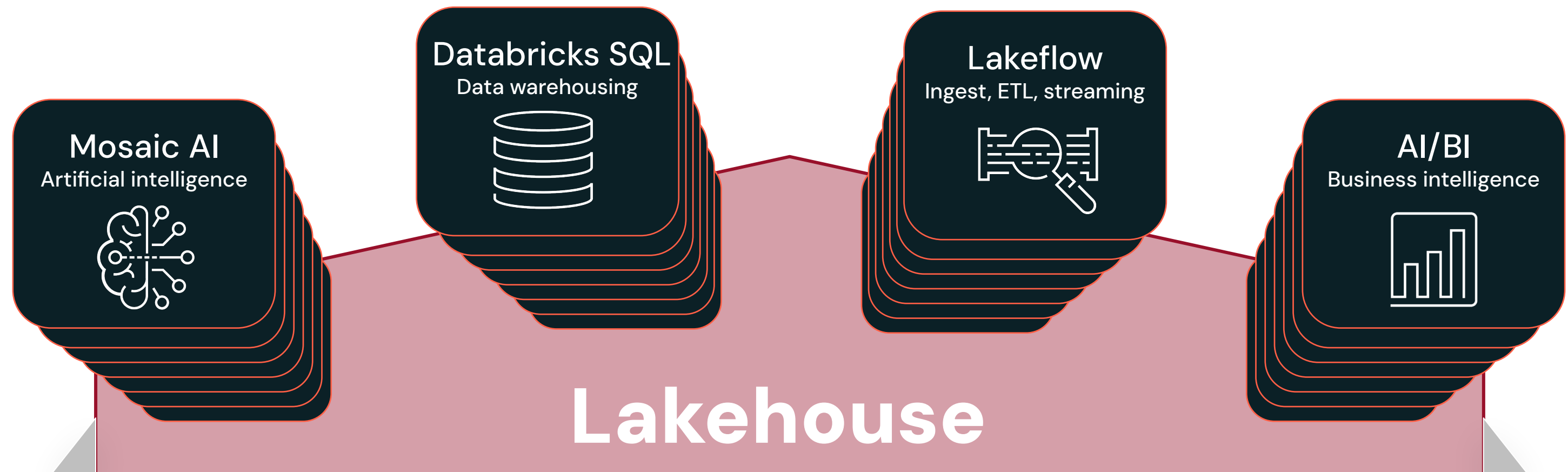


**With Databricks, data warehousing
becomes simpler, faster, and
smarter – enabling you to
overcome traditional challenges.**



Databricks Data Intelligence Platform

100% serverless



 **Unity Catalog**

 **DELTA LAKE**

ICEBERG

 **Parquet**

What Makes Databricks Different?

Databricks Data Warehousing: Simplified, Centralized, and Powerful

Unified Data Management:

- Combines structured and semi-structured data using Delta Tables for centralized storage.

Transform with Delta:

- Leverages Databricks' compute for ETL and real-time transformations.

Powered by Unity Catalog:

- Ensures governance, metadata management, and seamless collaboration across teams.

Optimized Outputs:

- Enables data marts, machine learning models, and applications for actionable insights.



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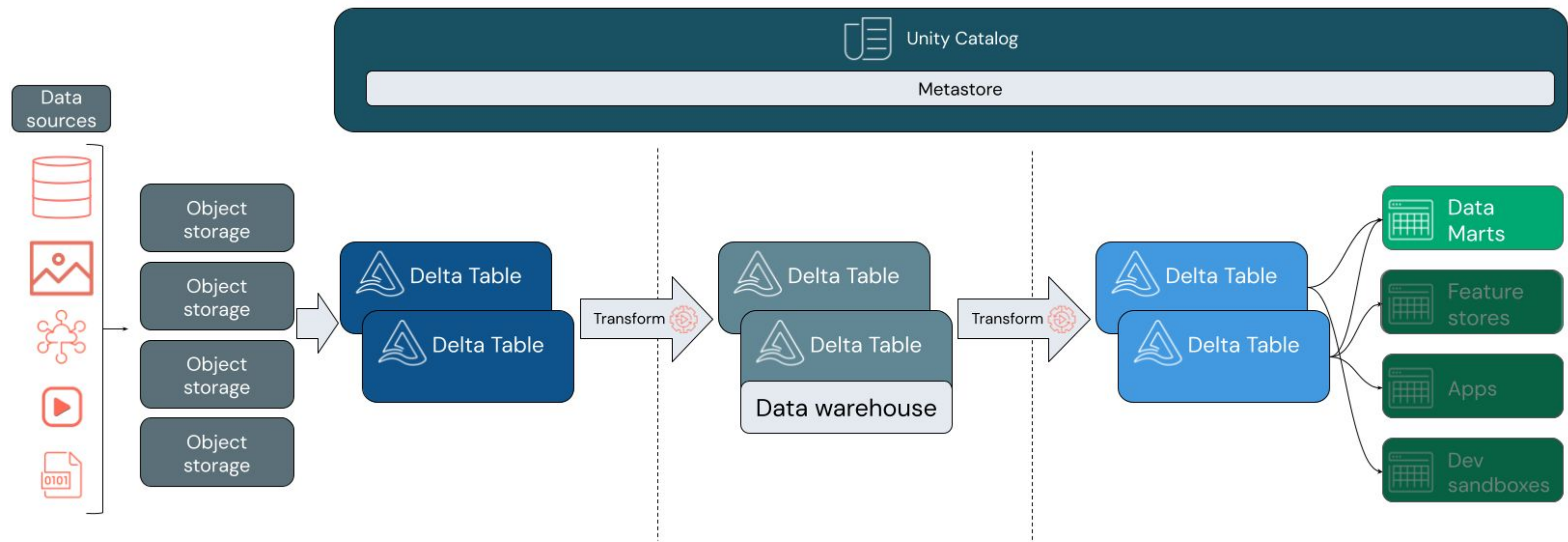
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Databricks Approach to Data Warehousing

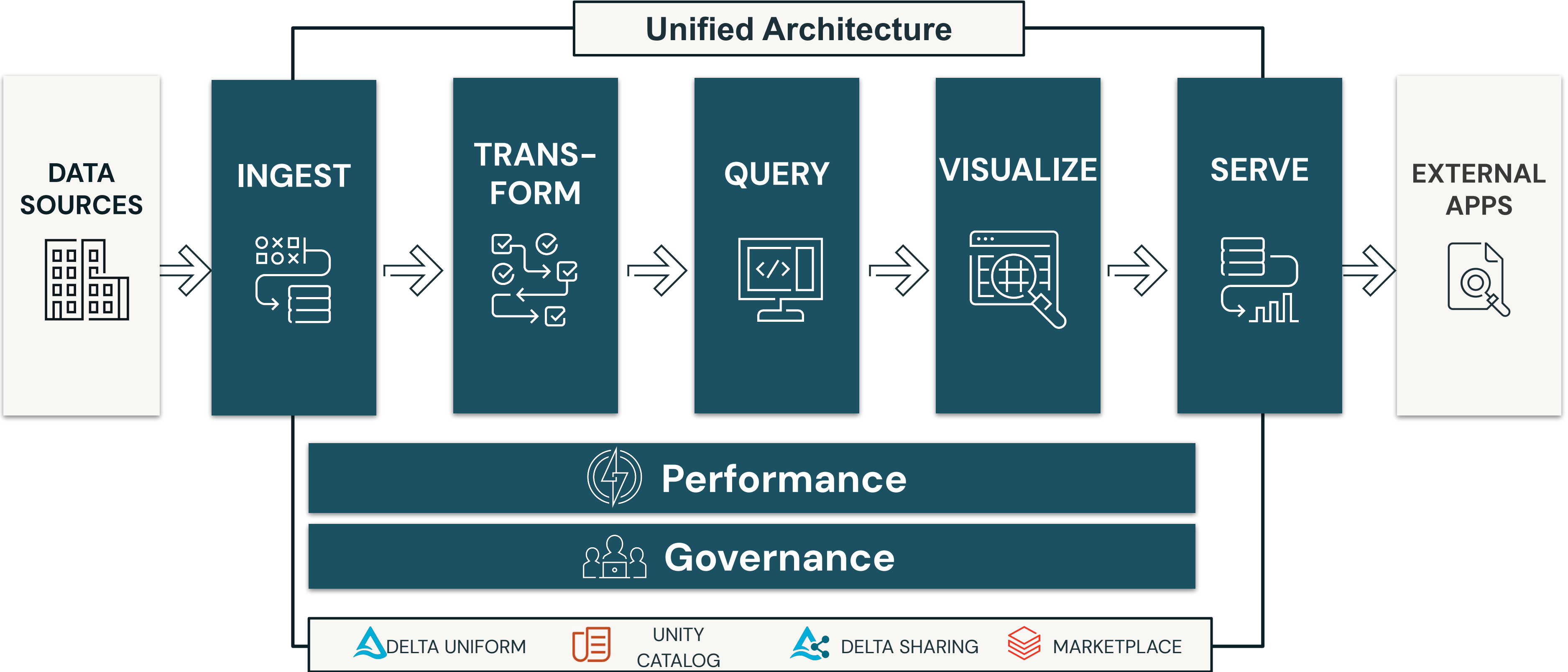
From raw data to insights: Leveraging Delta Tables and Unity Catalog.



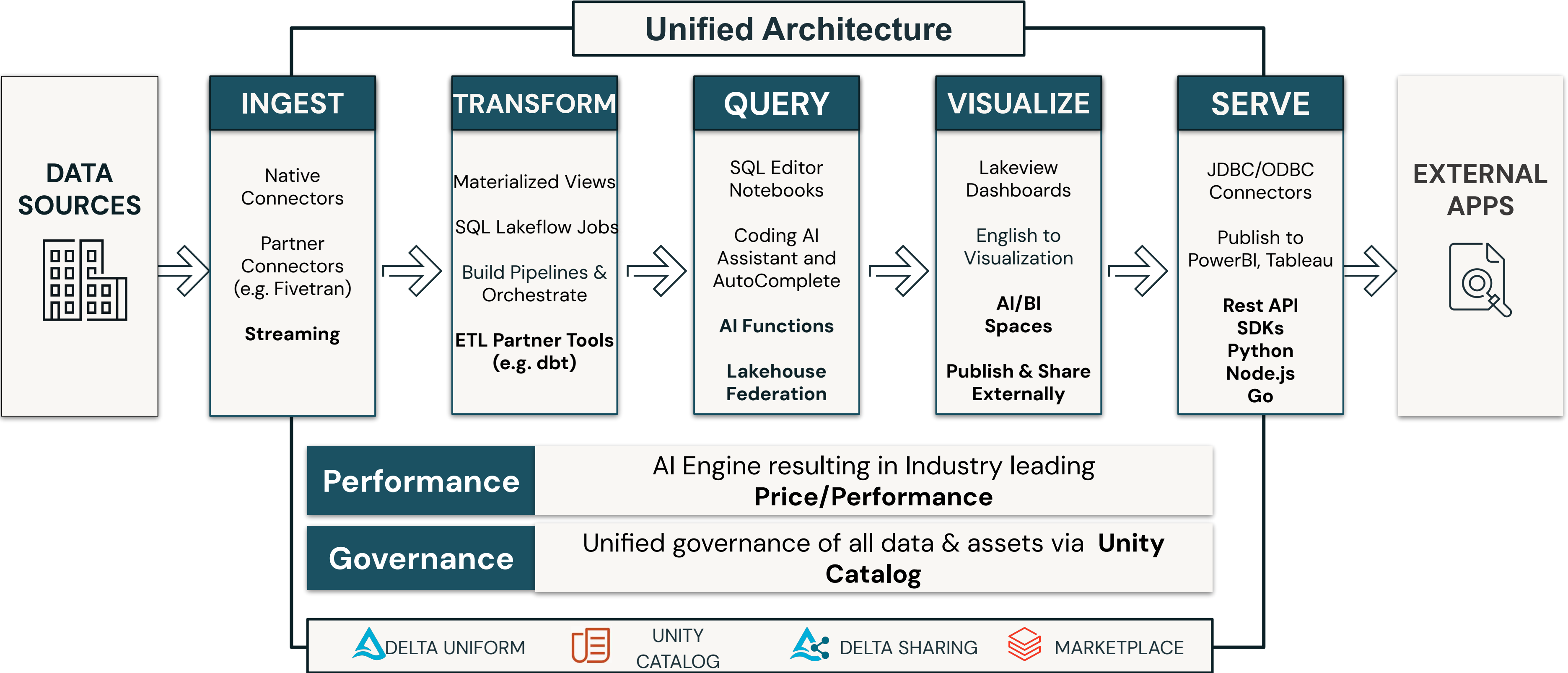
Databricks combines unified storage, real-time transformations, and governance to enable smarter data-driven insights.



Complete Data Warehousing Solution



Complete Data Warehousing Solution



The Future of Data Warehousing: Smarter, Faster, and AI-Powered



User experiences with
natural language and
data intelligence



Empower everyone,
regardless of skill level, to
query data using natural
language and intuitive
tools.



Predictive optimizations
for your infrastructure



World-class price
performance



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AI-powered automation that predicts and optimizes infrastructure performance in real-time.



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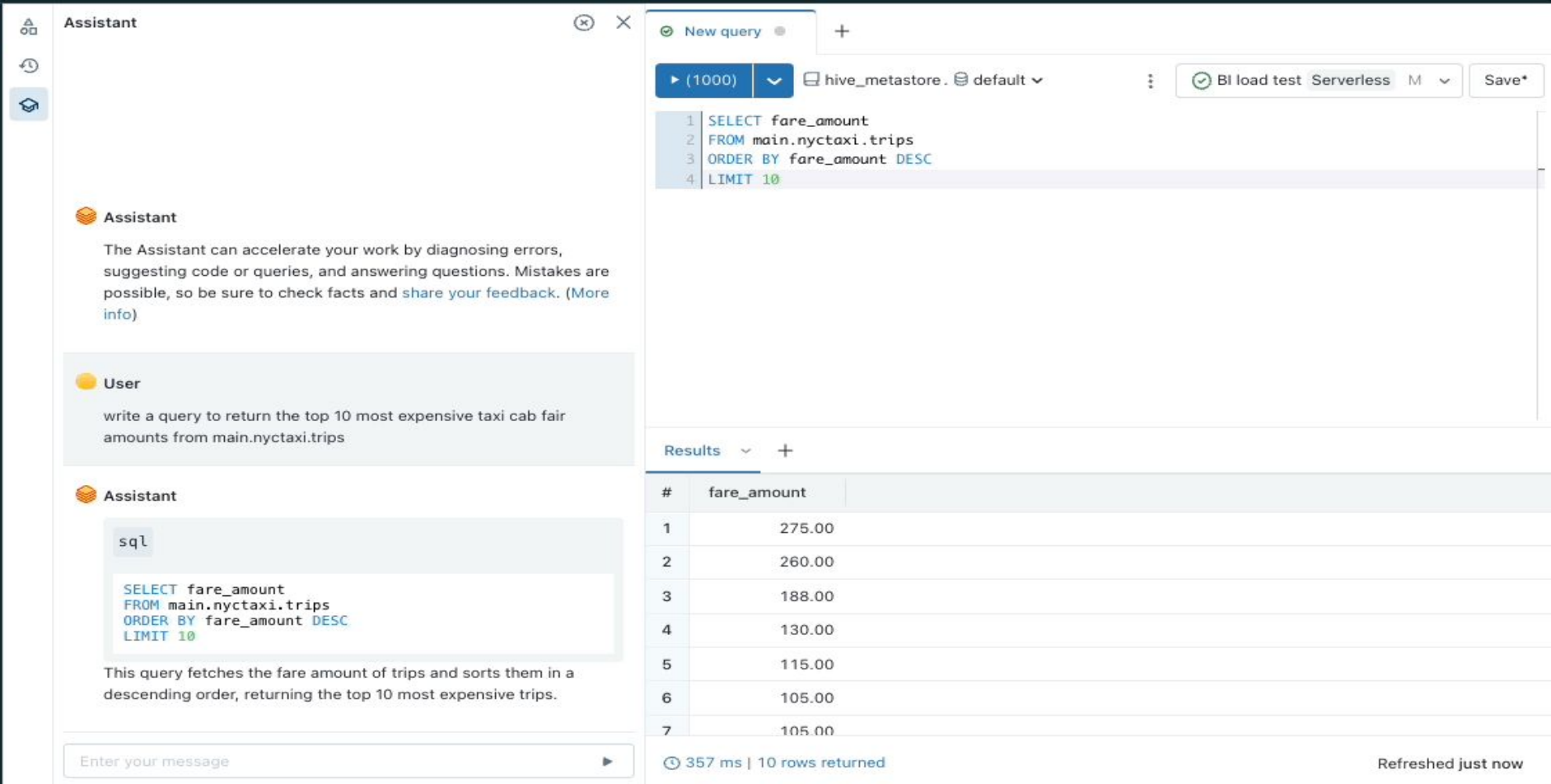


Achieve unmatched cost efficiency with best-in-class performance for every workload.



Tools for Intelligent Data Warehousing

Powered by Intelligence Engine



The screenshot displays the Databricks Unified SQL Editor interface. On the left, the 'Assistant' panel shows a conversation where a user asks for a query to return the top 10 most expensive taxi cab fare amounts from the 'main.nyctaxi.trips' table. The assistant provides the following SQL query:

```
sql
SELECT fare_amount
FROM main.nyctaxi.trips
ORDER BY fare_amount DESC
LIMIT 10
```

The assistant explains: 'This query fetches the fare amount of trips and sorts them in a descending order, returning the top 10 most expensive trips.'

On the right, the 'New query' editor shows the same SQL query. Below it, the 'Results' section displays a table with 10 rows of data:

#	fare_amount
1	275.00
2	260.00
3	188.00
4	130.00
5	115.00
6	105.00
7	105.00
8	
9	
10	

At the bottom of the results section, it indicates '357 ms | 10 rows returned' and 'Refreshed just now'.

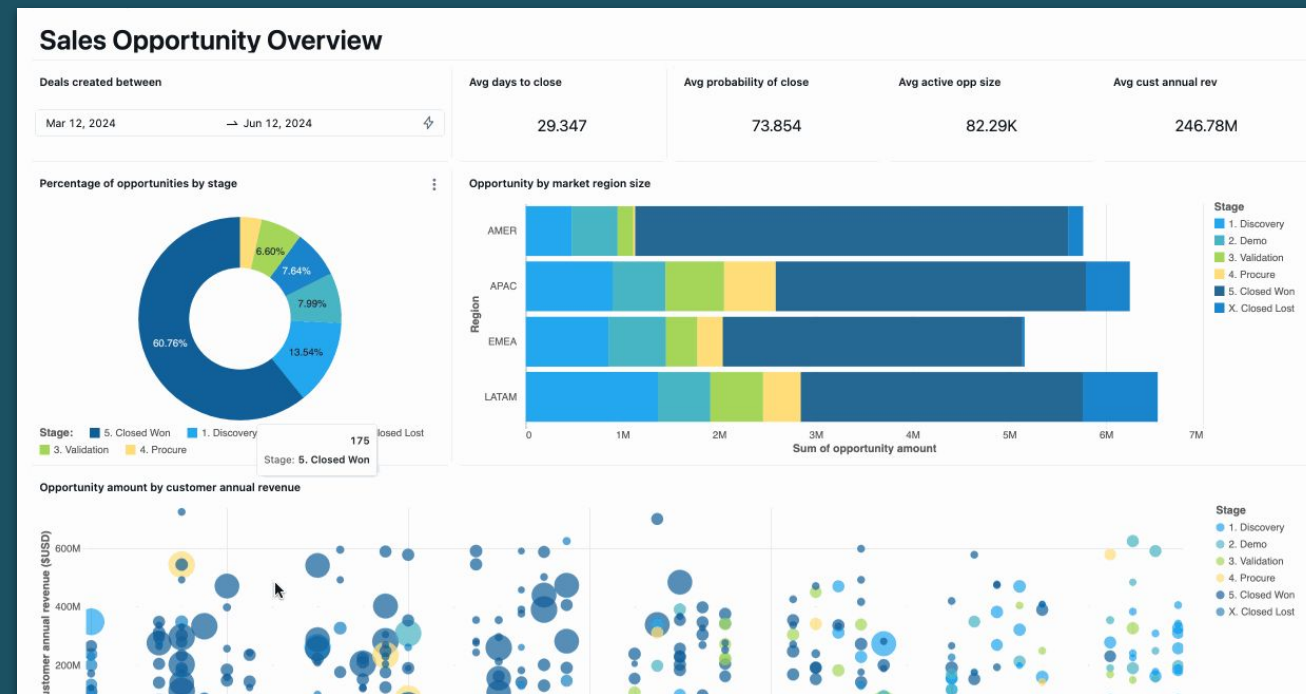
Unified SQL Editor

Data Science / Engineers



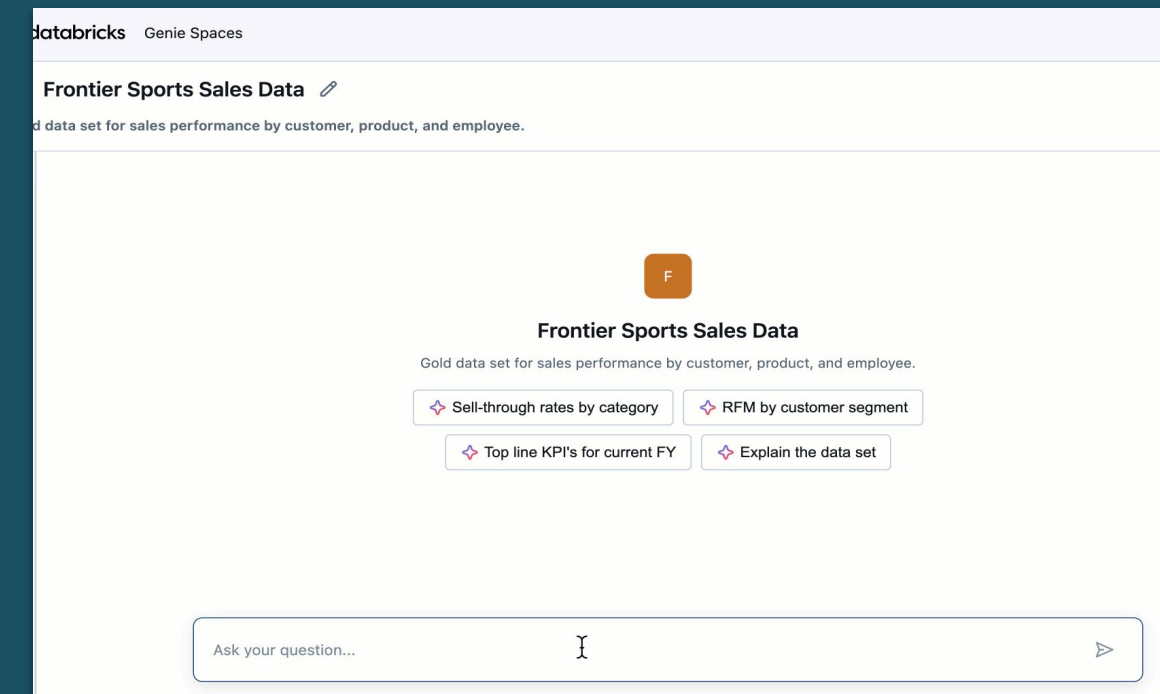
Tools for Intelligent Data Warehousing

Powered by Intelligence Engine



AI/BI Dashboards

Analysts



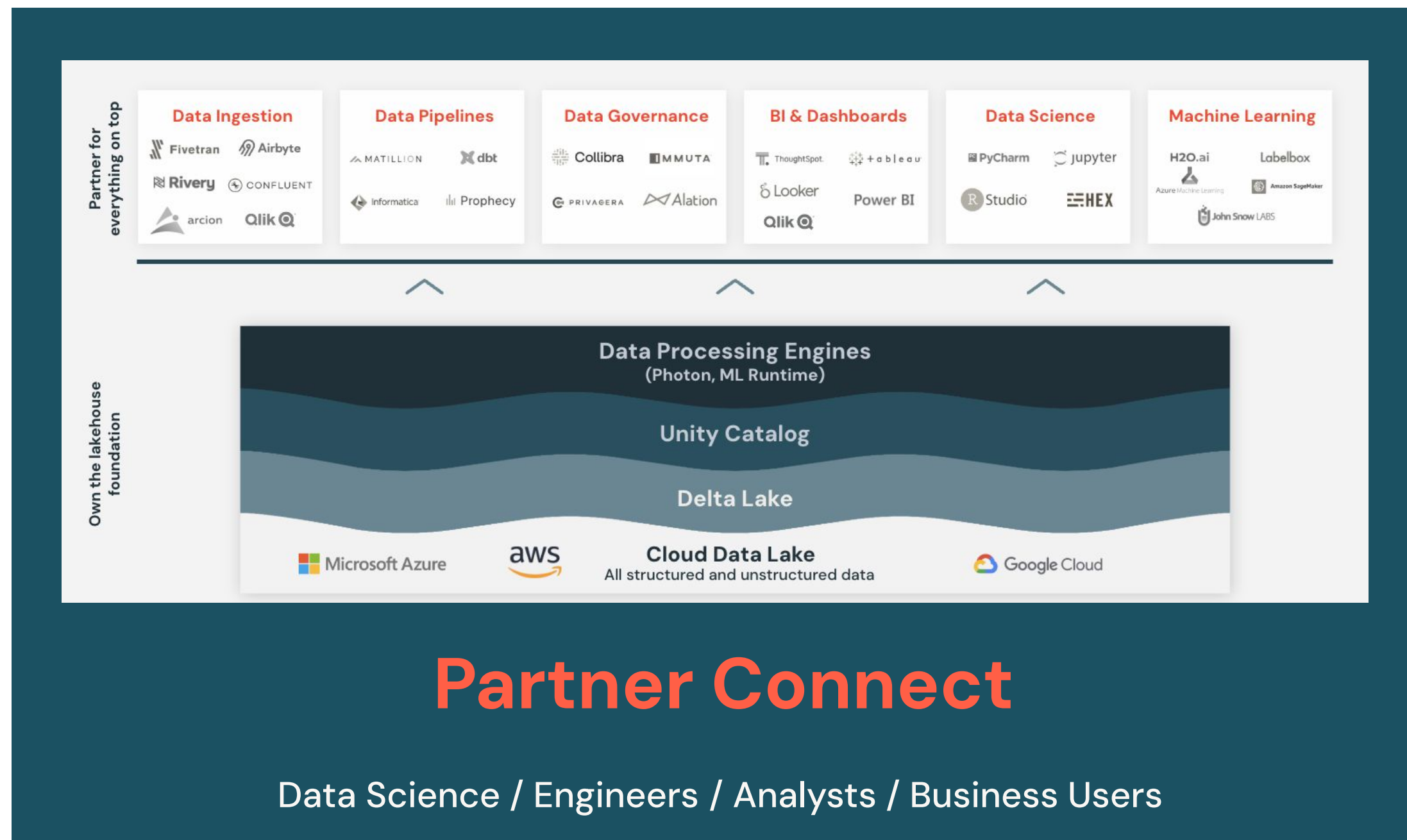
AI/BI Genie

Business Users



Tools for Intelligent Data Warehousing

Powered by Intelligence Engine





Databricks Overview

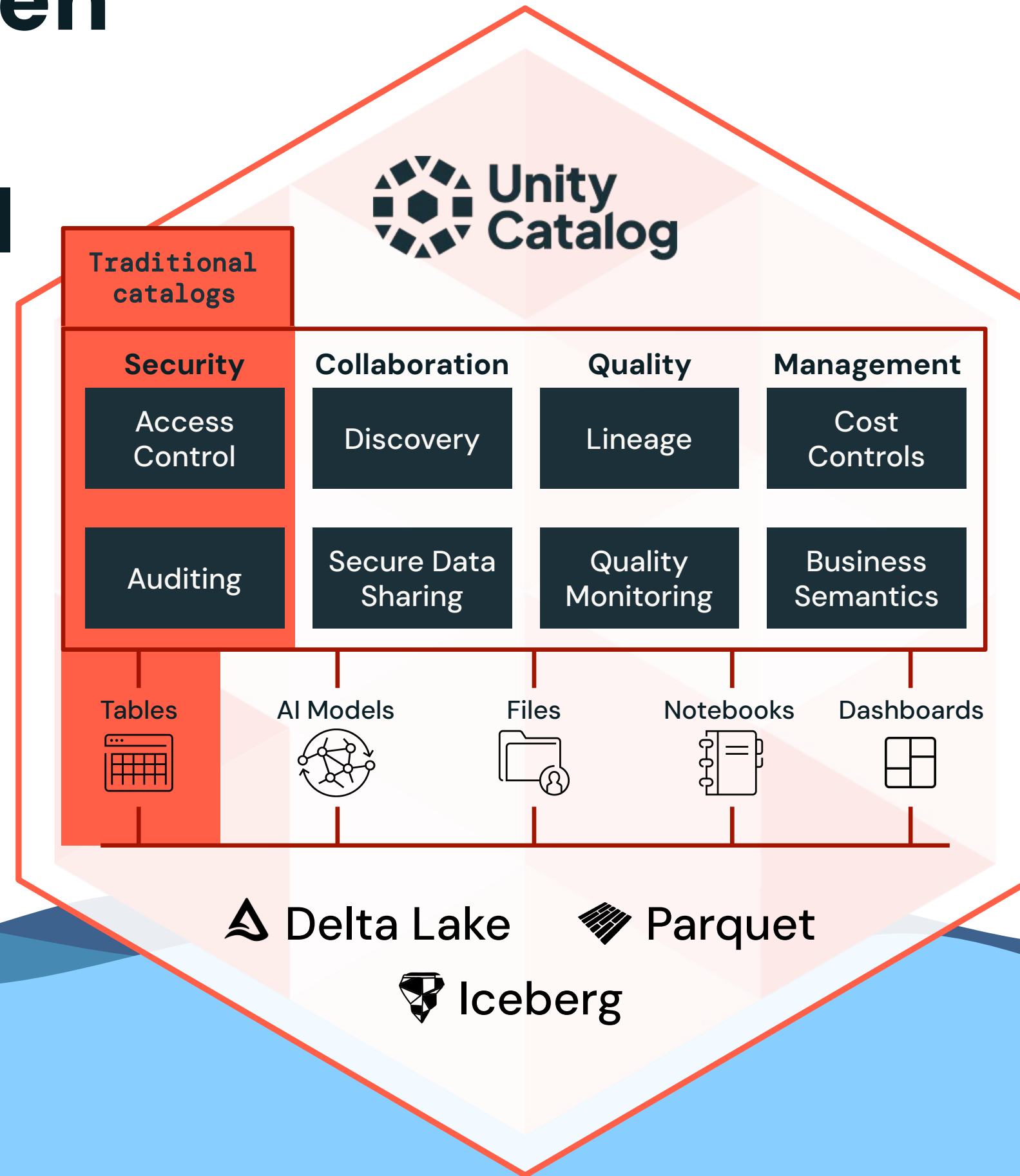
LECTURE

Unity Catalog Overview



Unified and open governance for all data + AI

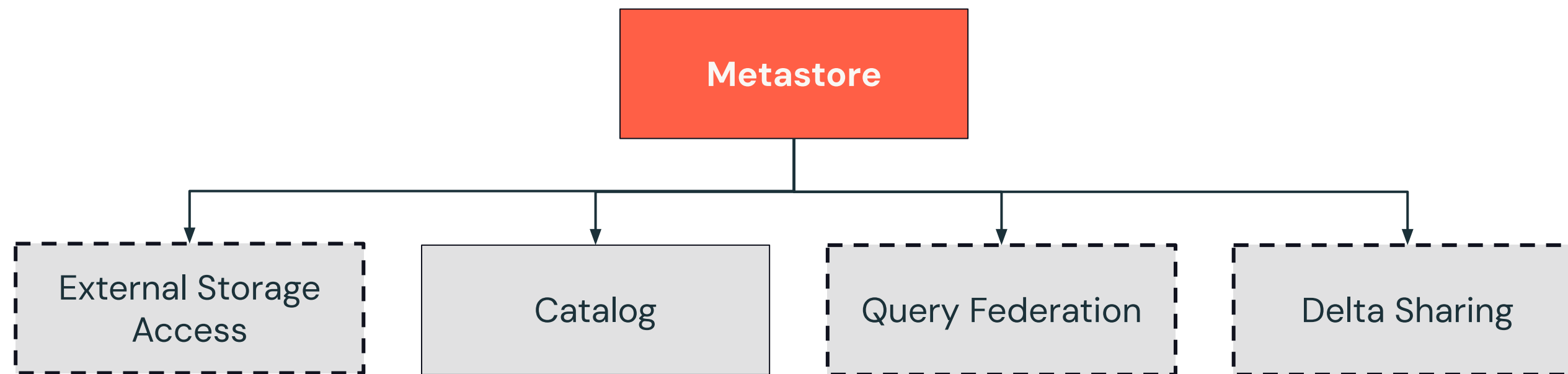
Connect to any external data source



Open access and collaboration with any tool, engine or platform

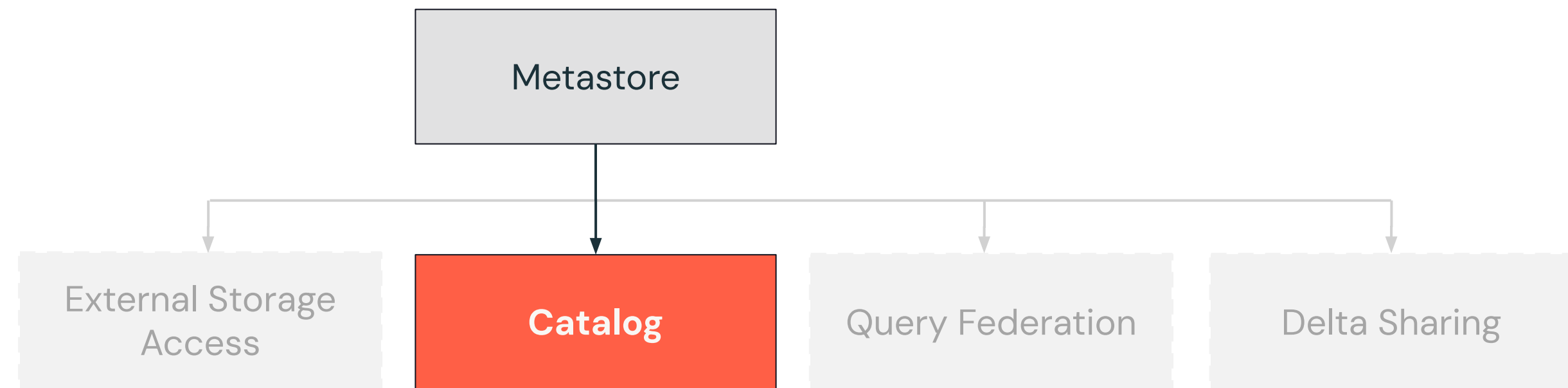
Unity Catalog Overview

Metastore



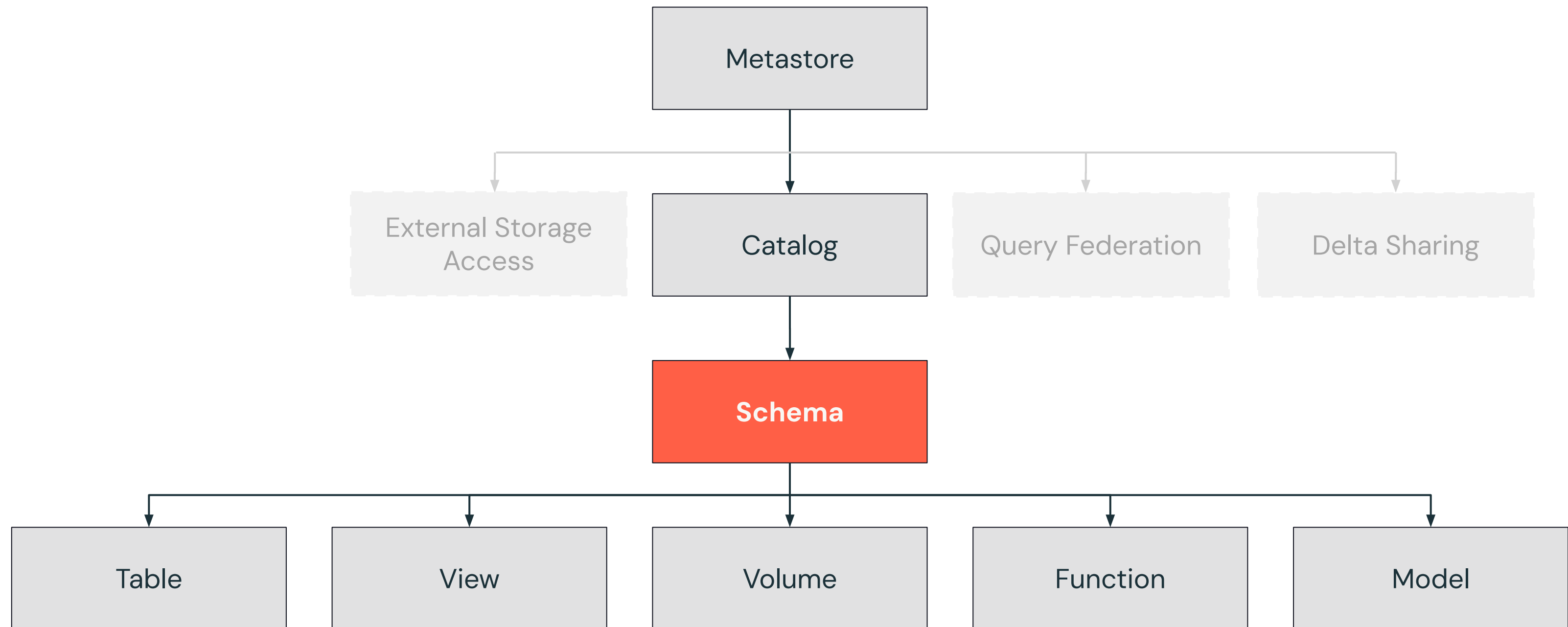
Unity Catalog Overview

Hierarchy of Objects in Unity Catalog

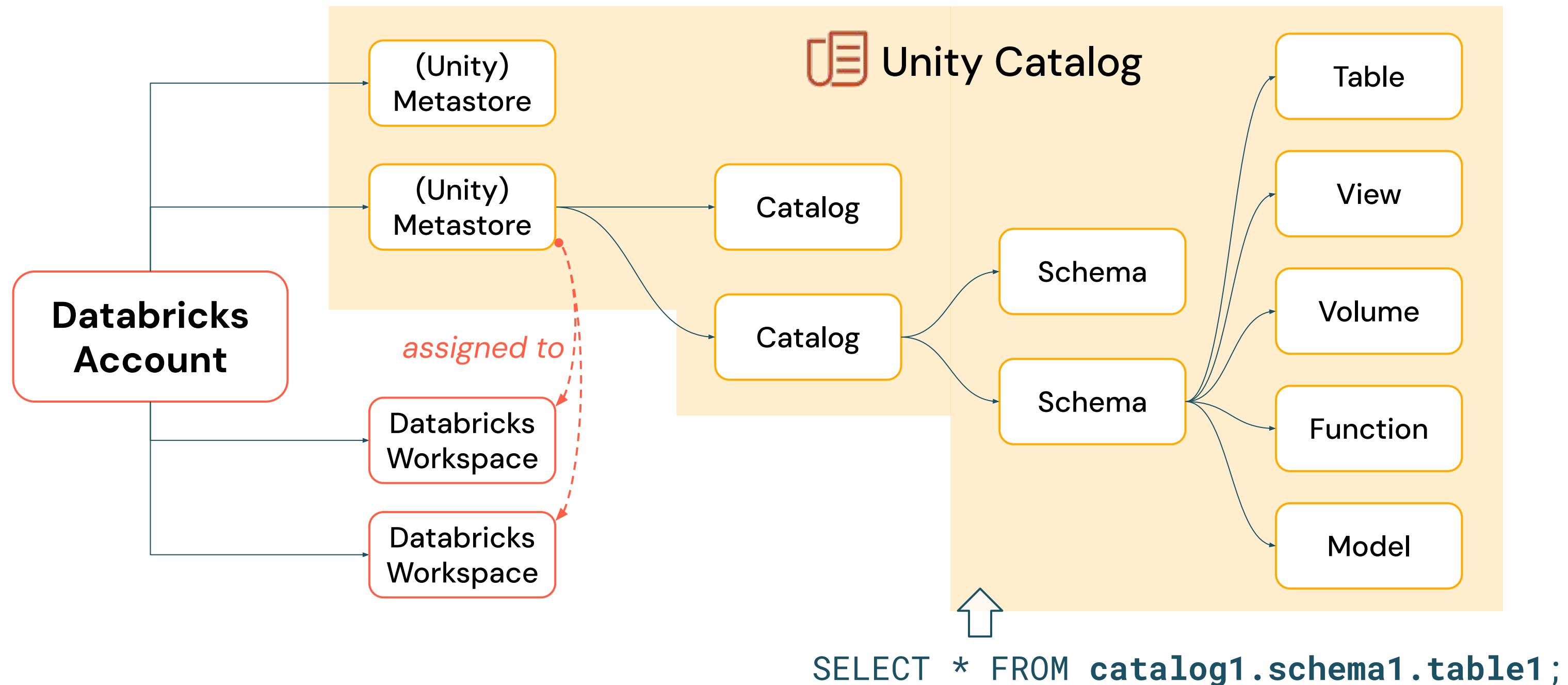


Unity Catalog Overview

Hierarchy of Objects in Unity Catalog



Unity Catalog Overview





Databricks Overview

LECTURE

Databricks SQL Warehouses



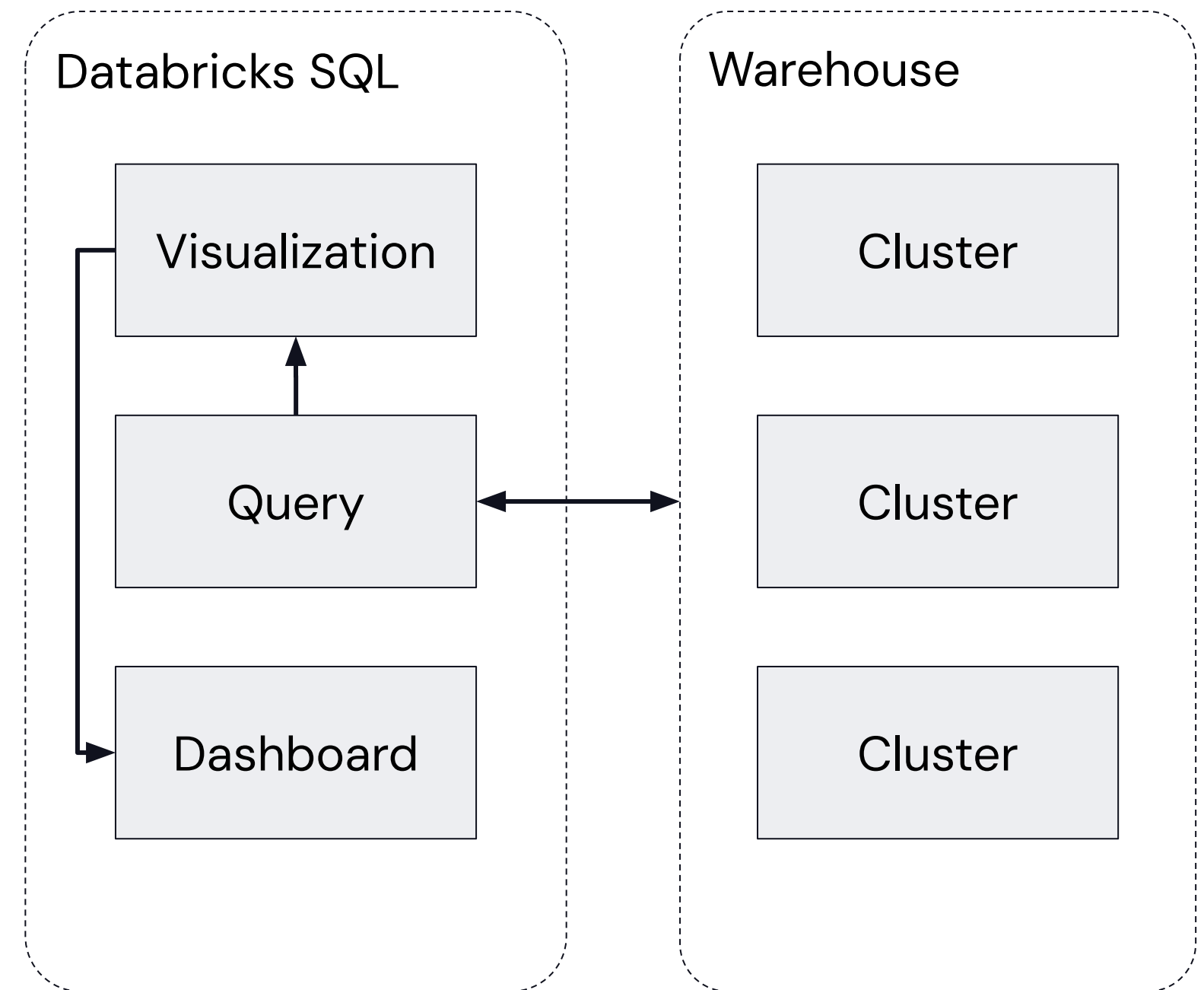
SQL Warehouses

Overview

SQL Warehouse in Databricks is a scalable compute resource optimized for executing SQL queries, analyzing data, creating visualizations, and supporting business intelligence (BI) Lakeflow Jobs.

Key Features:

- Scalable compute for distributed SQL Lakeflow Jobs.
- Optimized for data queries and BI Lakeflow Jobs.
- Supports integration with dashboards and analytics tools.



SQL Warehouses

Cluster Size:

- Refers to the computational power allocated to a SQL Warehouse.
- Expressed in “**t-shirt sizes**” such as **Small**, **Medium**, **Large**, and **X-Large**.
- Each size defines the number of compute nodes and capacity for parallel processing.
- **Example:**
 - **Small:** Suitable for light queries.
 - **X-Large:** Ideal for heavy, high-concurrency Lakeflow Jobs.

New SQL warehouse

Name

SQL warehouse name

Cluster size ⓘ

X-Large 80 DBU / h ▾

Auto stop

Scaling ⓘ

Type

2X-Small4 DBU / h

X-Small6 DBU / h

Small12 DBU / h

Medium24 DBU / h

Large40 DBU / h

✓ X-Large80 DBU / h

2X-Large144 DBU / h

3X-Large272 DBU / h

4X-Large528 DBU / h

Advanced options

Cancel

Create



SQL Warehouses

Scaling:

- Ensures SQL Warehouses adapt to varying Lakeflow Jobs by dynamically allocating resources.
- **Users can configure:**
 - **Minimum clusters:** Base capacity for consistent performance.
 - **Maximum clusters:** Auto-scale limit to handle peak demand.
- **Benefits:**
 - **Cost Efficiency:** Only pay for resources used.
 - **Performance Optimization:** Prevents resource bottlenecks.

New SQL warehouse

Name

SQL warehouse name

Cluster size ⓘ

X-Large 80 DBU / h / cluster ▾

Auto stop

☒ After 10 minutes of inactivity.

Scaling ⓘ

Min. 1 Max. 2 clusters (80 to 160 DBU)

Type

☒ Serverless ⓘ ☐ Pro ⓘ ☐ Classic

Advanced options

▼

Cancel

Create



SQL Warehouse Types

Databricks SQL Warehouses are designed to handle a wide variety of Lakeflow Jobs by offering three distinct types:

SQL Classic

- Provides a straightforward solution for running SQL queries.
- Operates entirely within the customer's cloud account, giving users complete control over their compute resources.

SQL Pro

- Balances performance and flexibility.
- Ideal for data engineering and analytics Lakeflow Jobs, with enhanced integration options for Lakeflow Jobs and tools.

SQL Serverless

- Offers the most advanced and modern architecture.
- Managed entirely by Databricks, ensuring seamless scaling, minimal setup time, and zero infrastructure maintenance.



Classic, Pro, and Serverless SQL Warehouses

Key Differences

	Classic SQL Warehouses	Pro SQL Warehouses	Serverless SQL Warehouses
	"GOOD"	"BETTER"	"BEST"
Compute Management:	Self-managed by the customer	Self-managed by the customer	Fully managed by Databricks, providing instant and elastic compute
Performance Features:	Includes the Photon Engine for massively parallel processing	Advanced features like Predictive I/O, integration with notebooks/jobs, Python UDFs, query federation, and materialized views	Optimized for high-concurrency BI Lakeflow Jobs with serverless query result caching and intelligent workload management
Use Cases:	Suitable for exploratory SQL Lakeflow Jobs or predictable scheduled Lakeflow Jobs where performance is not critical	Suited for SQL ETL/ELT and DSML use cases on the Databricks Data Intelligence Platform	Ideal for high-concurrency BI, SQL ETL, and DSML using SQL, with zero management overhead and instant scalability
Cluster Sizes:	Fixed compute capacity per unit, with scaling based on query throughput and incoming queries	Fixed compute capacity per unit, with dynamic scaling	Dynamic scaling based on real-time resource demands



Serverless Compute for Databricks SQL

Instant, elastic SQL compute decoupled from storage

- Quickly setup **instant, elastic SQL warehouse** – decoupled from storage – Powered by Photon
- Automatically determines instance types and configuration for best price/perf (up to 12x)
- **High concurrency** built-in, automatic load balancing
- Intelligent **workload management** and faster reads from cloud storage
- Instant startup, greater availability, and **40% average lower overall costs with serverless**

New SQL warehouse ×

Name

Cluster size ⓘ

Auto stop ☒ After minutes of inactivity.

Scaling ⓘ Min. Max. clusters (80 to 160 DBU)

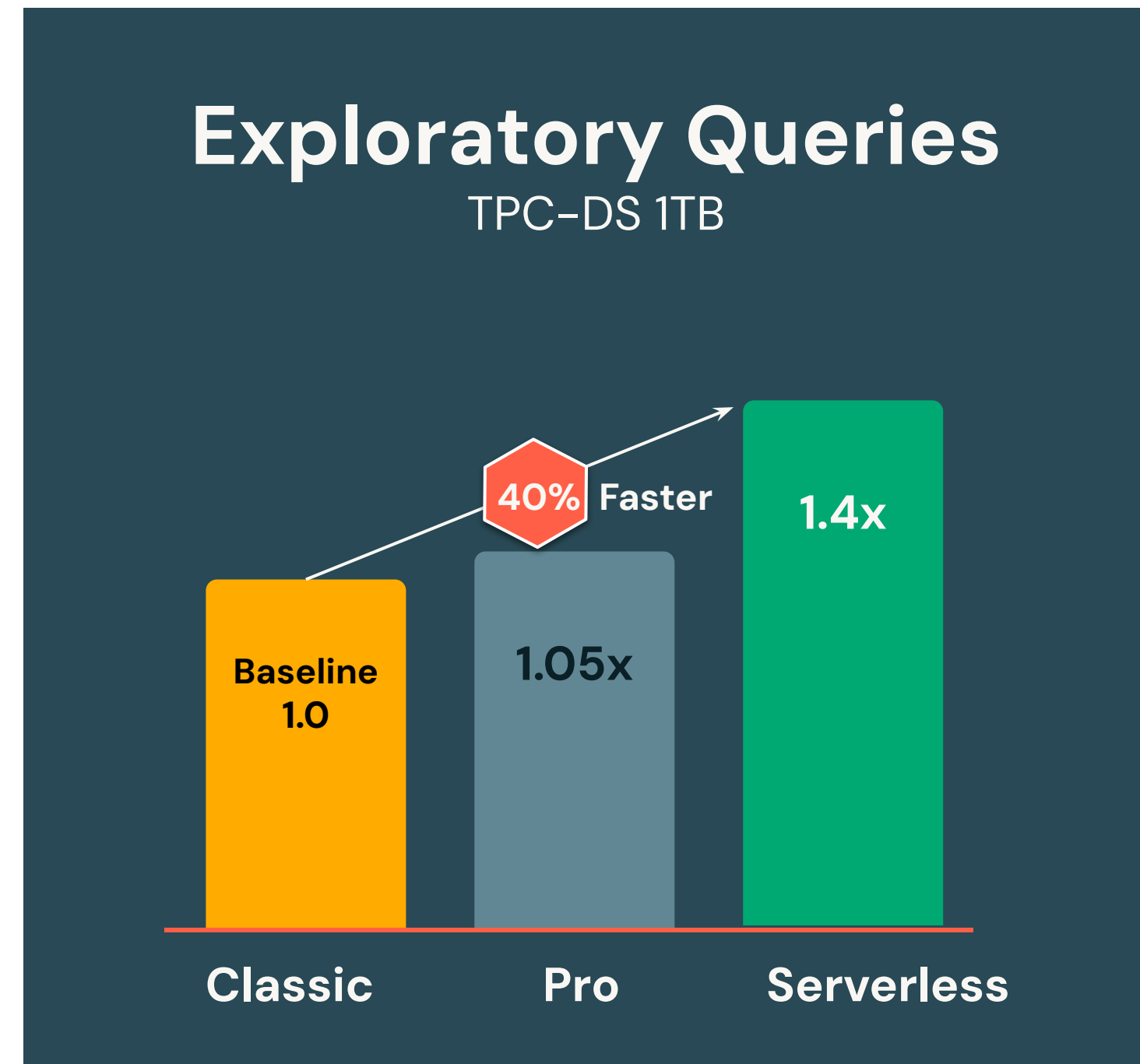
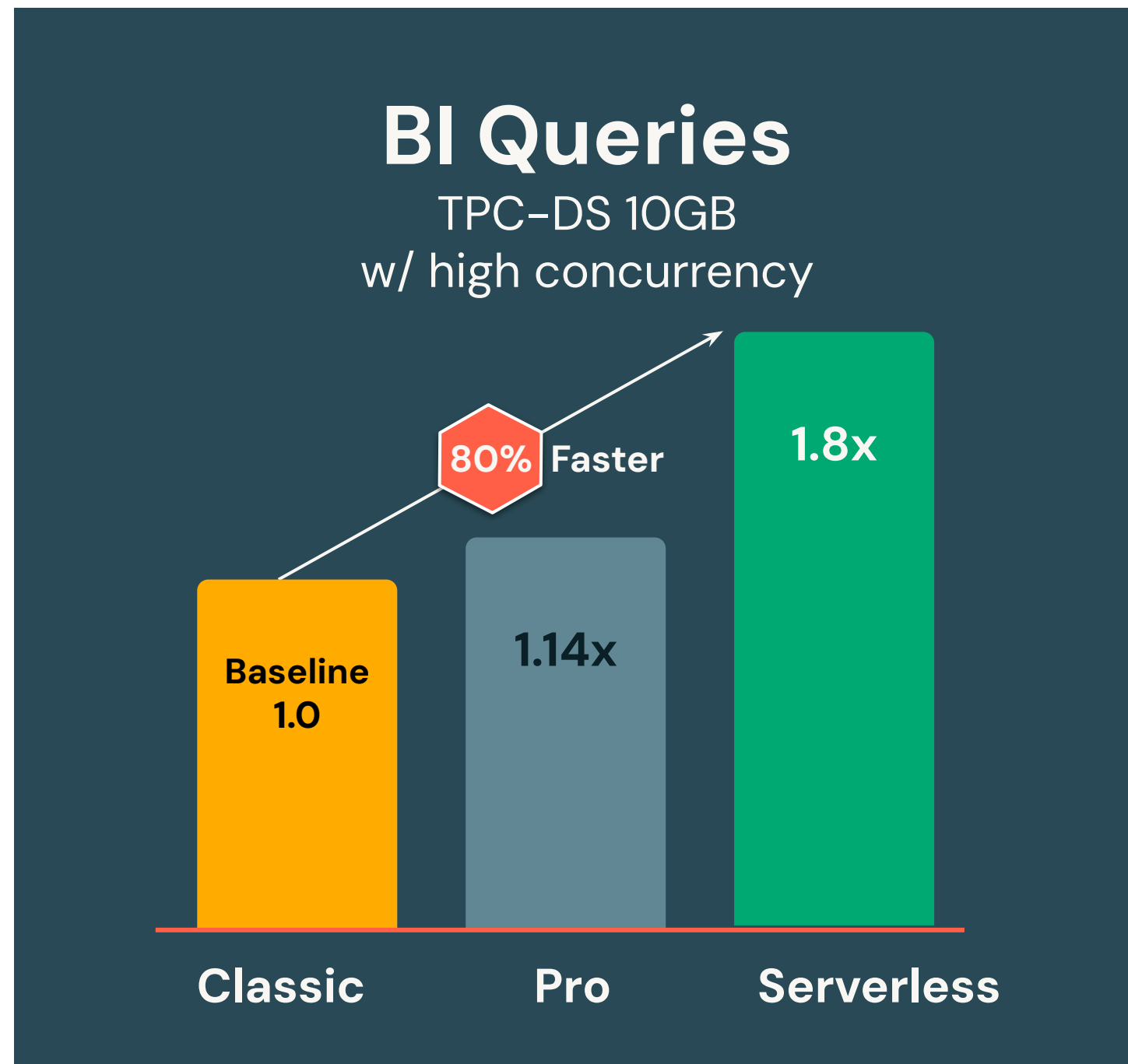
Type ☒ Serverless ⓘ ☐ Pro ⓘ ☐ Classic

Advanced options ▼



Serverless SQL has the Best Performance!

Let's see how they perform in real-world scenarios:

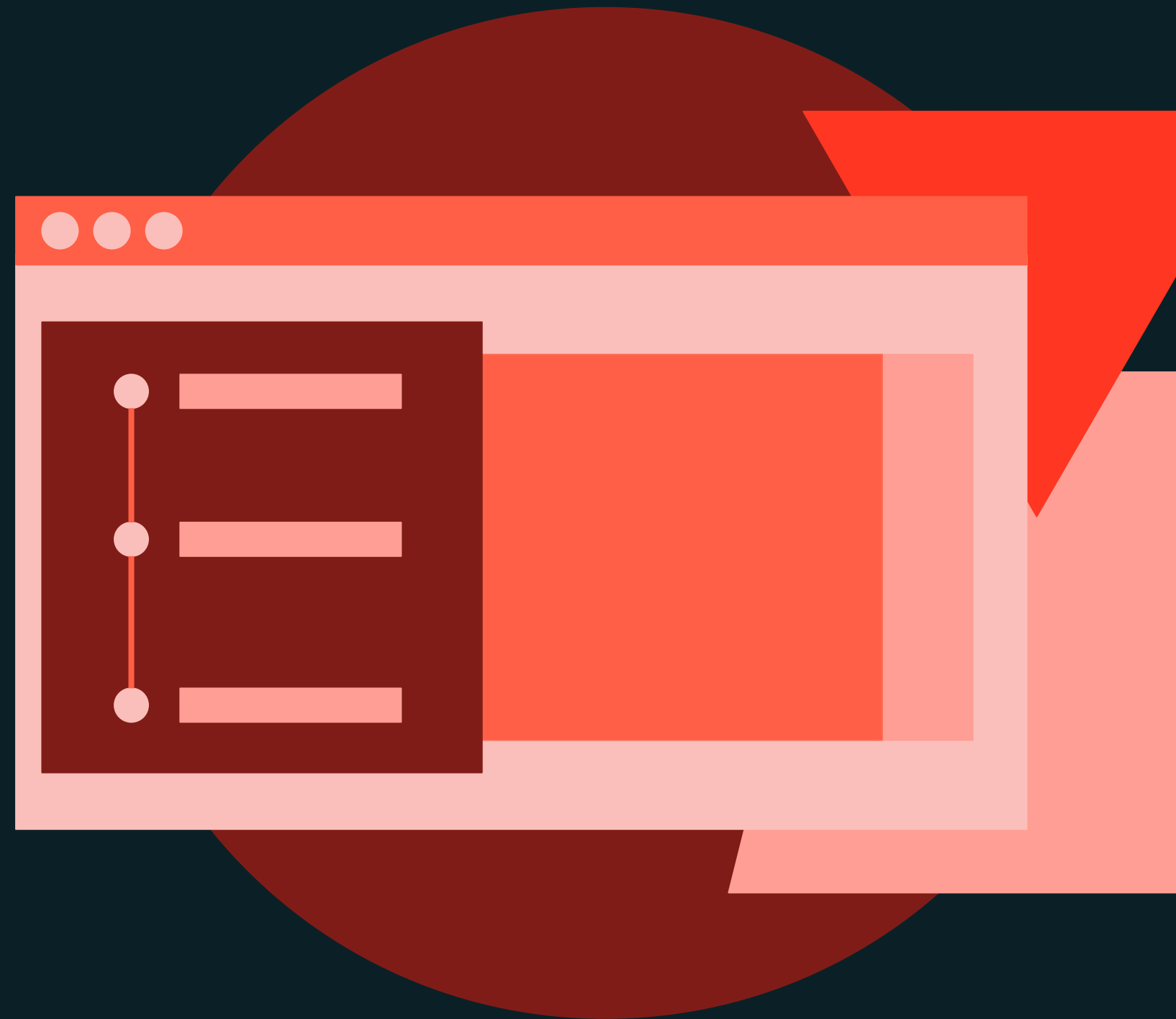




Databricks Overview

DEMONSTRATION

Preparing TPC-DI



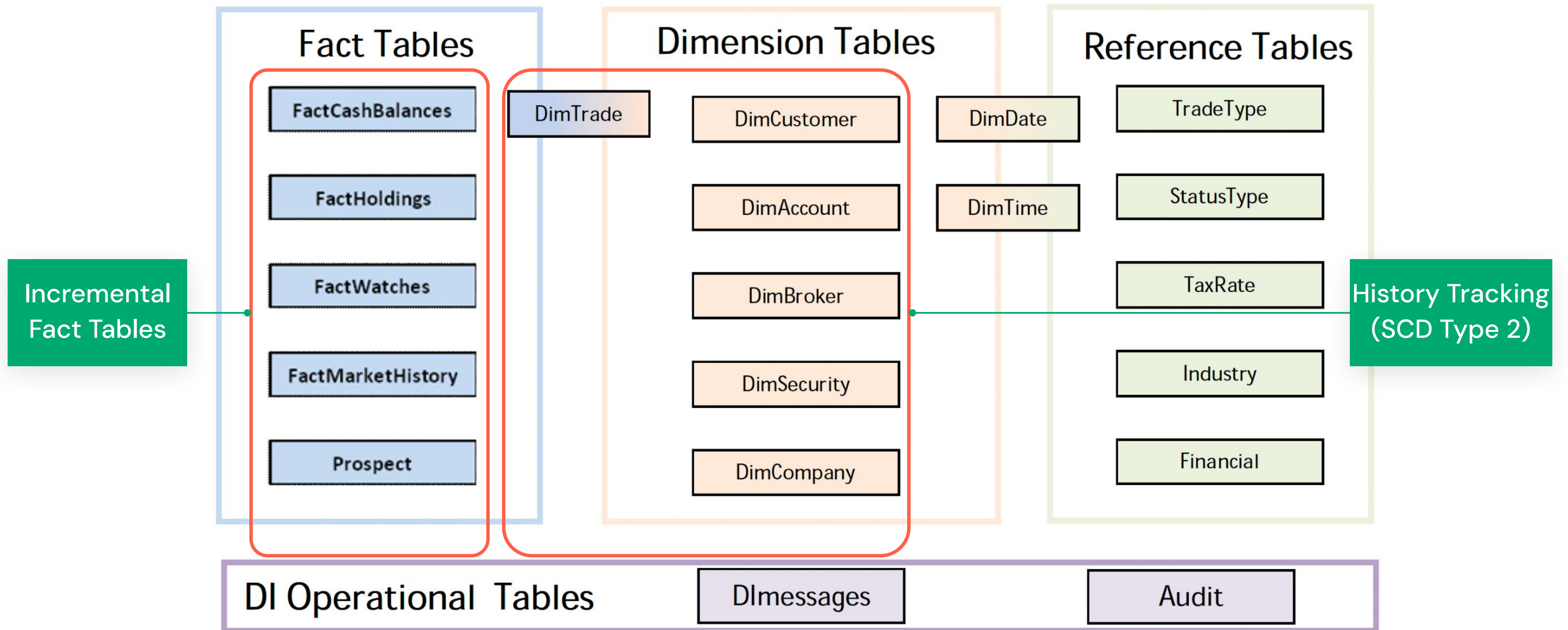
TPC's ETL/Data Integration Benchmark

- Extract and combine data from a variety of data source formats, transforming that data into a unified data model representation, and loading it into a data store.
- The TPC-DI benchmark combines and transforms data extracted from an On-Line Transaction Processing (OLTP) system along with other sources of data, and loads it into a data warehouse.
- The source and destination data models, transformations, and rules were designed by TPC to be broadly representative traditional/modern data integration requirements.



Warehouse Data Model

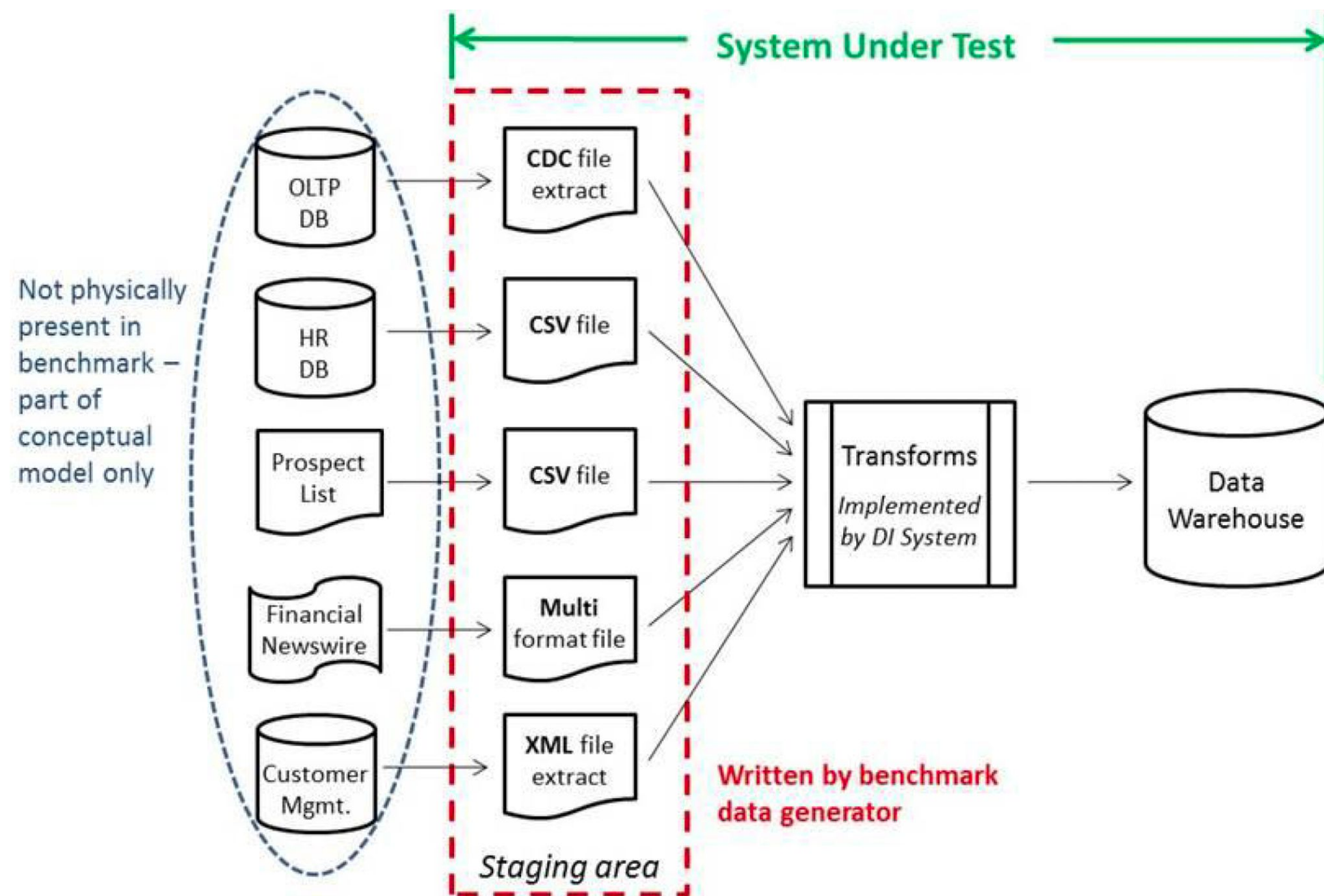
These tables are exposed to users to consume downstream



What are we ingesting?

Depends on target table

- Mostly TXT files
 - Most files come from Financial Newswire data that are fixed length TXT
 - Most data, however, comes from fact table data from OLTP as TXT files
- 1 file is XML (customer & account history)
- A handful are CSV



Follow-along Instructions

Estimated Time: 20 minutes

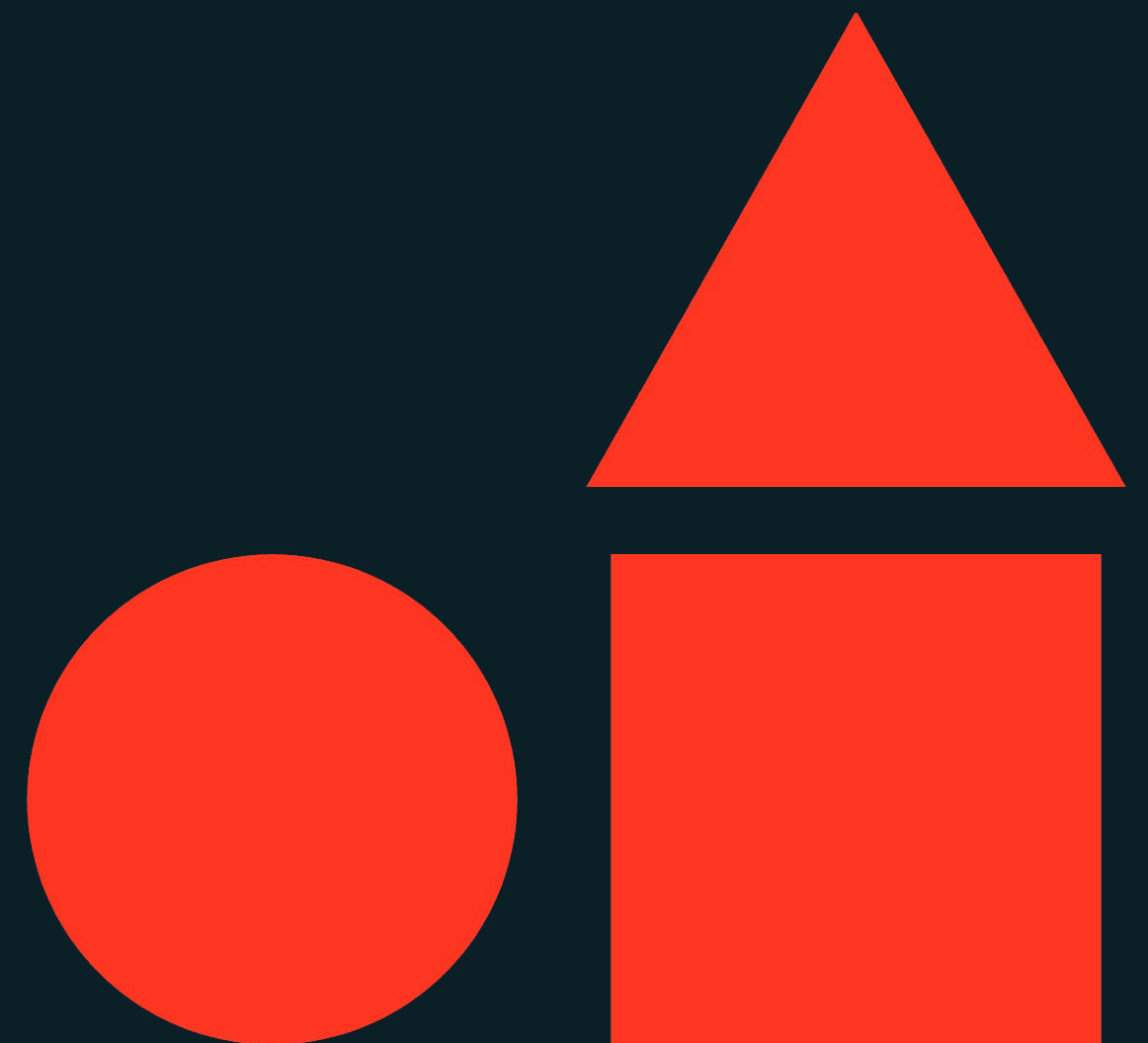
For this demonstration, the instructor will take you on a brief guided tour of the platform, highlighting the notebooks and dataset to be used throughout the rest of the course.





Ingesting and Transforming Data

Data Warehousing with Databricks



Objectives

- Describe the different options for ingesting data into Databricks
- Leverage Databricks capabilities to ingest, transform, and explore your data





Ingesting and Transforming Data

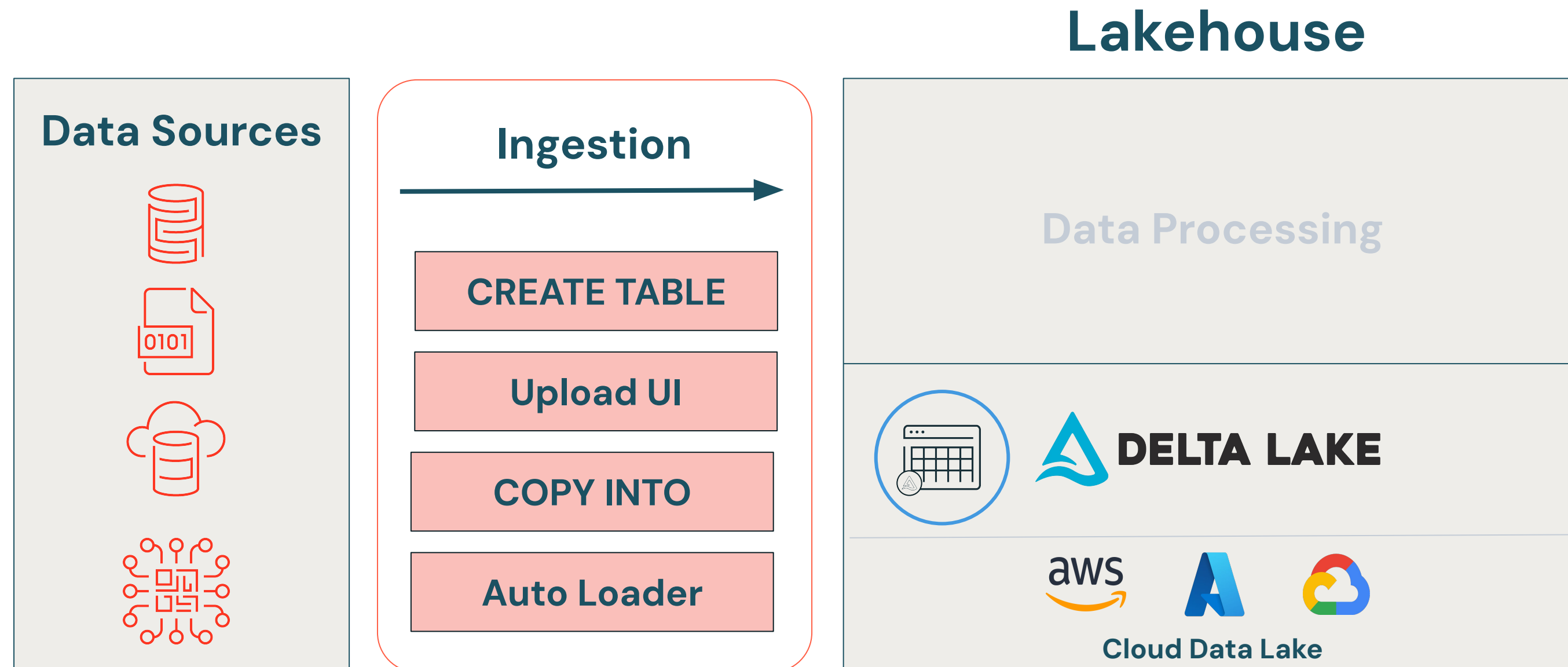
LECTURE

Data Ingestion Techniques



Delta Lake Data Ingestion Techniques

Overview



Delta Lake Data Ingestion Techniques

Overview

CREATE TABLE

Upload UI

COPY INTO

Auto Loader



Delta Lake Data Ingestion Techniques

Overview

CREATE TABLE

Upload UI

COPY INTO

Auto Loader

```
CREATE TABLE mydeltatable  
USING DELTA  
AS  
<query>
```

USING DELTA is optional.

- CREATE TABLE AS (CTAS) creates a table by selecting data from an existing table or data source
- Default table format is **Delta**



Delta Lake Data Ingestion Techniques

Overview

CREATE TABLE

Upload UI

COPY INTO

Auto Loader

- The UPLOAD UI provides a **point-and-click interface** to upload files and create tables
- Allows you to upload CSV, TSV, JSON, Avro, Parquet, or text files
- The UI also enables you to upload files directly to a Unity Catalog volume



Delta Lake Data Ingestion Techniques

Overview

CREATE TABLE

Upload UI

COPY INTO

Auto Loader

```
COPY INTO mydeltatable  
FROM 'your-path'  
FILE_FORMAT = 'format'  
FILE_OPTIONS = ('format-options')
```

- Loads files from a **file location** into a table
- Supports **various file formats** and cloud storage locations
- Automatically can handle **schema changes**
- **Idempotent** operation—Already-loaded files in the source are skipped



Delta Lake Data Ingestion Techniques

Overview

CREATE TABLE

Upload UI

COPY INTO

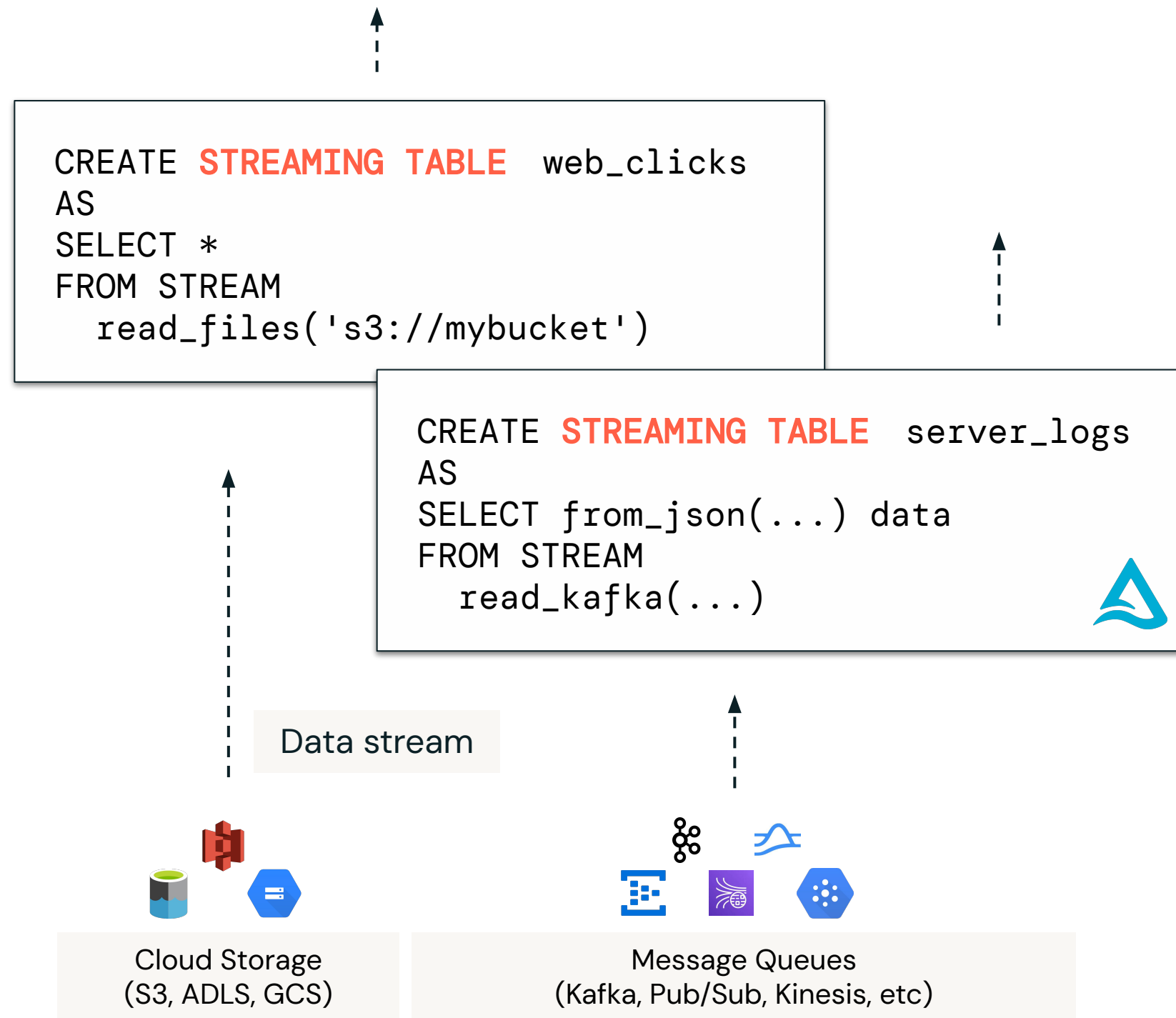
Auto Loader

- Auto Loader **incrementally (streaming) processes** new data files as they arrive in cloud storage
- Automatically **infers schemas** and accommodates **schema changes**
- Includes a **rescue data** column for data that does not adhere to the schema



Streaming Tables in Delta Lake

Real-Time and Incremental Data Processing Made Simple

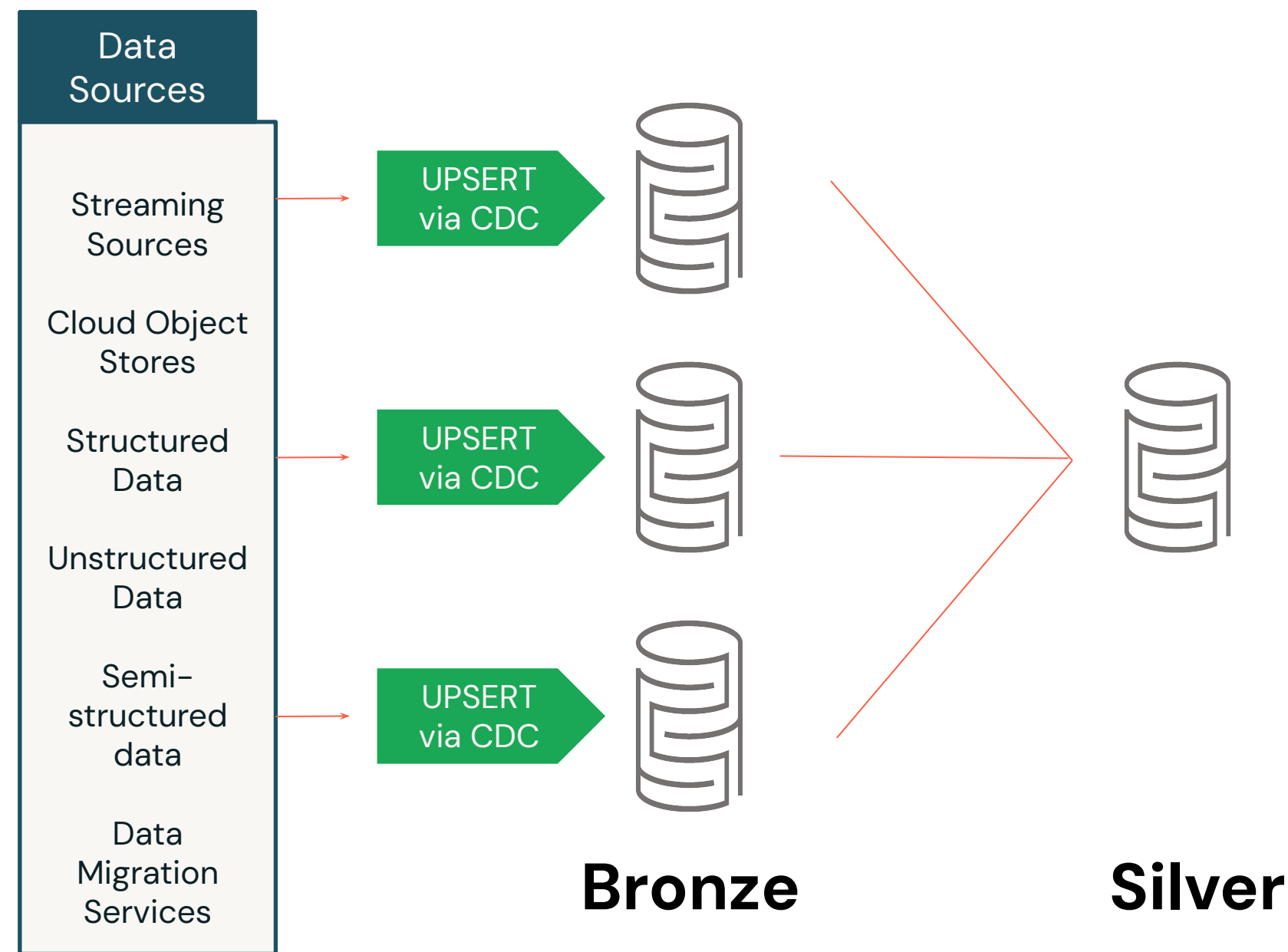


Benefits:

1. **Unlock real-time use cases.** Ability to support real-time analytics/BI, machine learning and operational use cases with streaming data.
2. **Better scalability.** More efficiently handle high volumes of data via incremental processing vs large batches.
3. **Enable more practitioners.** Simple SQL syntax makes data streaming accessible to all data engineers and analysts.



Change data capture (CDC)



Stream row-level changes from any supported data source

Handles out-of-order events

Schema evolution

SCD type 1 and 2 support

Process change records incrementally using a simple API:

APPLY CHANGES INTO





Databricks LakeFlow Connect

Native ingestion connectors for the Data Intelligence Platform

Efficient ingestion pipelines

Simple setup and maintenance

Unified orchestration, observability, and governance



Setting the scene



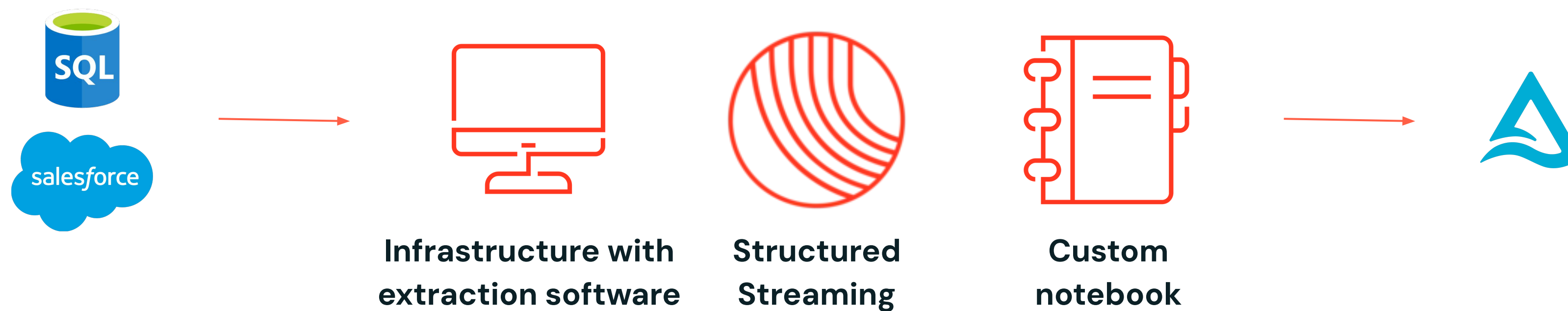
I'm a data engineer at a company



My data lives in several places, including Salesforce and SQL Server



Before



After





Ingesting and Transforming Data

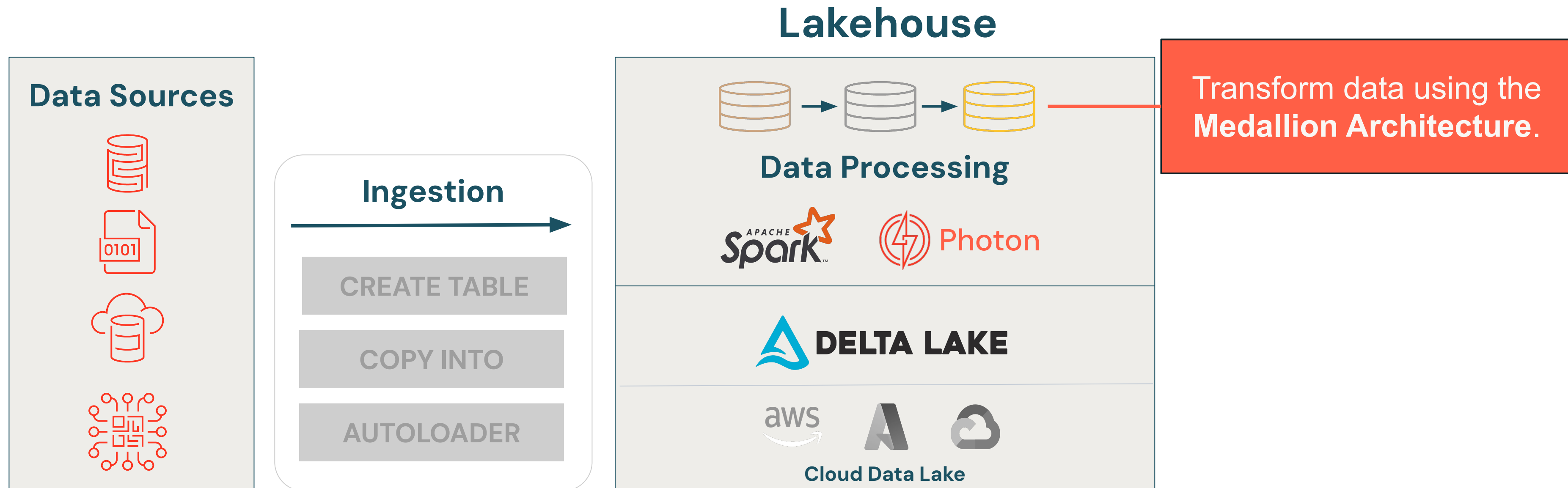
LECTURE

Data Transformation



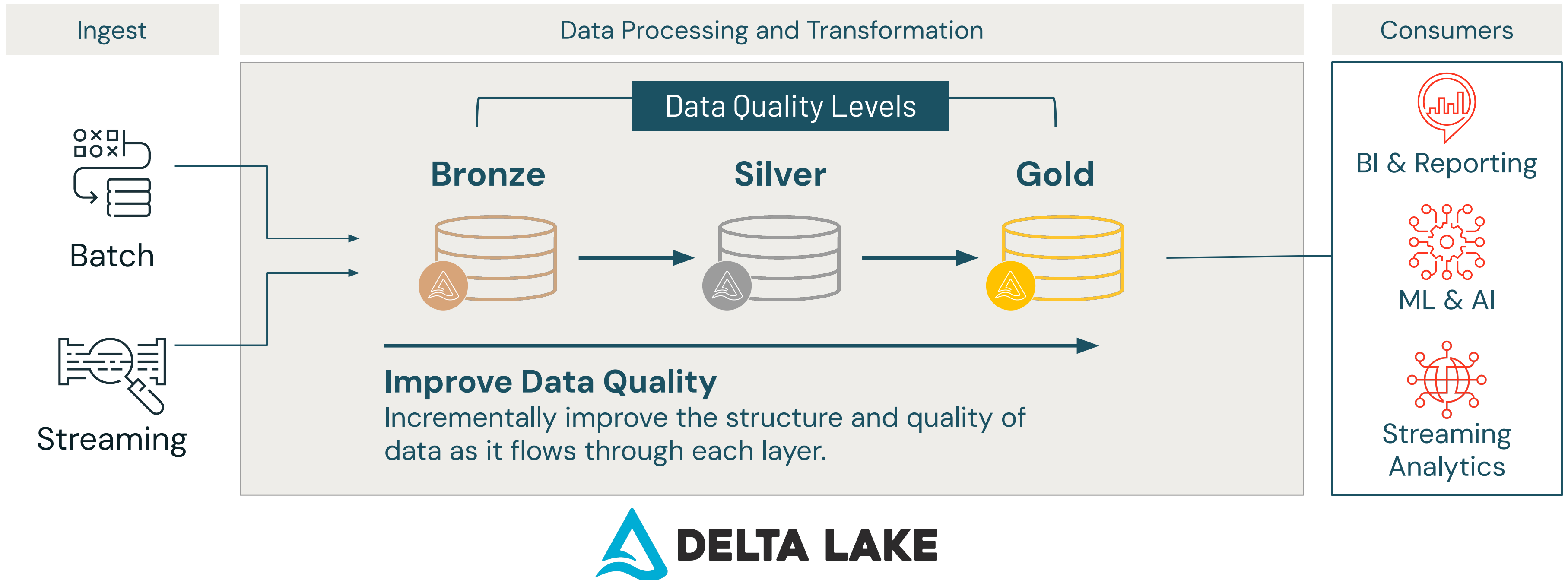
Data Transformation Overview

Medallion Architecture (Multi Hop)



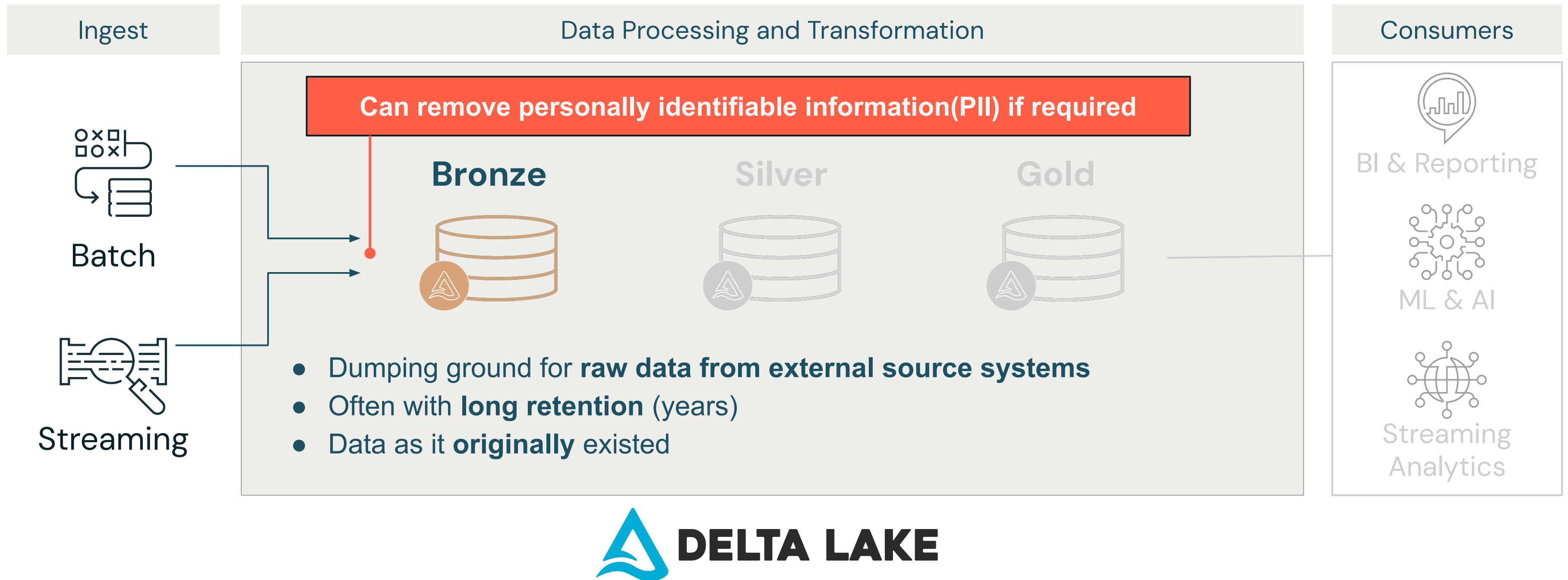
Data Transformation Overview

Medallion Architecture (Multi Hop)



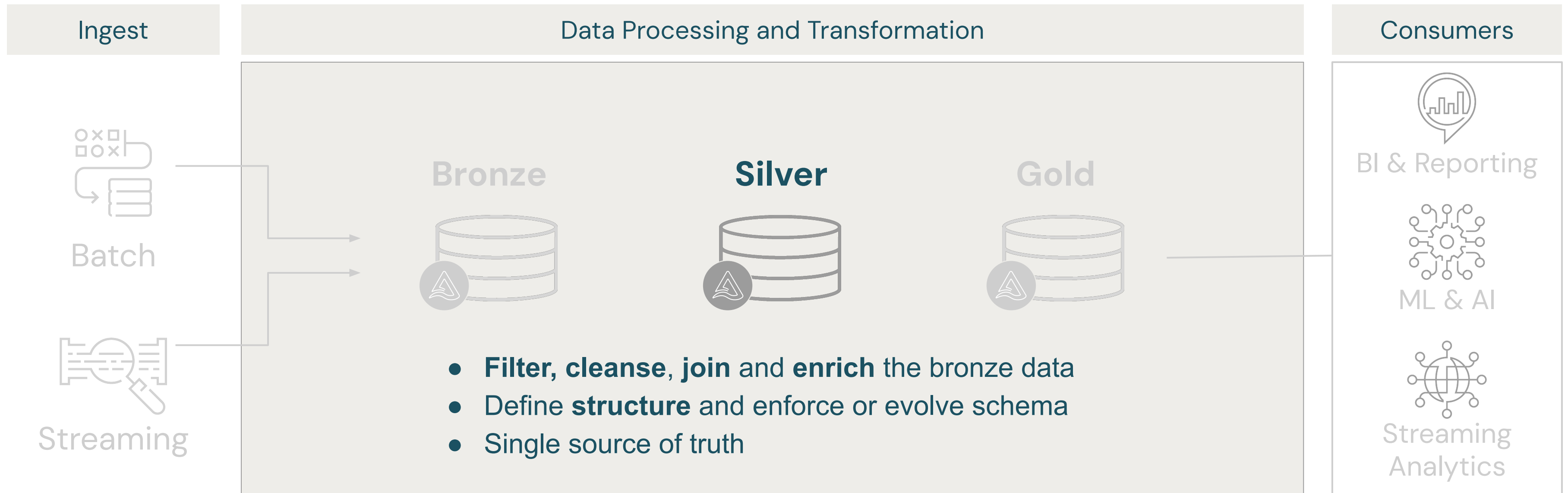
Data Transformation Overview

Medallion Architecture (Multi Hop)



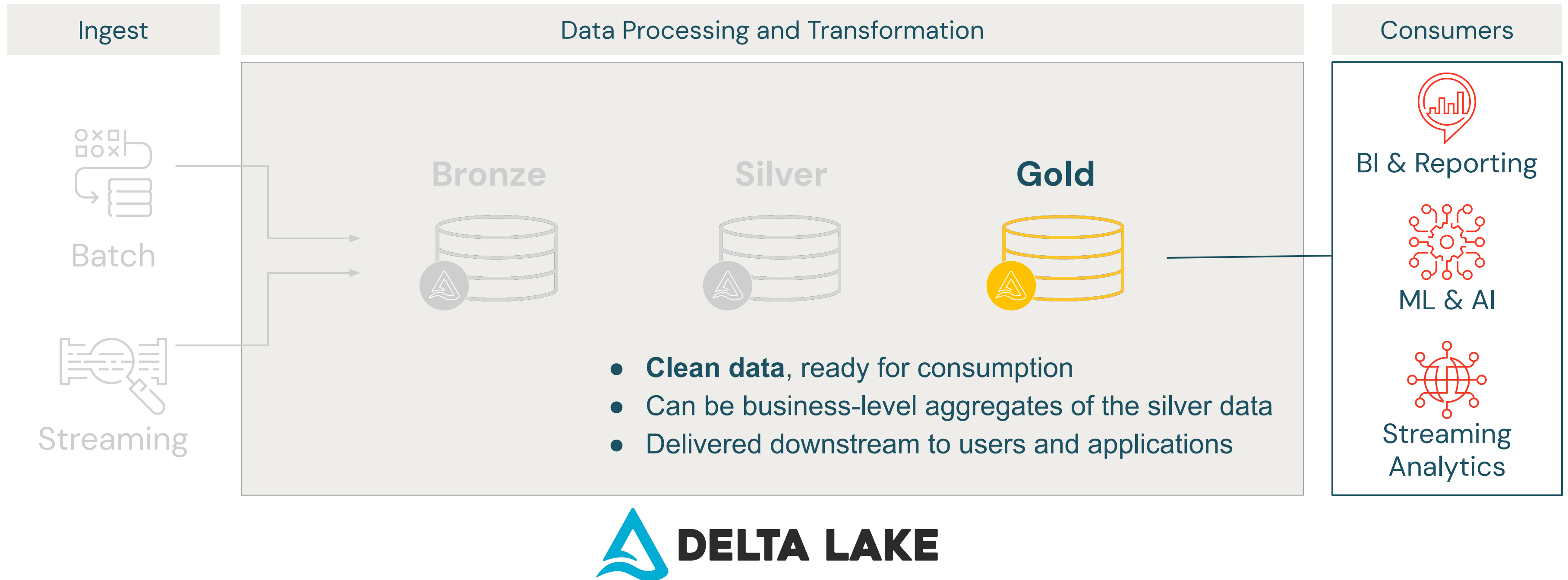
Data Transformation Overview

Medallion Architecture (Multi Hop)



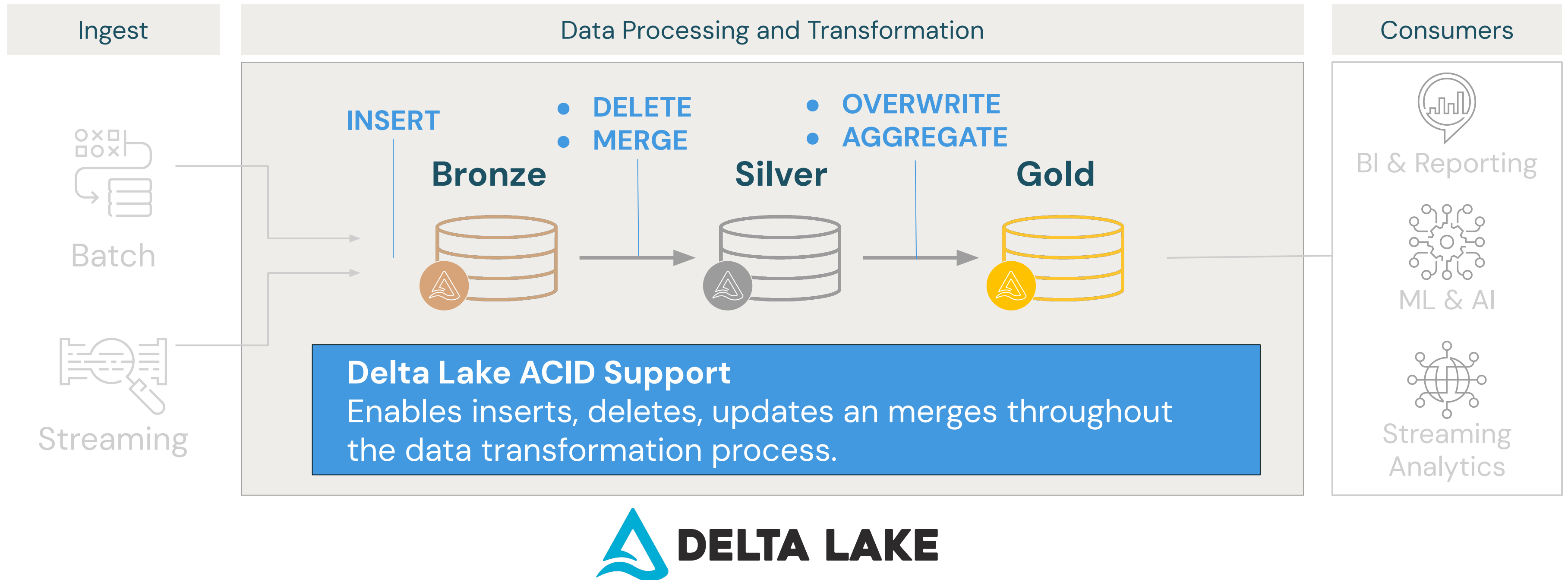
Data Transformation Overview

Medallion Architecture (Multi Hop)



Data Transformation Overview

Medallion Architecture (Multi Hop)

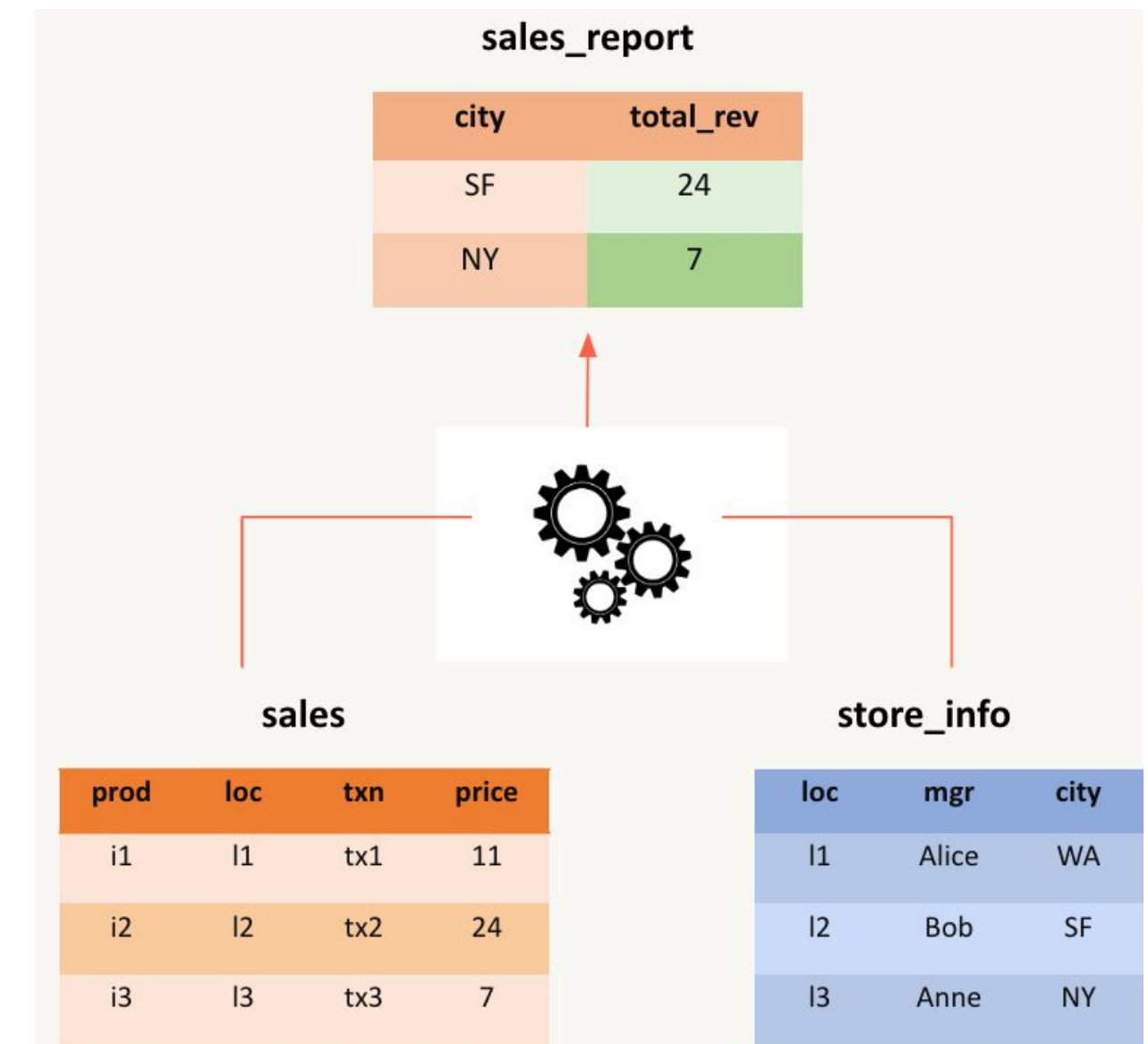


Materialized Views

Speed up queries with pre-computed results

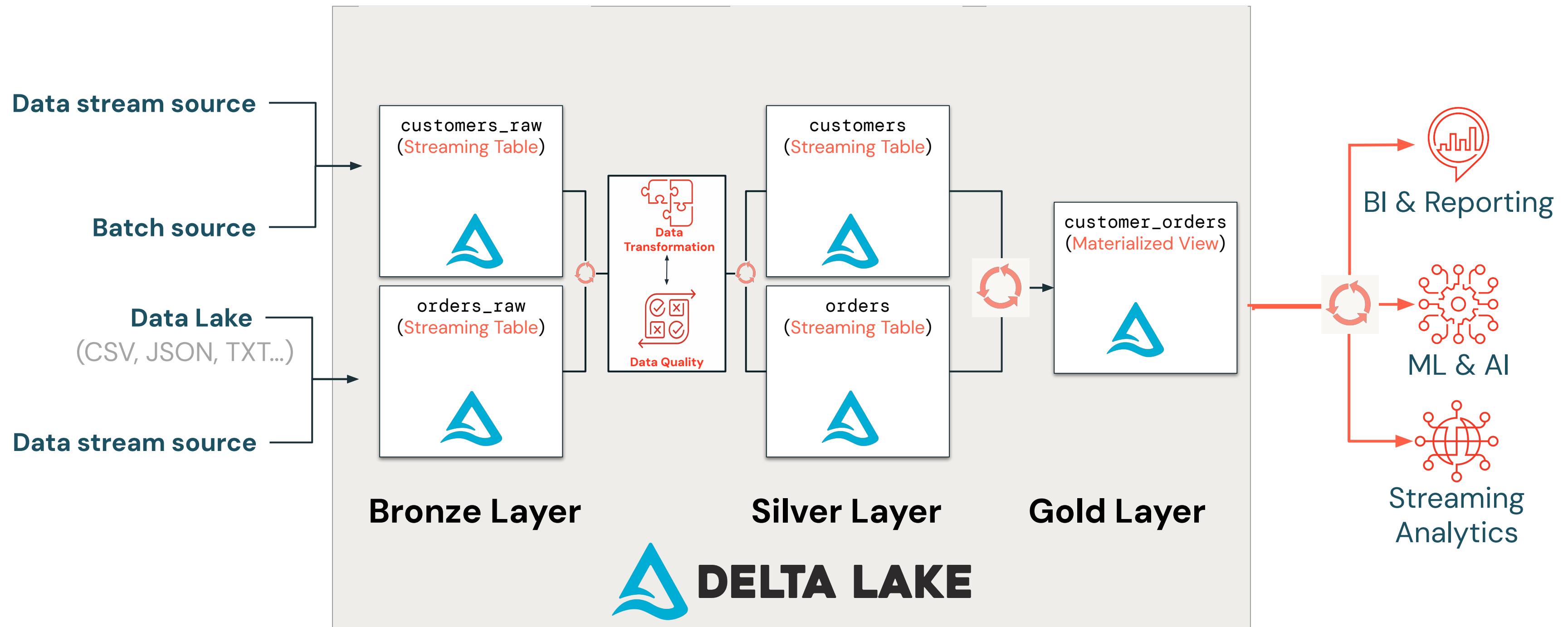
Accelerate end-user queries and reduce infrastructure costs with efficient, incremental computation

- Accelerate **BI dashboards** and **ETL queries**
- **Streaming**: build MVs on top of streaming tables
- **Easy ELT**: Simplify reporting by cleaning, enriching, denormalizing the base tables
- **Data Sharing & Access Control**: control what info can be seen by internal and external users and organizations



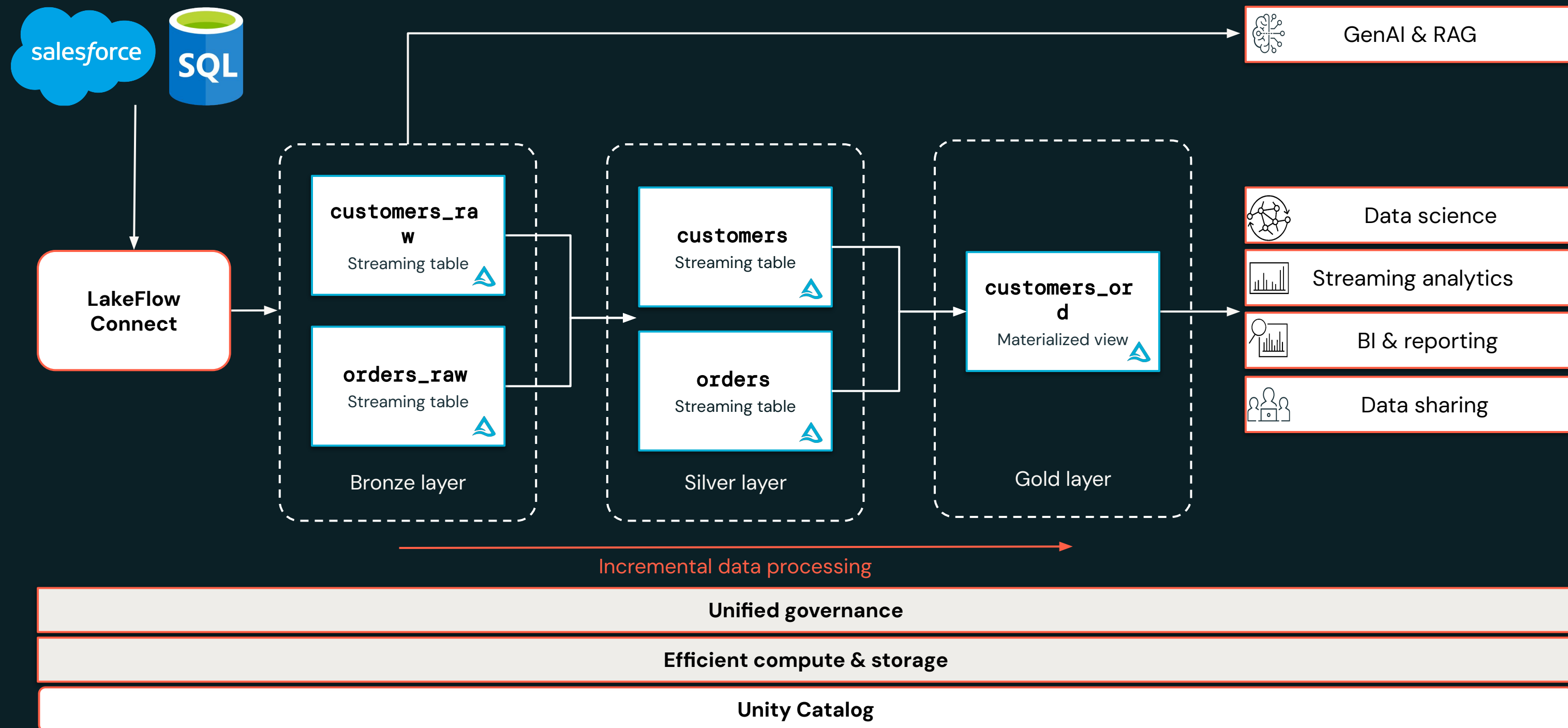
Constructing a Data Transformation Pipeline

Real World Architecture is Typically More Complex



LakeFlow Connect in Action

Medallion Architecture powered by Delta Lake ensures reliable, scalable, and high-performance data storage and analytics.

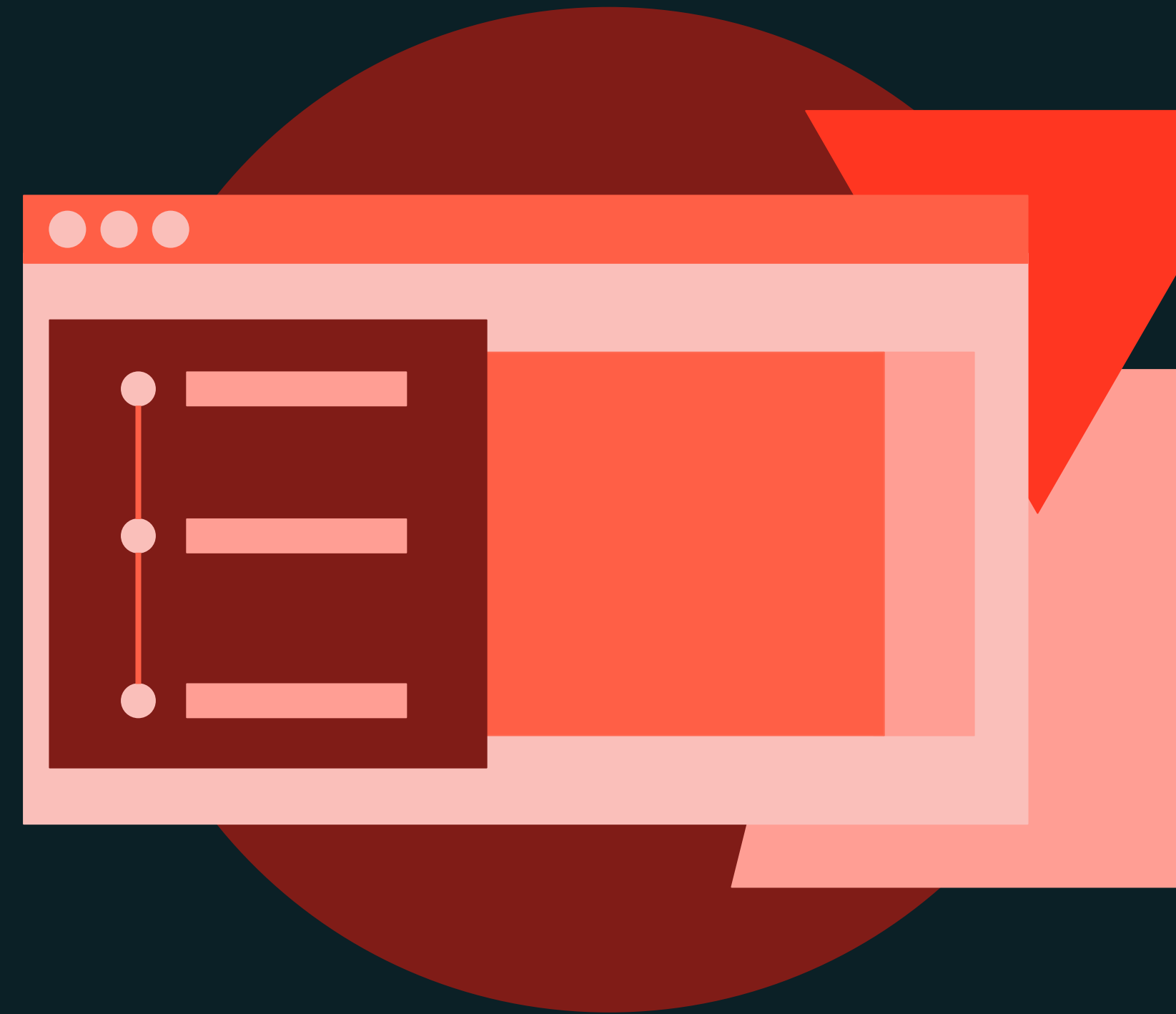




Ingesting and Transforming Data

DEMONSTRATION

Ingesting and Exploring Data



Follow-along Instructions

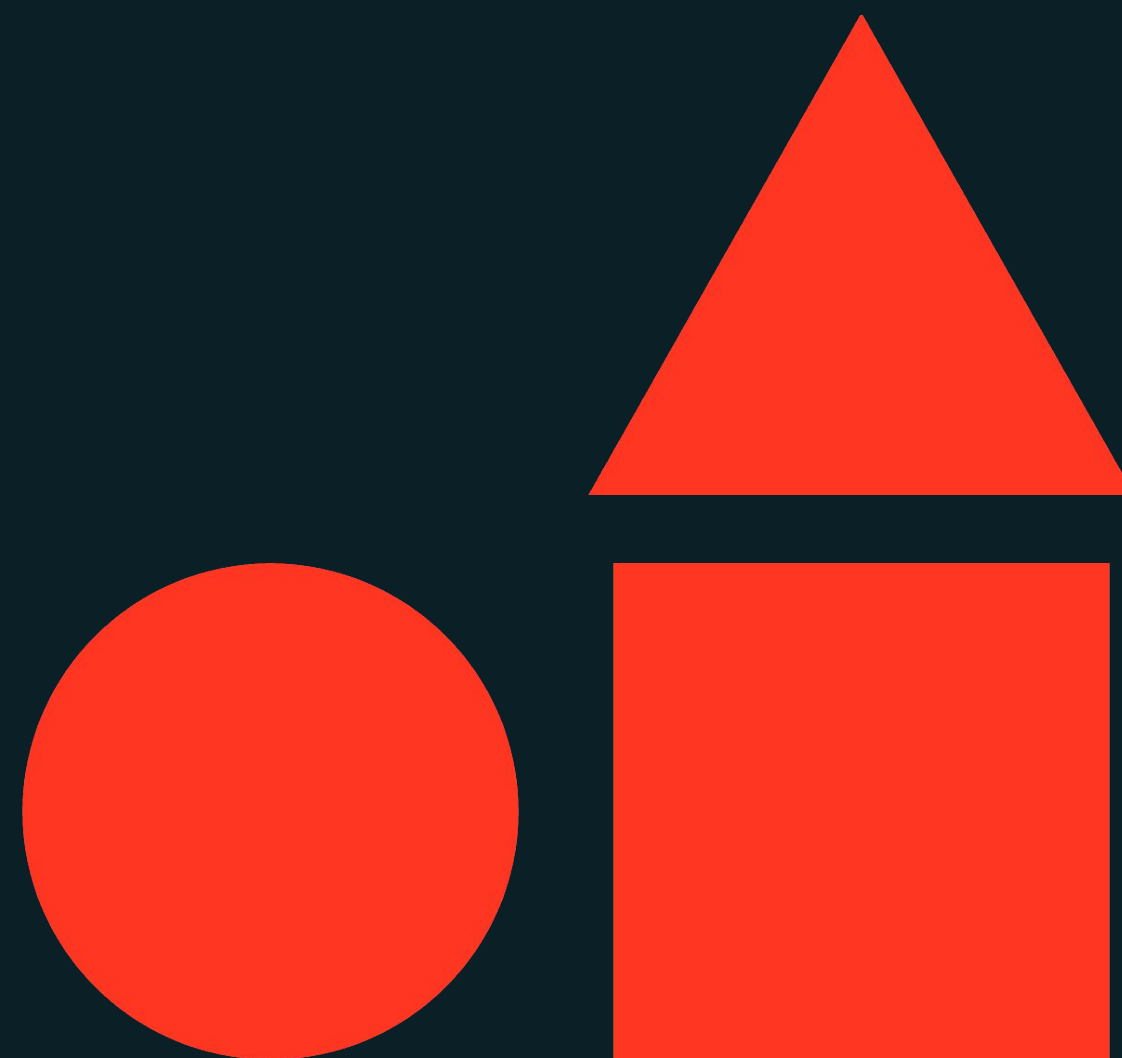
Estimated Time: 20 minutes

For this demonstration, the instructor will take you through an example ETL process exemplifying the medallion architecture. If you have access to the Vocareum lab environment, feel free to follow along.



Orchestration

Data Warehousing with Databricks



Objectives

- Describe Databricks Lakeflow jobs and the benefits of serverless
- Describe the types of tasks supported
- Understand and apply advanced job features
- Use jobs to automate data ingestion and transformation
- Trigger and schedule jobs, and set up notifications





Orchestration

LECTURE

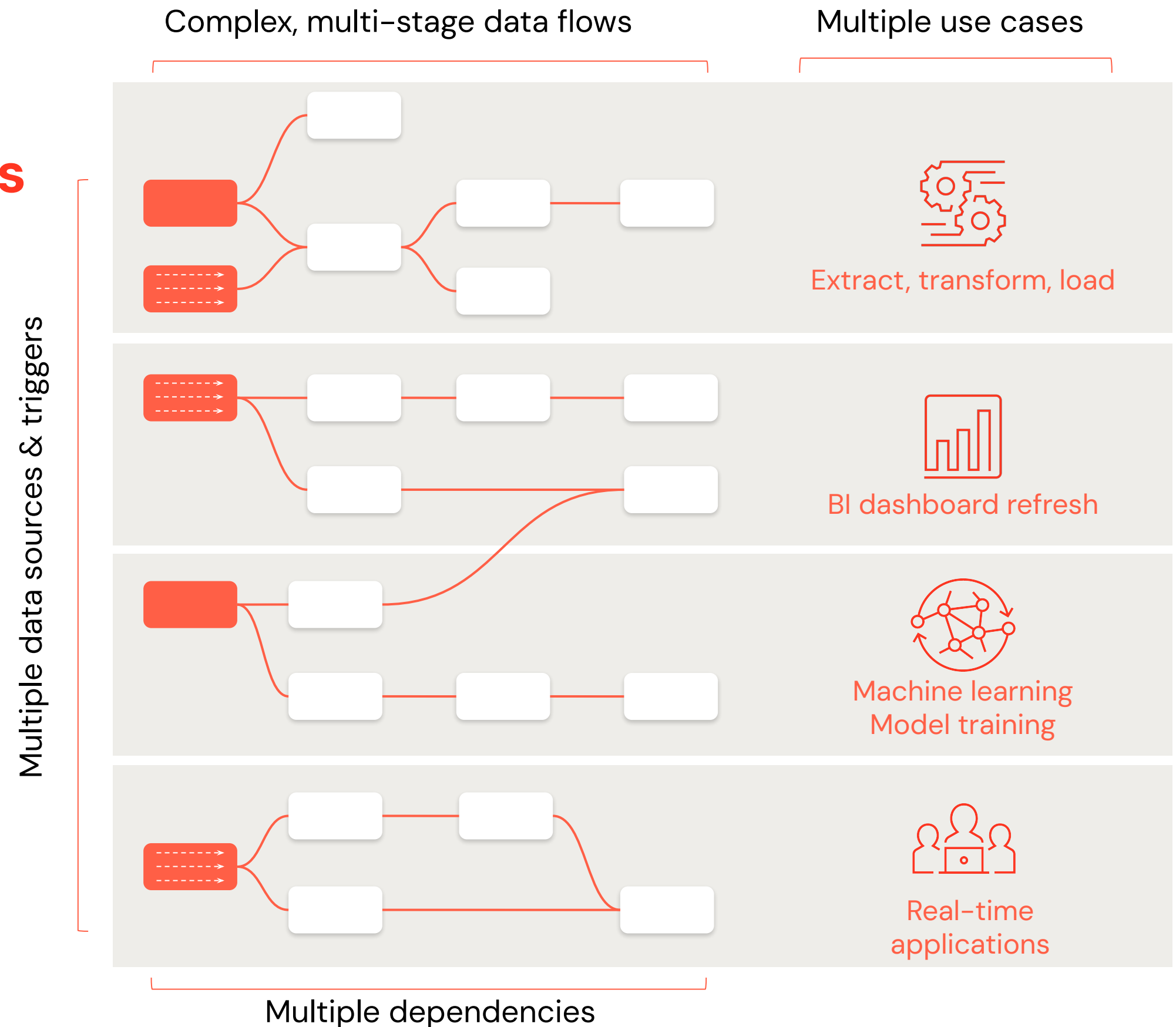
Introduction to Lakeflow Jobs



Modern Data Warehousing Requires Modern Orchestration

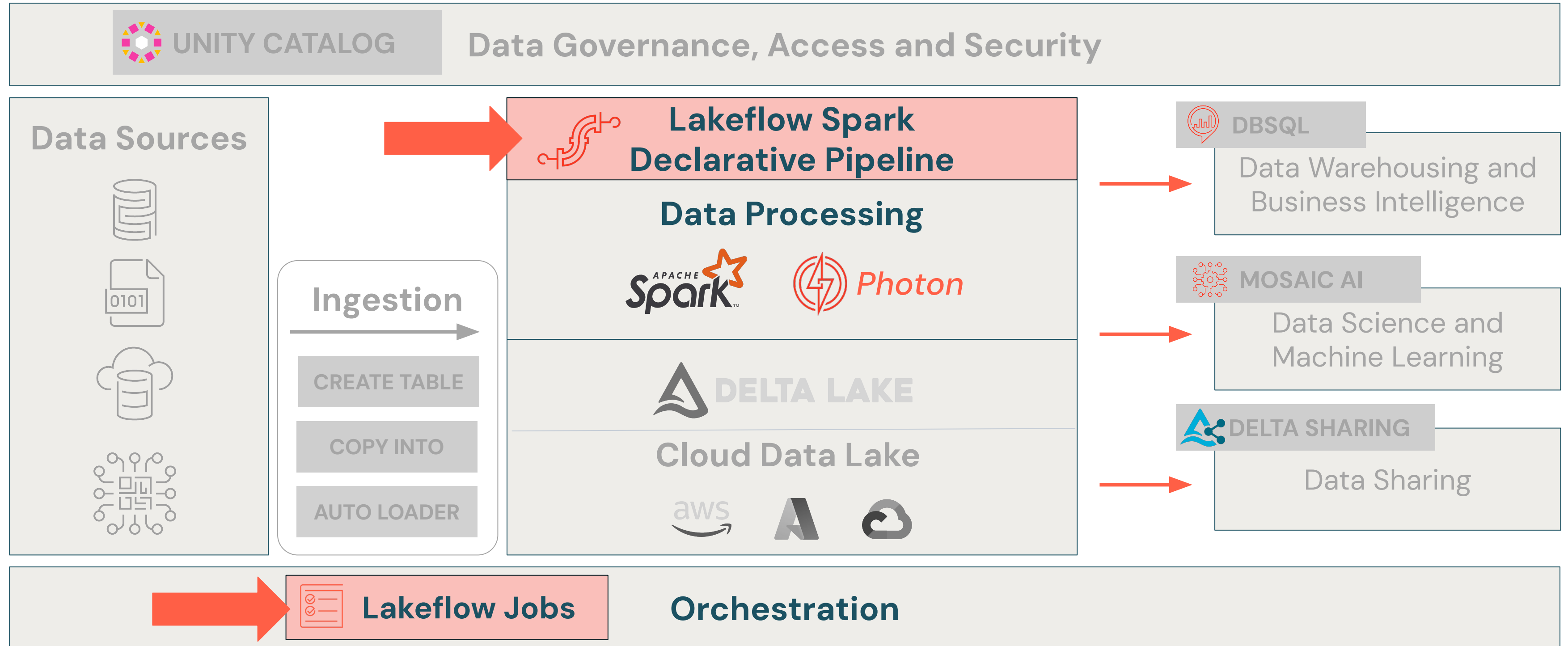
Orchestrating processes across all data, analytics and AI use cases is business critical

- Handles complex, multi-stage data flows.
- Manages multiple dependencies in Lakeflow Jobs.
- Supports diverse use cases, including:
 - Extract, transform, load (ETL).
 - BI dashboard refresh.
 - Machine learning model training.
 - Real-time applications.

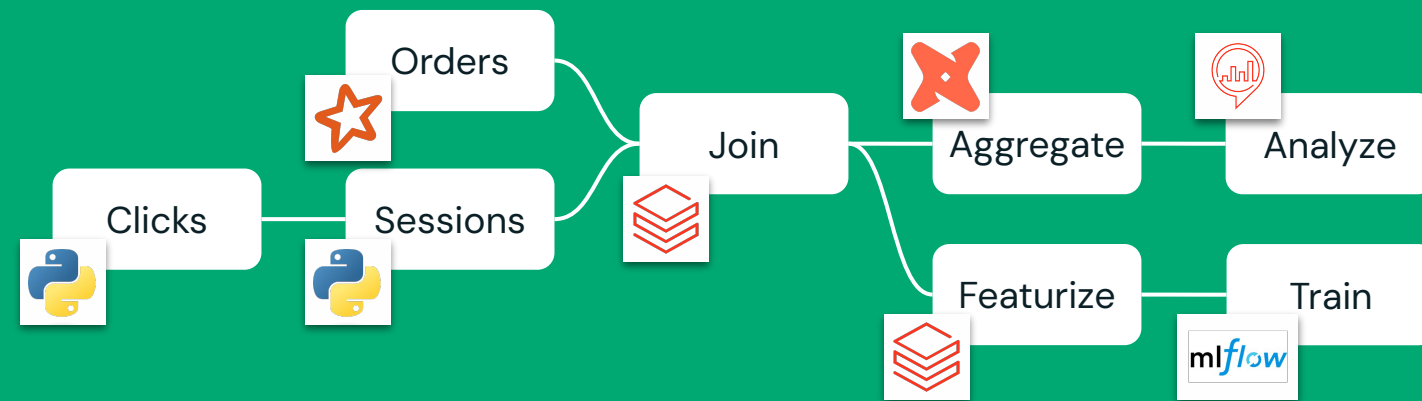


Unified Orchestration Using Databricks

Overview



Lakeflow Jobs



BI & Data
Warehousing



Data
Engineering



Data
Streaming



Data
Science & ML

Unity Catalog

Delta Lake



Databricks Lakeflow Jobs

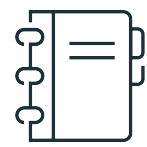
Unified orchestration for data,
analytics, and AI on the
Lakehouse Platform

- Simple authoring
- Actionable insights
- Proven reliability

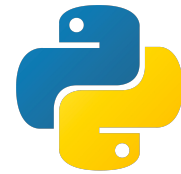


Building Blocks of Databricks Lakeflow Jobs

Jobs consist of one or more **Tasks**



Databricks Notebooks



Python Scripts



Python Wheels



SQL Files/Queries



DBSQL Dashboards



Spark Declarative Pipeline



dbt

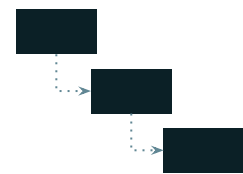


Java JAR file

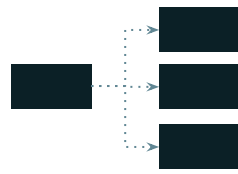


Spark Submit

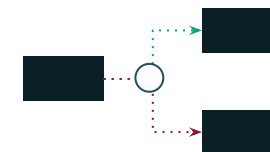
Control flows can be established between **Tasks**.



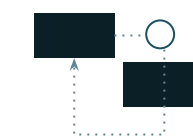
Sequential



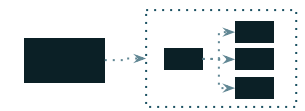
Parallel



Conditionals (If/else)



Loops (For each)



Jobs as a Task (Modular)

Jobs supports different **Triggers**



Manual Trigger



Scheduled (Cron)



API Trigger



File Arrival



Table Update



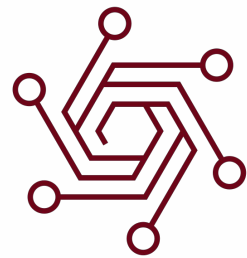
Continuous (Streaming)



Jobs can be executed on different compute

Development, Ad-Hoc Analysis & Exploration

Interactive Clusters



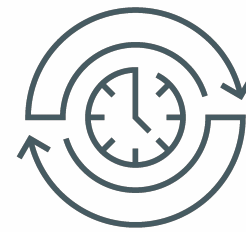
Interactive or all-purpose clusters can be shared by multiple users.

It is best for performing **ad-hoc analysis, data exploration, or development**.

Interactive **should not be used in Production** as they are not cost-efficient.

Production Grade & Operational Use-Cases

Job Clusters



Job clusters are approximately **50% cheaper** as they terminate when the job ends, reducing resource usage and costs.

However, Job clusters are subject to **cloud providers start-up times**.

With Databricks Lakeflow Jobs, you can **re-use the same cluster across Tasks** for **better price performance!**

Simple, Fast, Reliable, Cost Efficient!

Serverless Lakeflow Jobs



Serverless Lakeflow Jobs are a **fully managed service** that are operationally simpler and more reliable

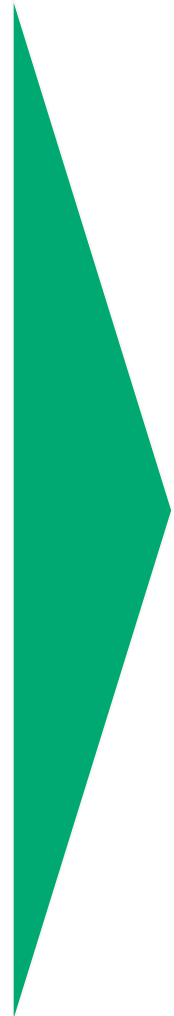
They provide you with **faster clusters & auto-scaling capabilities** providing you with a **better user experience for a lower cost**.

With **out of the box performance optimizations**, Serverless provides you with a **lower overall TCO**.



When to leverage Databricks Lakeflow Jobs?

- ...if you want **simple, yet powerful** ETL/Spark Declarative pipeline/ML orchestration
- ...if you use **notebook Lakeflow Jobs (%run)**
- ...if you use Airflow and want to **reduce infrastructure overhead** and make orchestration easier
- ...to **consolidate** your toolset
- ...to **enable non-engineers** to orchestrate their code
- ...if you want **cloud-provider independent** orchestration



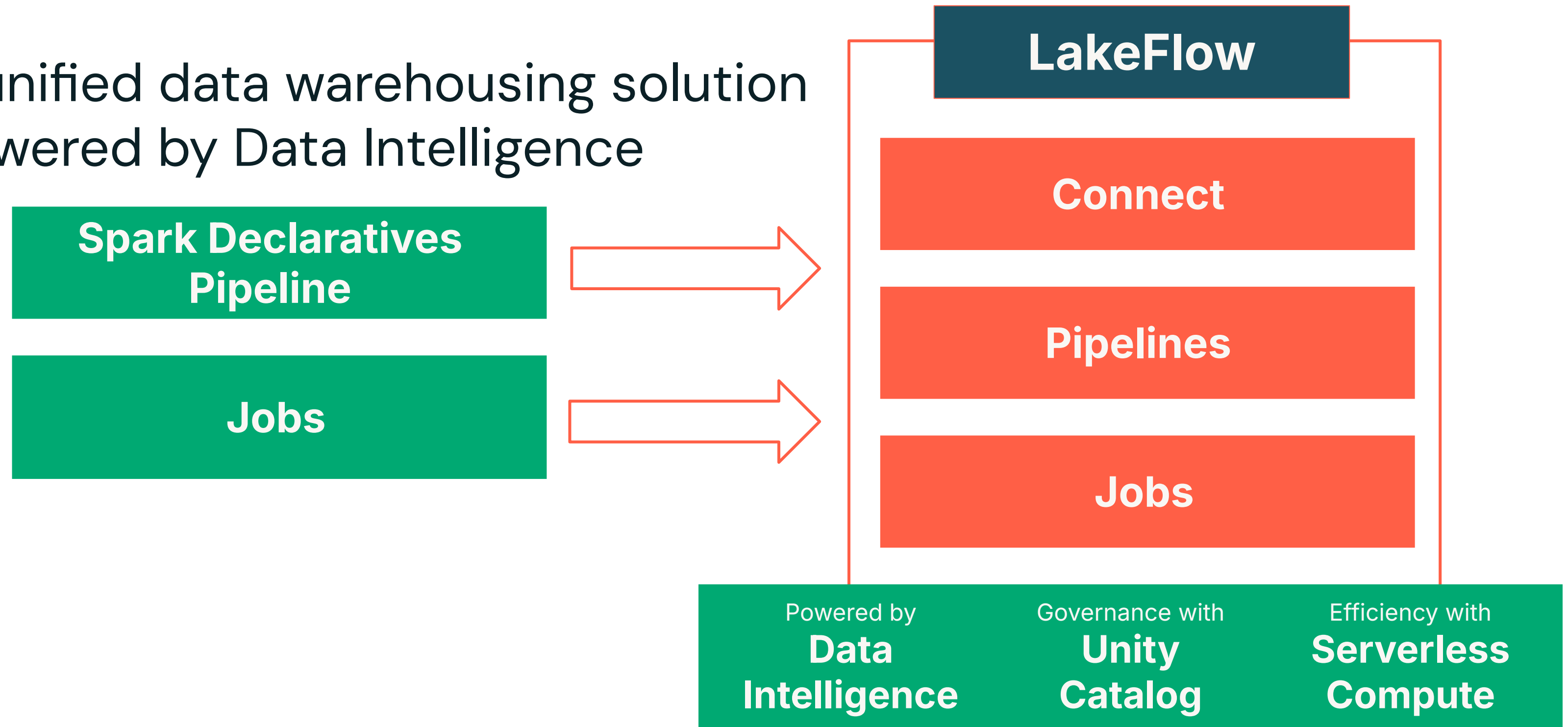
**...then
orchestrate
with Jobs**



LakeFlow

The evolution of data warehousing on Databricks

A unified data warehousing solution
powered by Data Intelligence

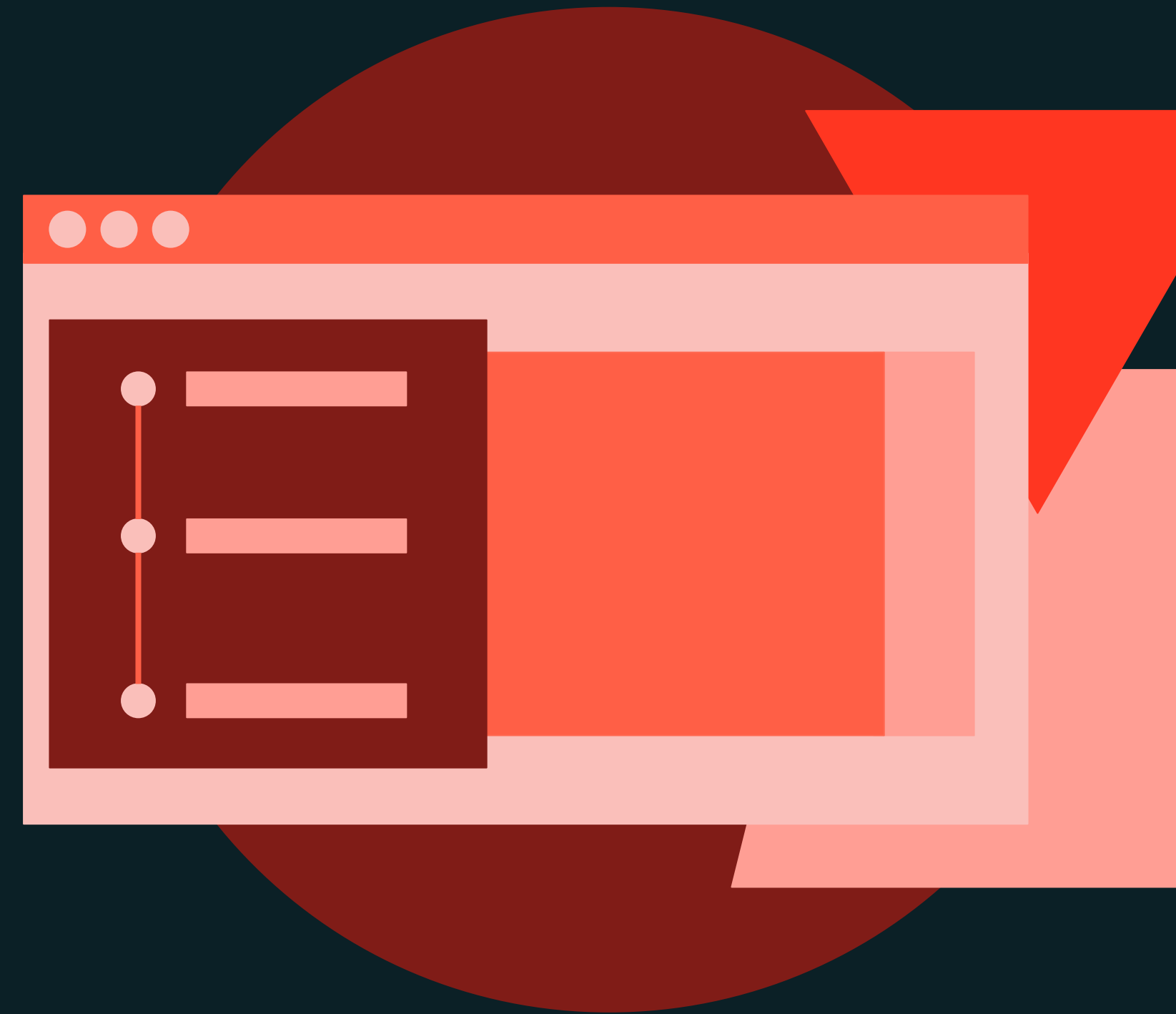




Ingesting and Transforming Data

DEMONSTRATION

Exploring Job Capabilities



Follow-along Instructions

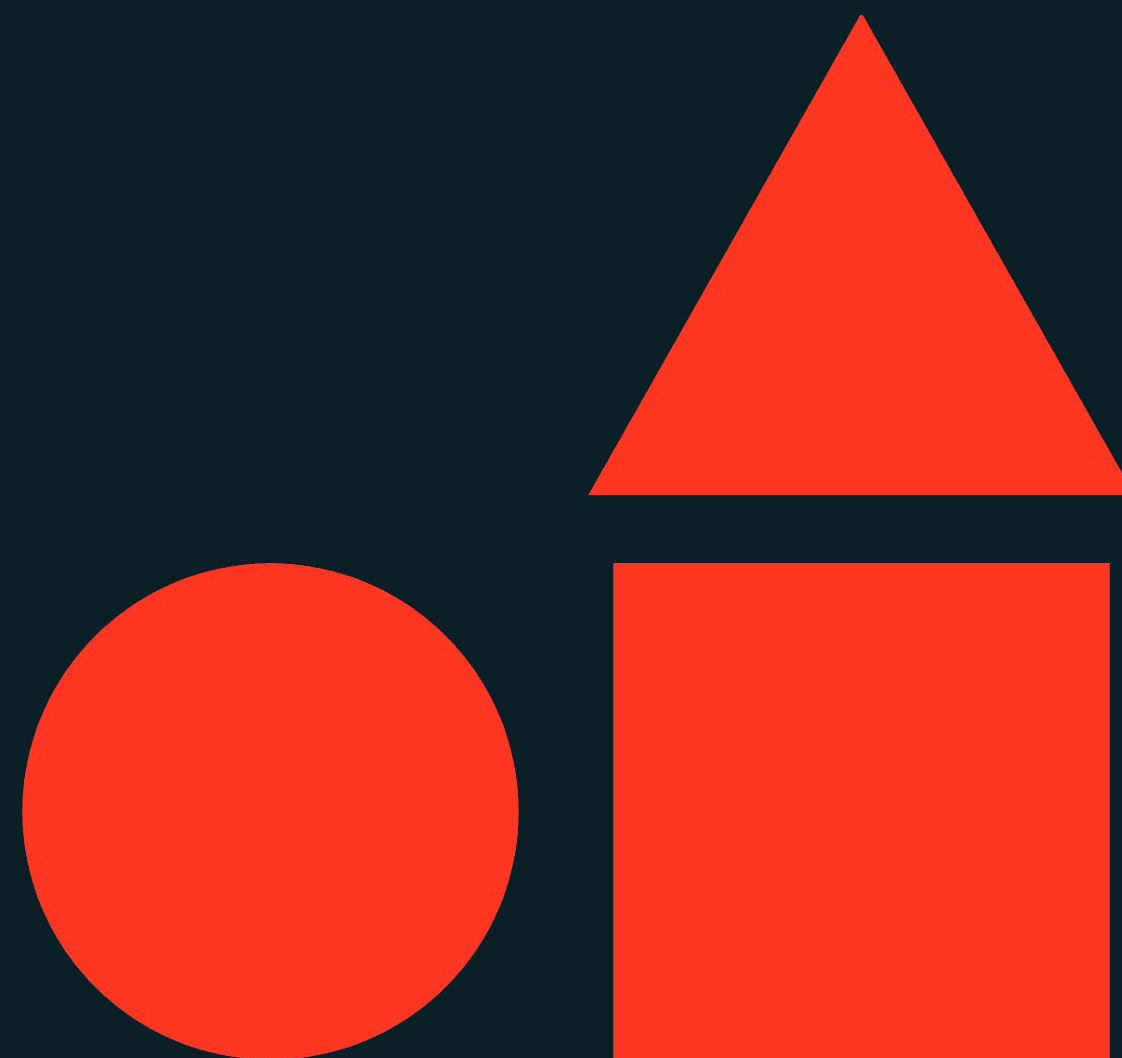
Estimated Time: 10 minutes

For this demonstration, the instructor will take you through an example ETL process highlighting the structure and control flow. If you have access to the Vocareum lab environment, feel free to follow along.



Monitoring

Data Warehousing with Databricks



Objectives

- Describe tagging and why it's useful
- Apply tagging and system tables for monitoring and controlling cost in your data warehouse
- Apply best practices for controlling cost and optimizing your data warehouse





Monitoring

LECTURE

Tagging



Tagging

Context

A tag is a piece of metadata consisting of a key/value pair

Tags are assigned to resources

Many uses for tagging:

- Cost tracking
- Resource management
- Policy enforcement
- Automation

Key	Value
Vendor	Databricks
Creator	f167293b-8316-41e0-a596-9765904a86d8
ClusterName	f167293b-8316-41e0-a596-9765904a86d8
Creator	f167293b-8316-41e0-a596-9765904a86d8
ClusterId	0225-221353-iopo6xqn



Tagging

Assigning to SQL Warehouses

Tags can be assigned to SQL warehouses any time

New SQL warehouse

Name

my_warehouse

Cluster size ⓘ

X-Large 80 DBU / h ▾

Auto stop

☒ After 10 minutes of inactivity.

Scaling ⓘ

Min. 1 Max. 1 clusters (80 DBU)

Type

☒ Serverless ⓘ ☐ Pro ⓘ ☐ Classic

Advanced options

Tags ⓘ

Key

Value

Unity Catalog

☒

Channel ⓘ

☒ Current ☐ Preview

Cancel

Create



Tagging

Best Practices

Standardize practice around key/value pairs

Use keys that attribute usage to a business unit or project

Assign more specific tags as needed

Incorporate tag enforcement where possible

Key	Value
business-unit	finance
project	annual-budget
customer	Acme Anvils LLC





Monitoring

LECTURE

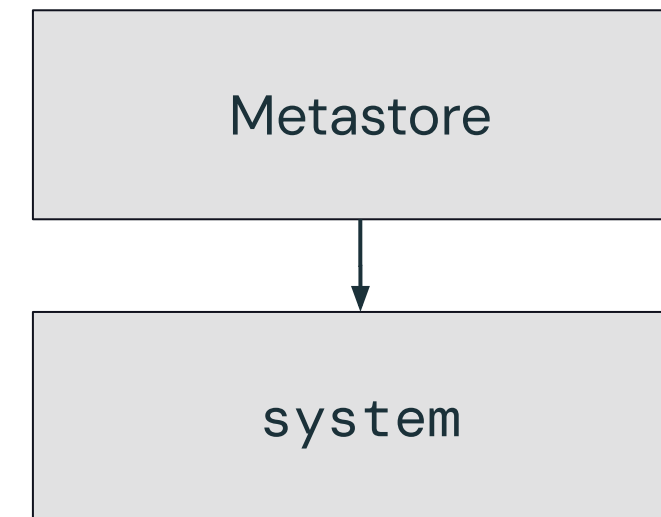
System Tables



System Tables

Overview

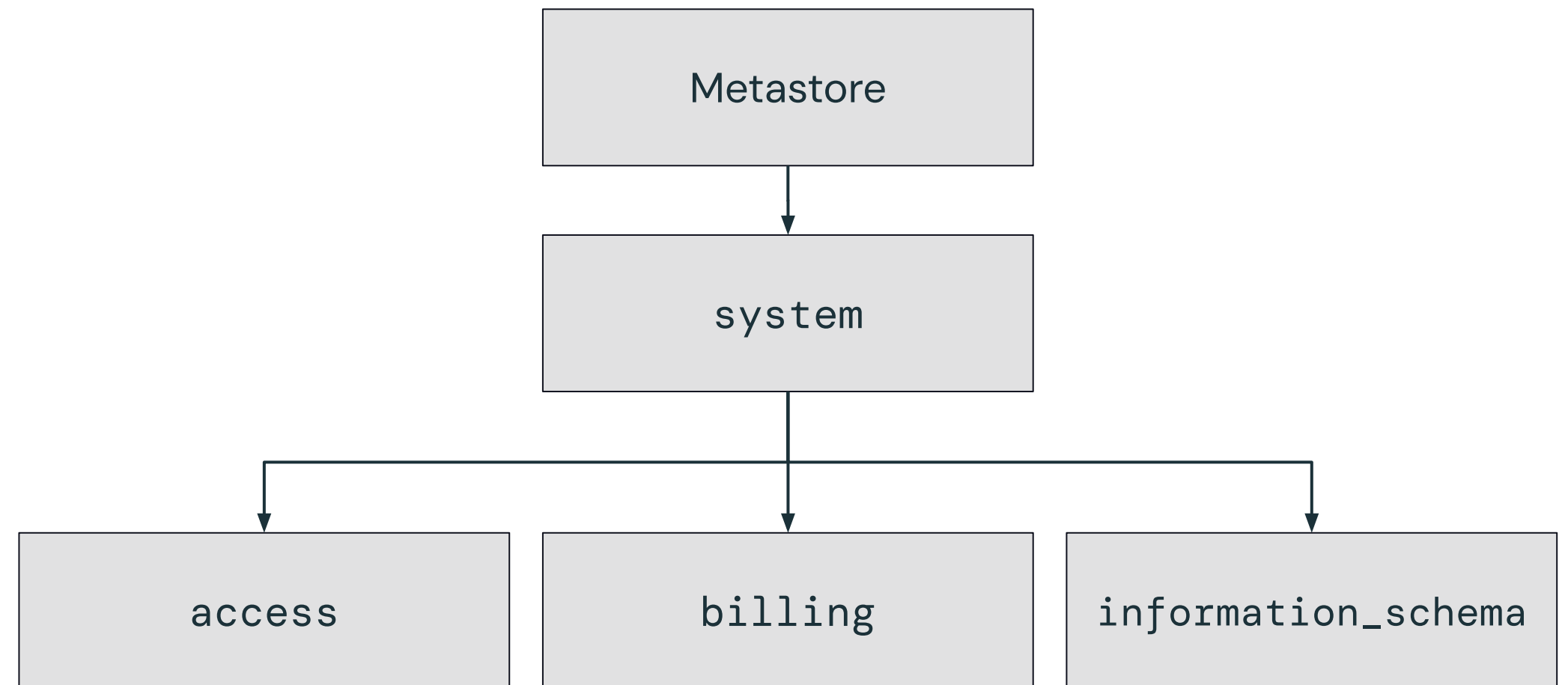
- Operational data for observability and analytics
- Comprehensive insight into many platform aspects
 - Debugging
 - Auditing
 - Cost control
 - Efficiency
 - Data quality
- Access through same namespace as any other data object



System Tables

Structure

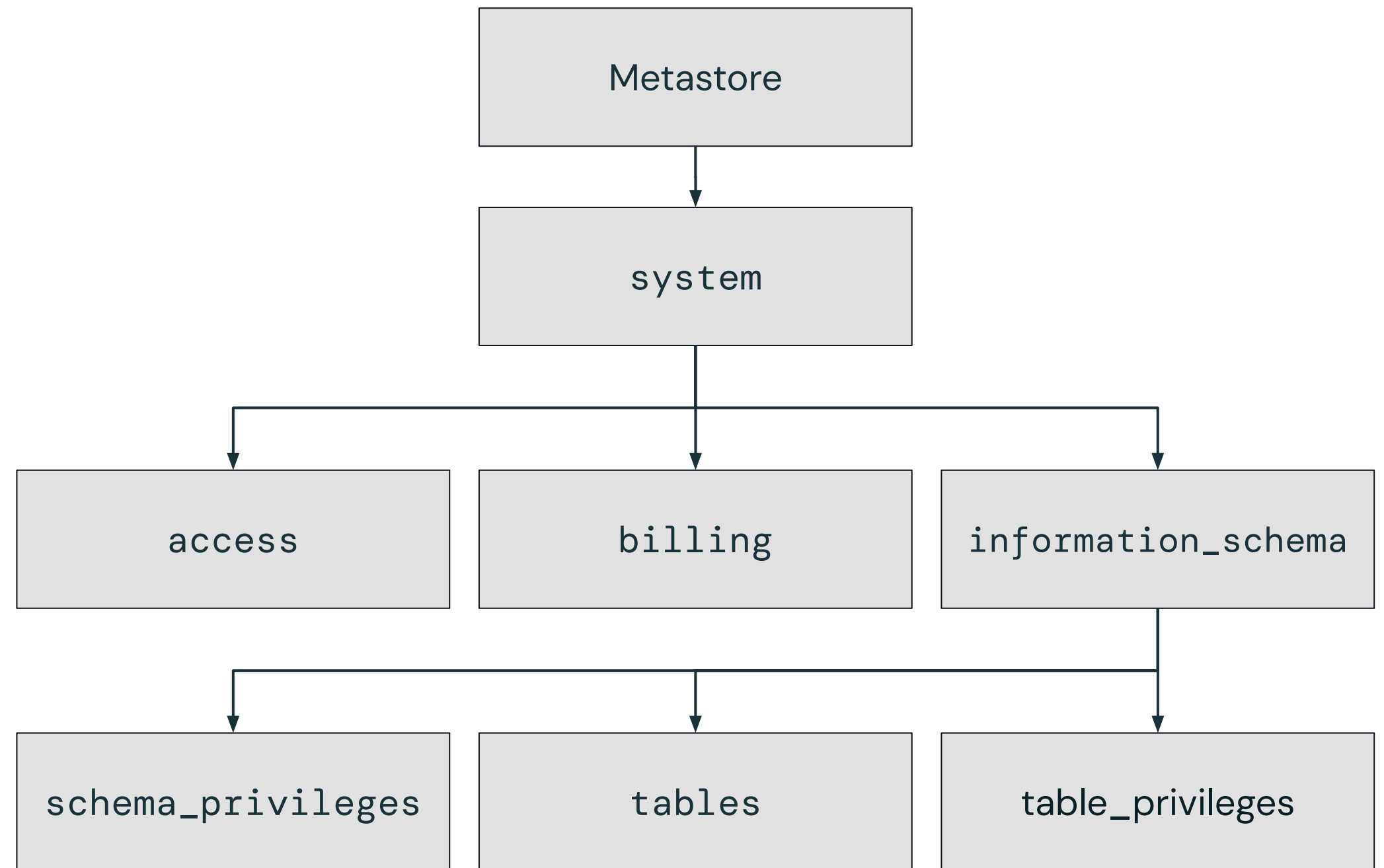
- **system** catalog
- Divided into schemas
 - Audit logs
 - Billing information
 - Object metadata



System Tables

Structure

- Example:
information_schema
- Metadata spanning entire metastore

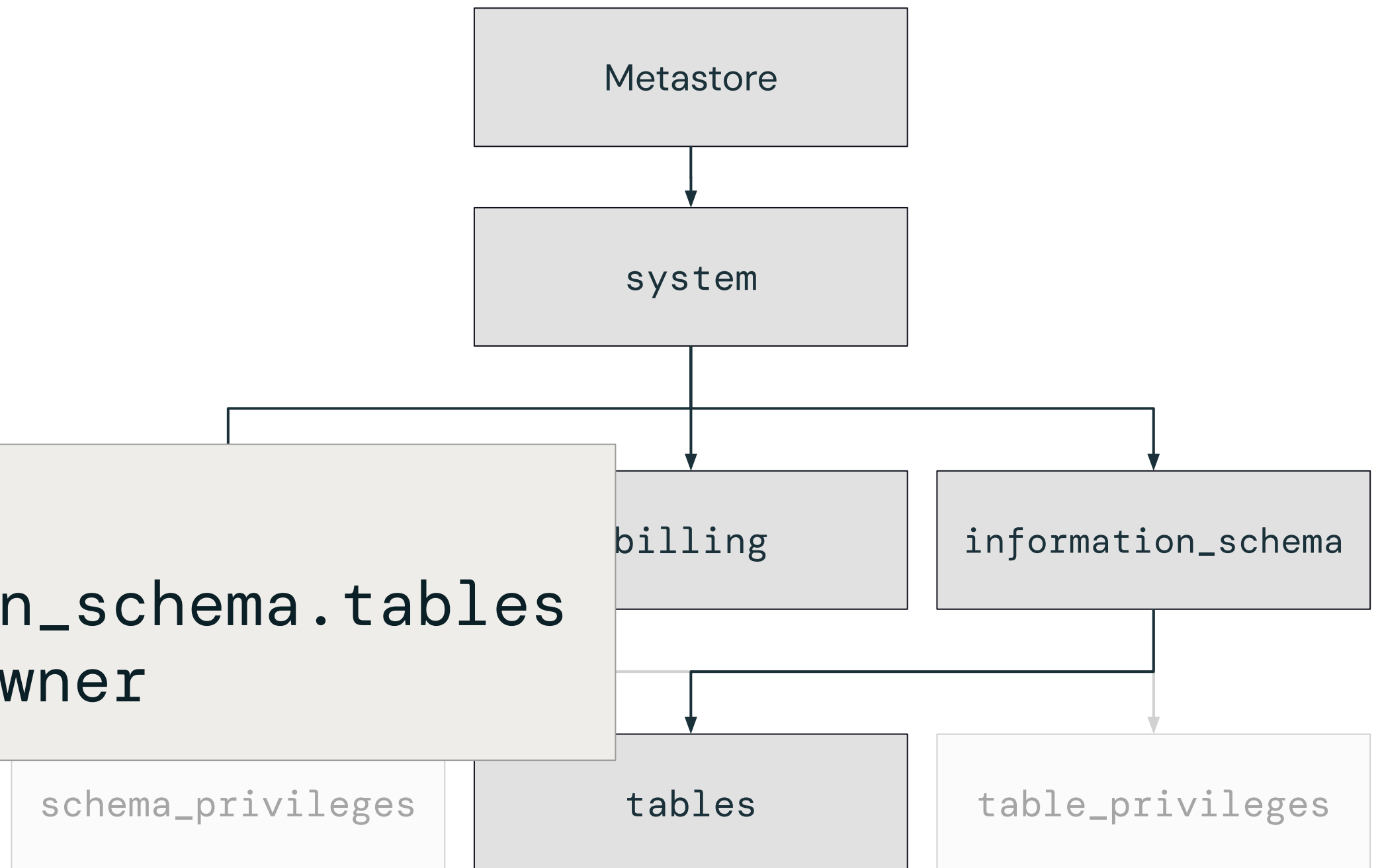


System Tables

Usage

- Example:
information_schema
- Metadata spanning entire metastore

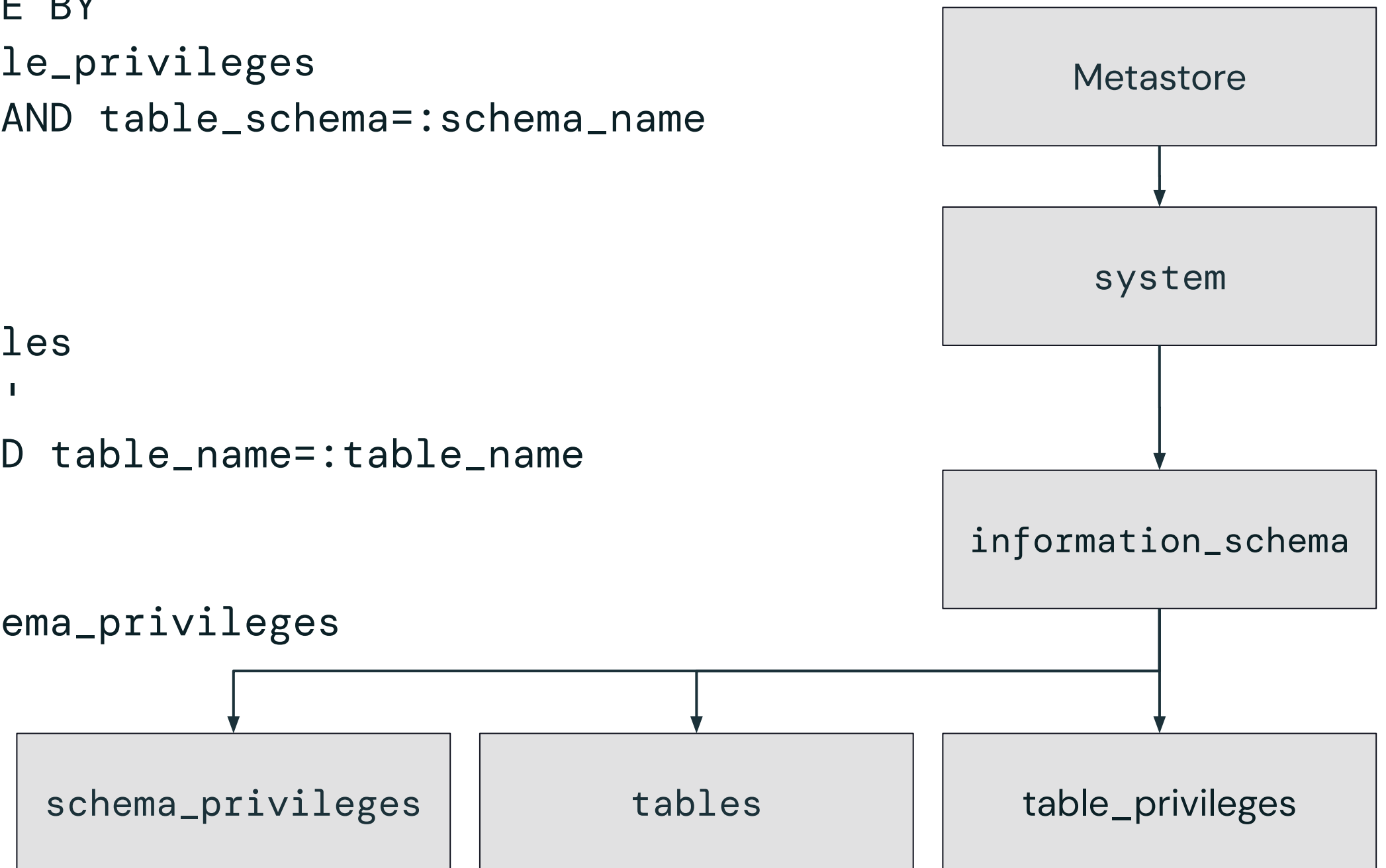
```
SELECT *  
FROM system.information_schema.tables  
WHERE table_owner = :owner
```



System Tables

Example: who has access to X?

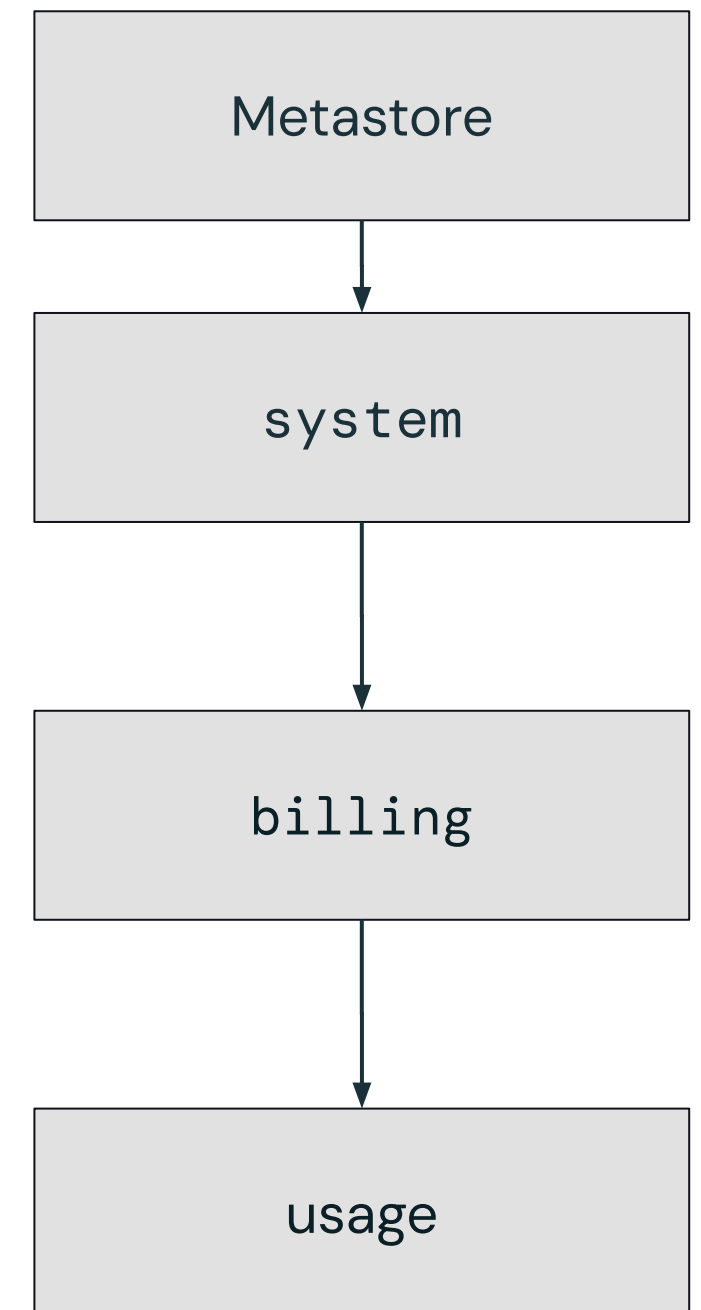
```
SELECT DISTINCT(grantee) AS `ACCESSIBLE BY`  
  FROM system.information_schema.table_privileges  
  WHERE table_catalog=:catalog_name AND table_schema=:schema_name  
        AND table_name=:table_name  
UNION  
  SELECT table_owner  
  FROM system.information_schema.tables  
  WHERE table_catalog='david_leblanc'  
        AND table_schema=:schema_name AND table_name=:table_name  
UNION  
  SELECT DISTINCT(grantee)  
  FROM system.information_schema.schema_privileges  
  WHERE catalog_name=:catalog_name  
        AND schema_name=:schema_name
```



System Tables

Example: which jobs consumed the most DBUs?

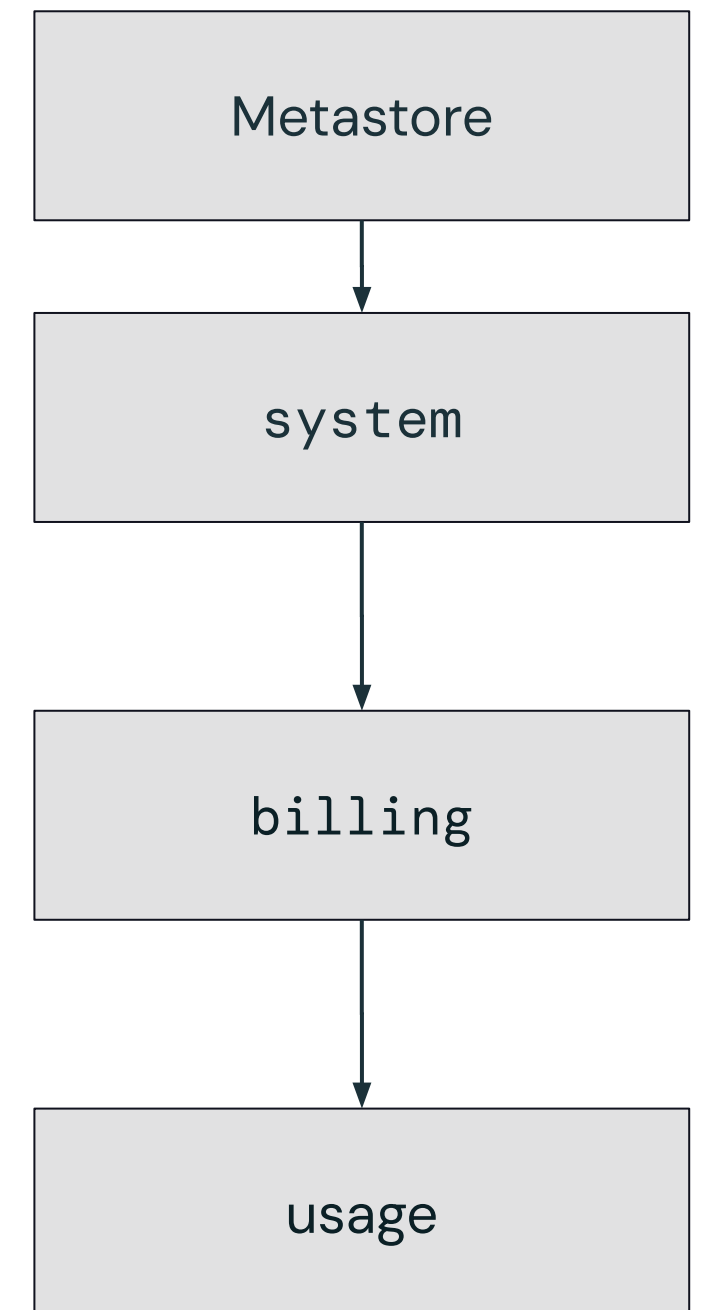
```
SELECT
  usage_metadata.job_id as `Job ID`, sum(usage_quantity) as `Usage`
FROM
  system.billing.usage
WHERE
  usage_metadata.job_id IS NOT NULL
GROUP BY
  `Job ID`
ORDER BY
  `Usage` DESC
```



System Tables

Example: resource attribution using tags

```
SELECT
  sku_name, usage_unit, SUM(usage_quantity) as `Usage`
FROM
  system.billing.usage
WHERE
  custom_tags [:key] = :value
GROUP BY 1, 2
```



System Tables

Usage dashboards

- Account admins can easily generate cost dashboards
- Leverages Databricks AI/BI dashboards for analyzing cost



System Tables

Usage dashboards

Workspaces

Catalog

Usage

User management

Security

Cloud resources

Previews

Settings

Usage

Consumption

Budgets

Preview

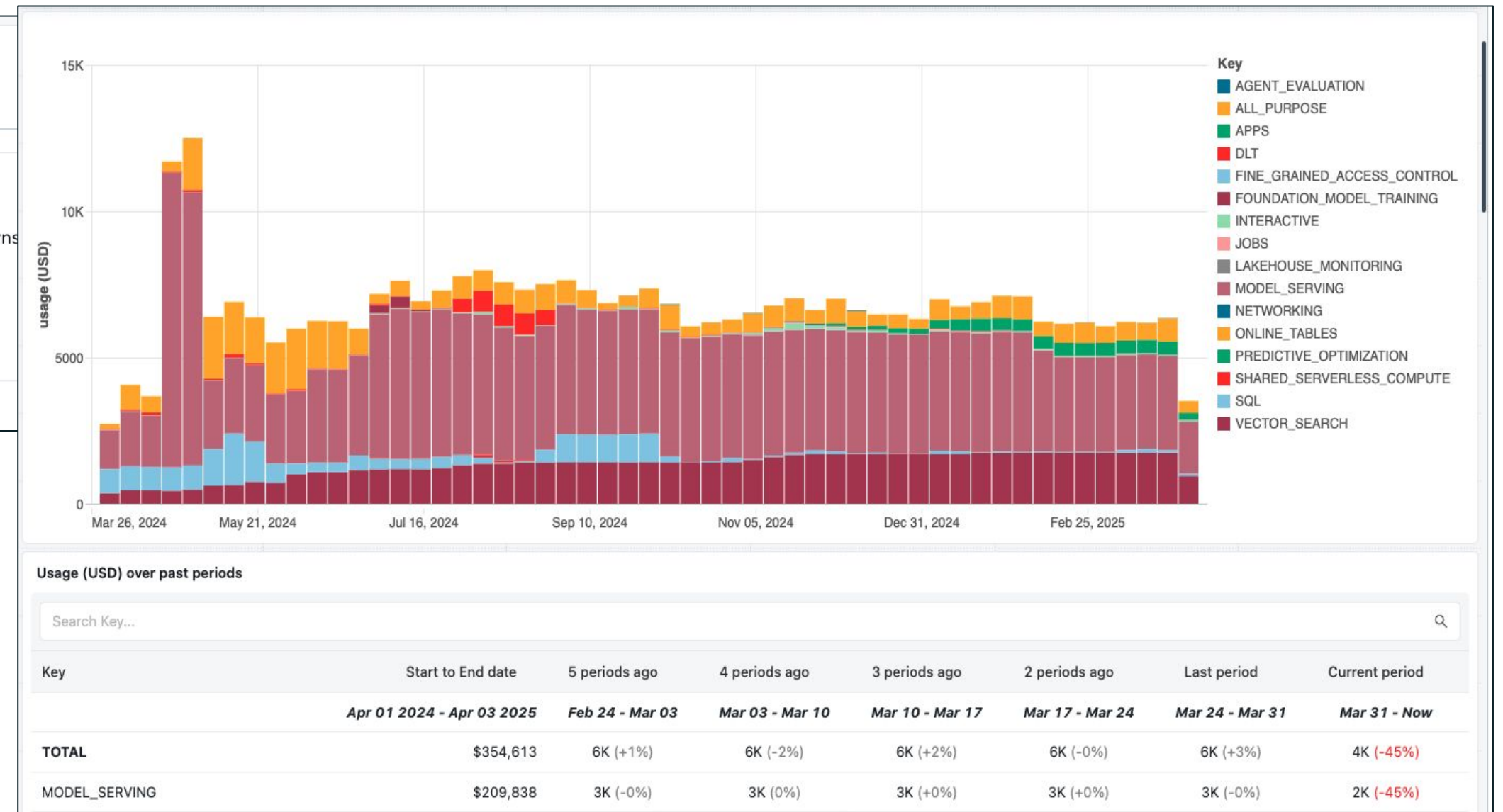
Consumption (Legacy)

Improved usage dashboards

Gain insights into your usage with pre-built dashboards: visualize breakdowns, tag attributions, and identify cost drivers.

Setup dashboard

Learn more



Setup usage dashboard

Account usage

Single workspace usage

Select a workspace

dddev-vf8zy1GD2zlj2q1

Cancel

Import





Monitoring

LECTURE

Cost Optimization



Cost Optimization

Best Practices

Use Serverless SQL Warehouses

Size warehouses properly

Autoscaling

Managed tables

Optimize SQL

Profile queries



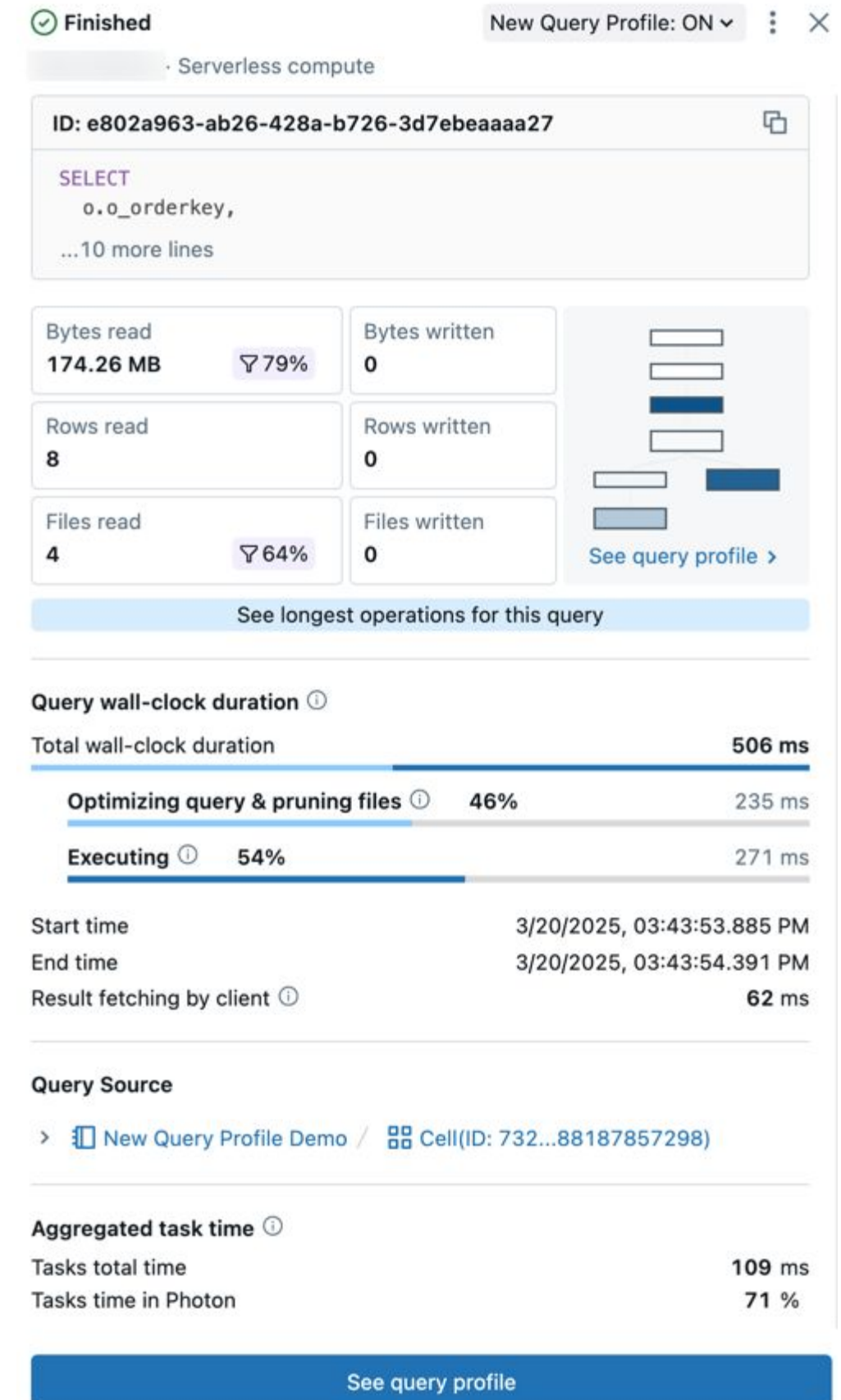
Query Profiling

Graphical depiction of query execution timing

Useful for tracing:

- Time spent
- Number of rows processed
- Memory consumption

Can use query history to display details for previous query executions



SQL Warehouse Monitoring

Live statistics

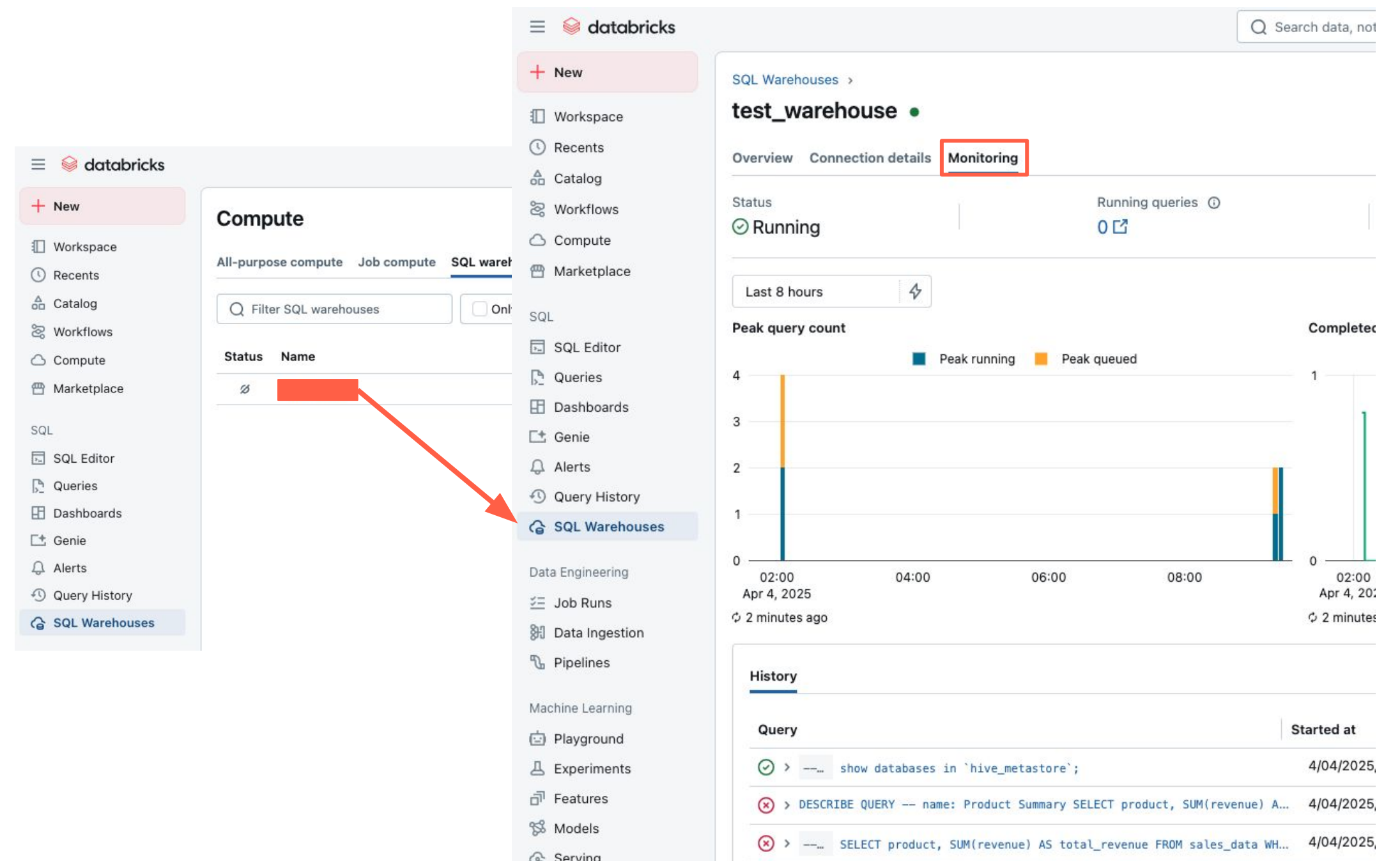
Time scale filtering

Peak query count chart

Completed query count chart

Running clusters chart

Query history table



SQL Warehouse Sizing

Best Practices

Start large and work down

Single SQL warehouse

Leverage Databricks to autoscale

Use tools like SQL warehouse monitor
or query profiler



SQL Warehouse Sizing

Best Practices

Start large and work down

Single SQL warehouse

Leverage Databricks to autoscale

Use tools like SQL warehouse monitor or query profiler

Decreasing query latency

Spilling to disk? Highly parallelizable queries?

Increase size

Multiple queries at a time?

Add more clusters for autoscaling



SQL Warehouse Sizing

Best Practices

Start large and work down

Single SQL warehouse

Leverage Databricks to autoscale

Use tools like SQL warehouse monitor
or query profiler

Decreasing query latency

Spilling to disk? Highly parallelizable queries?

Increase size

Multiple queries at a time?

Add more clusters for autoscaling

Reduce cost

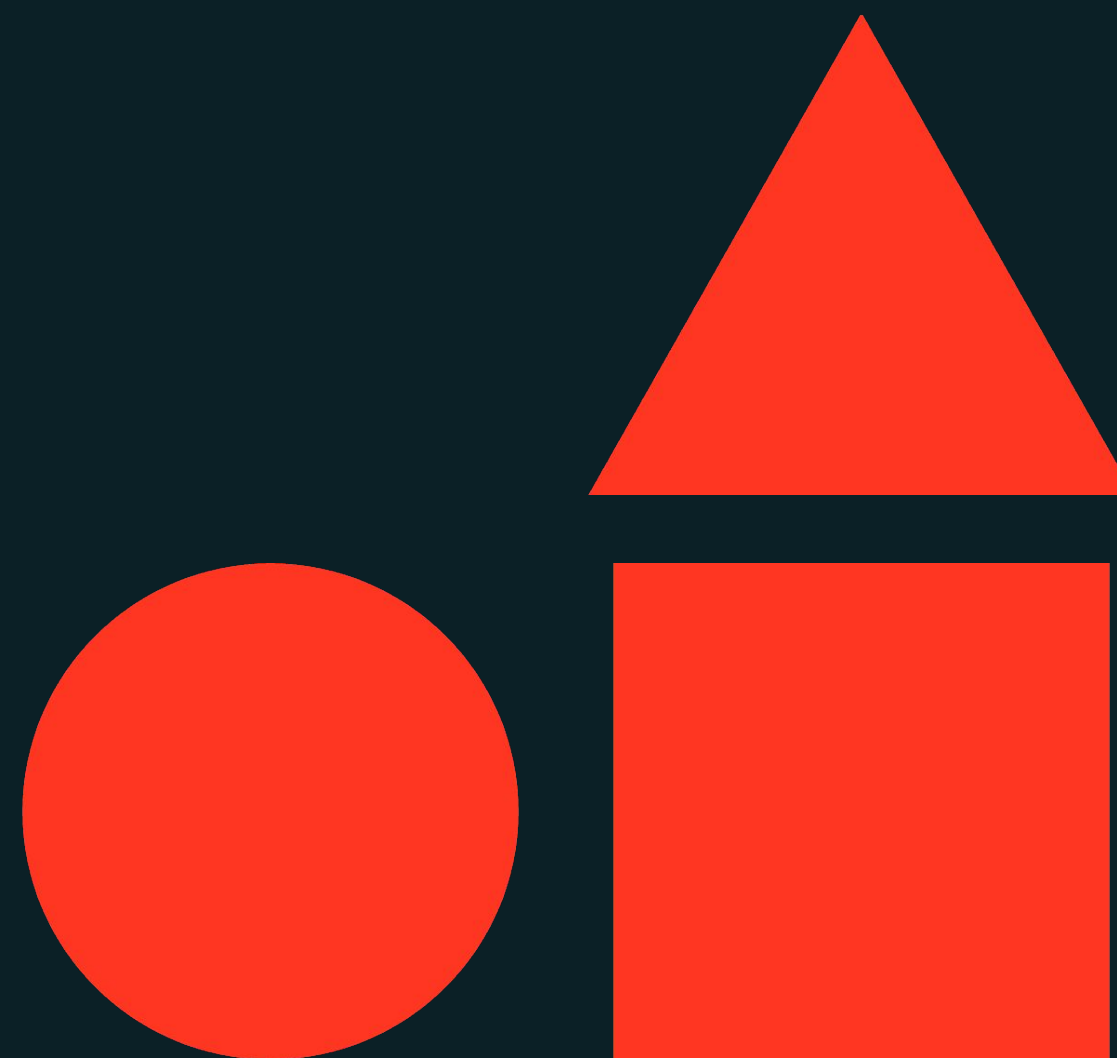
Decrease size





Visualization

Data Warehousing with Databricks



Objectives

- Creating meaningful visualizations of your data using AI/BI dashboards





Visualization

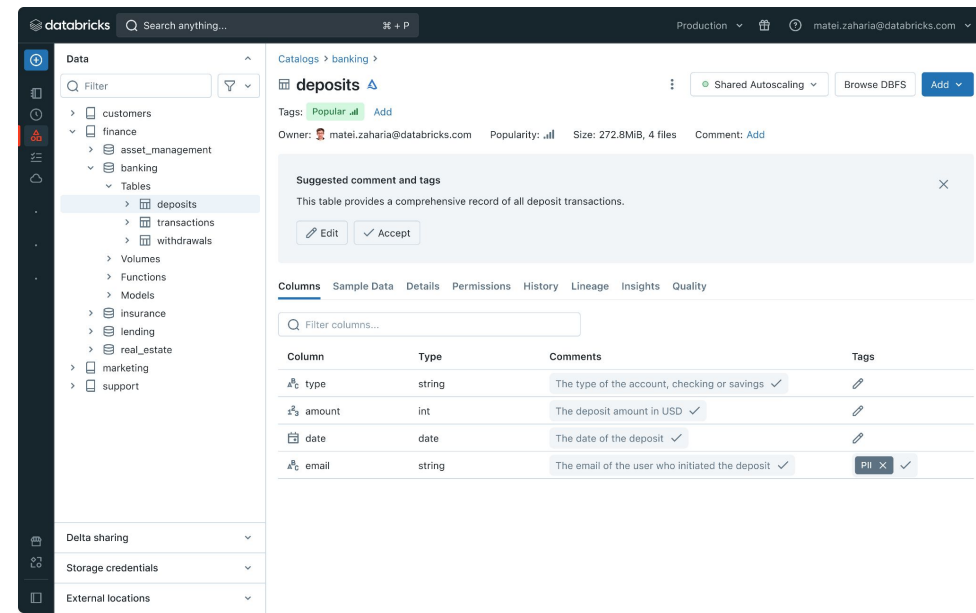
LECTURE

Introduction to AI/BI

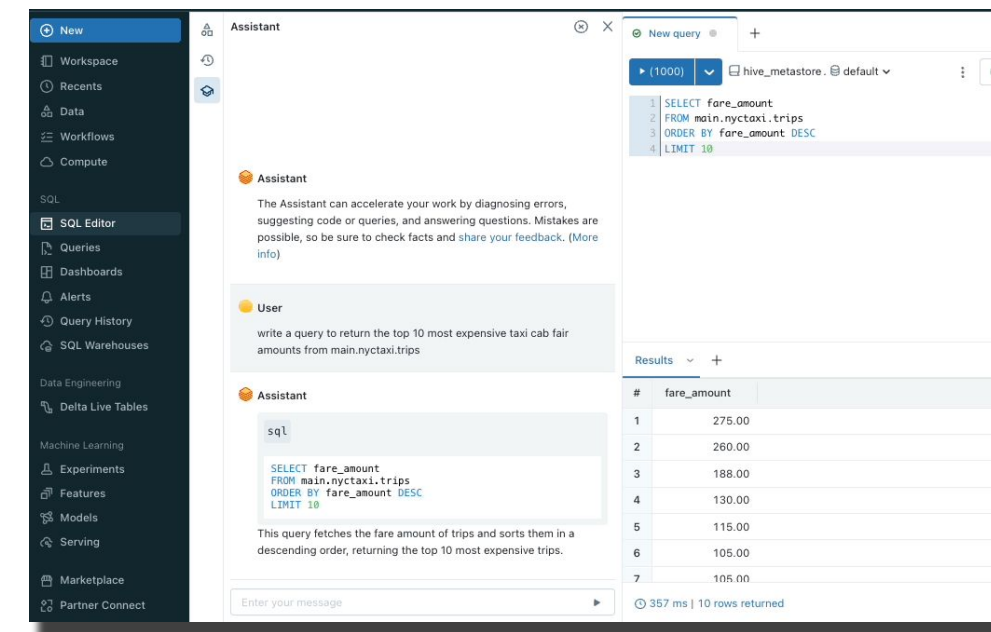


Infusing AI in our user experience

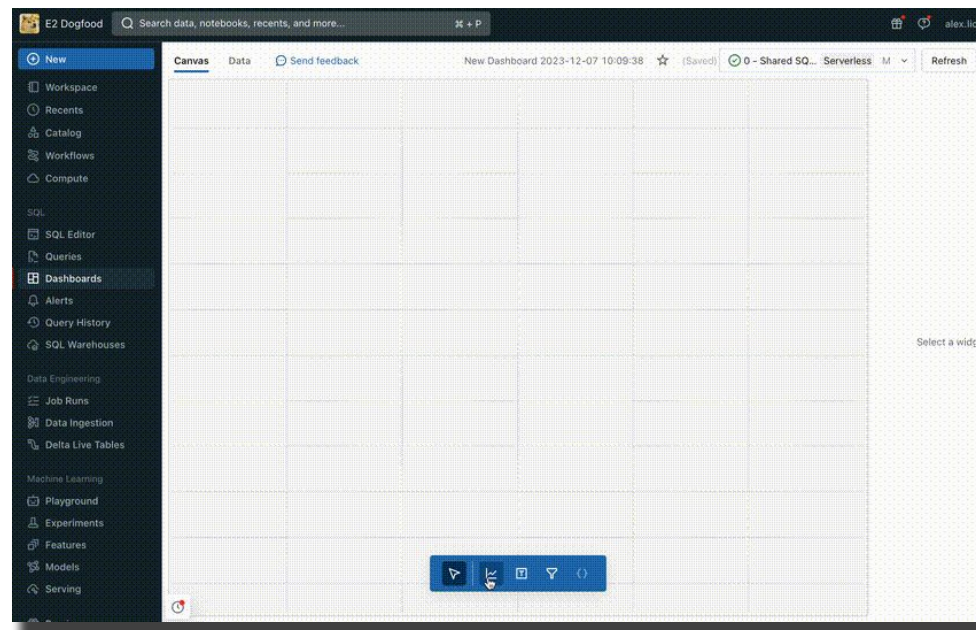
AI-generated docs for easier search and discoverability



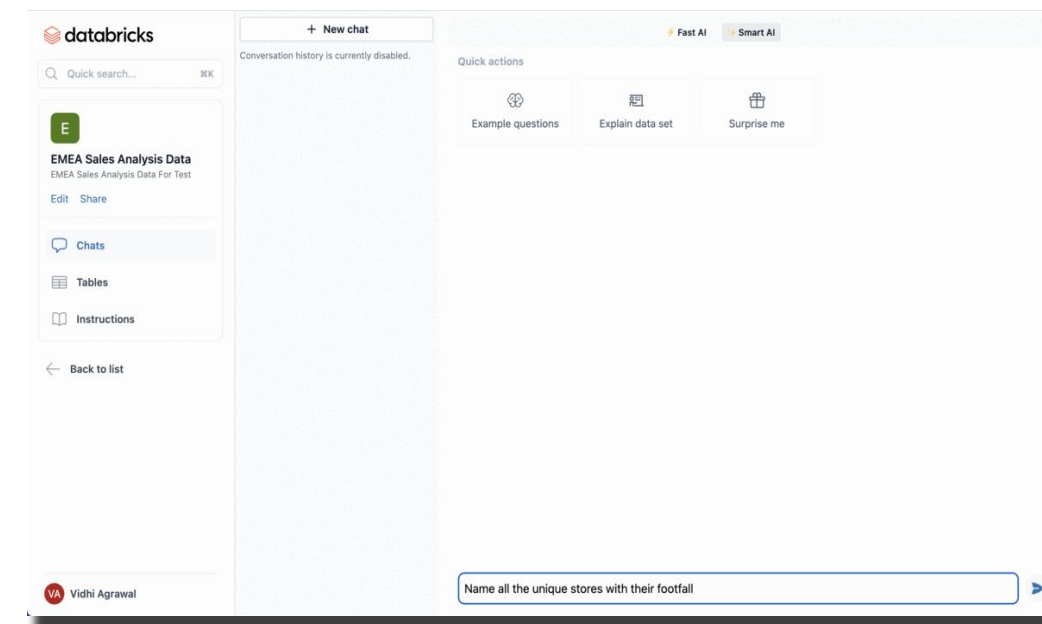
AI Assistant for quick code fixes, queries and descriptions



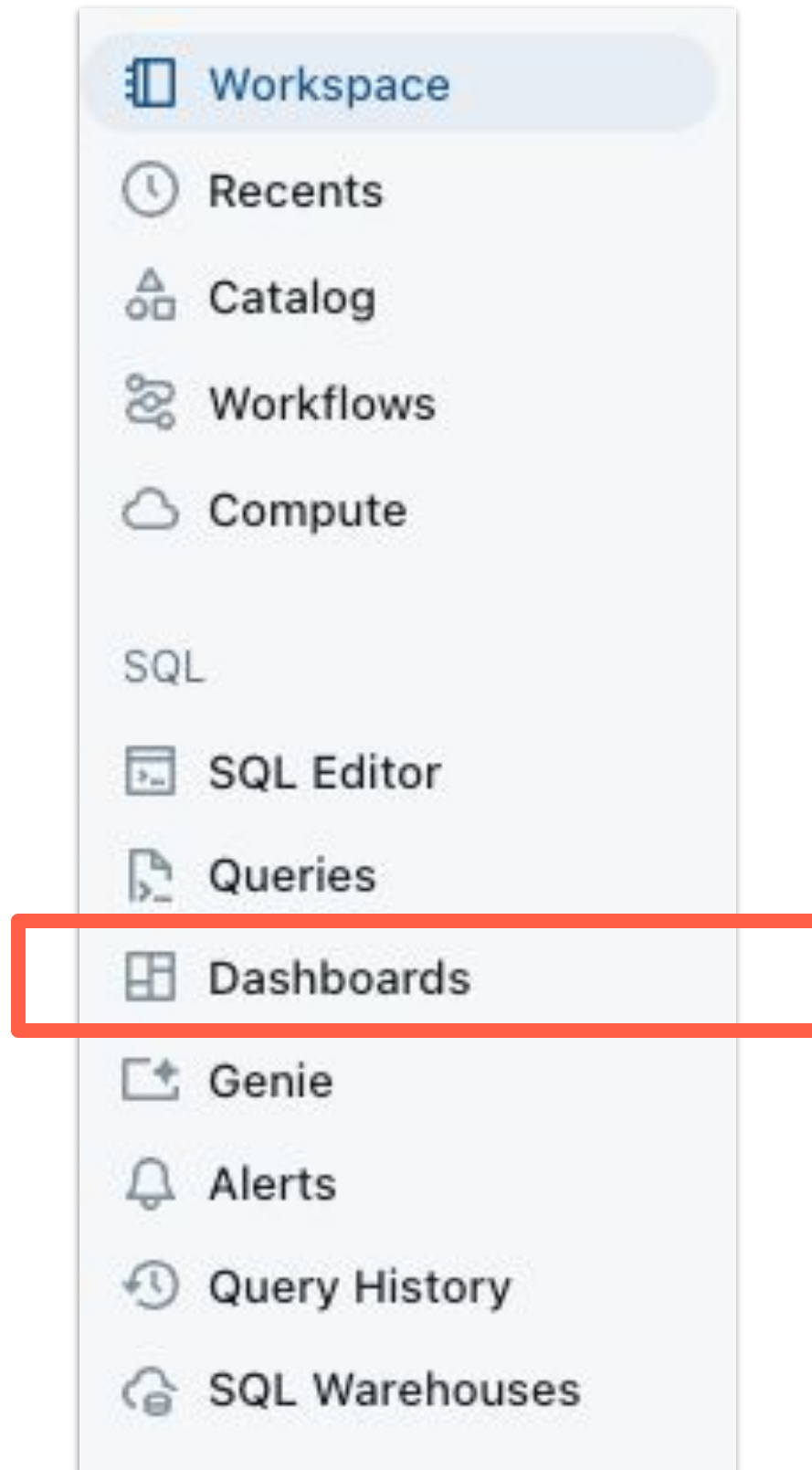
AI Assistant for quick creation of visualizations



AI-powered Genie Spaces to quickly understand your data using natural language



Databrick's AI/BI Offerings



Located under the SQL header in the navigation:

AI/BI Dashboards

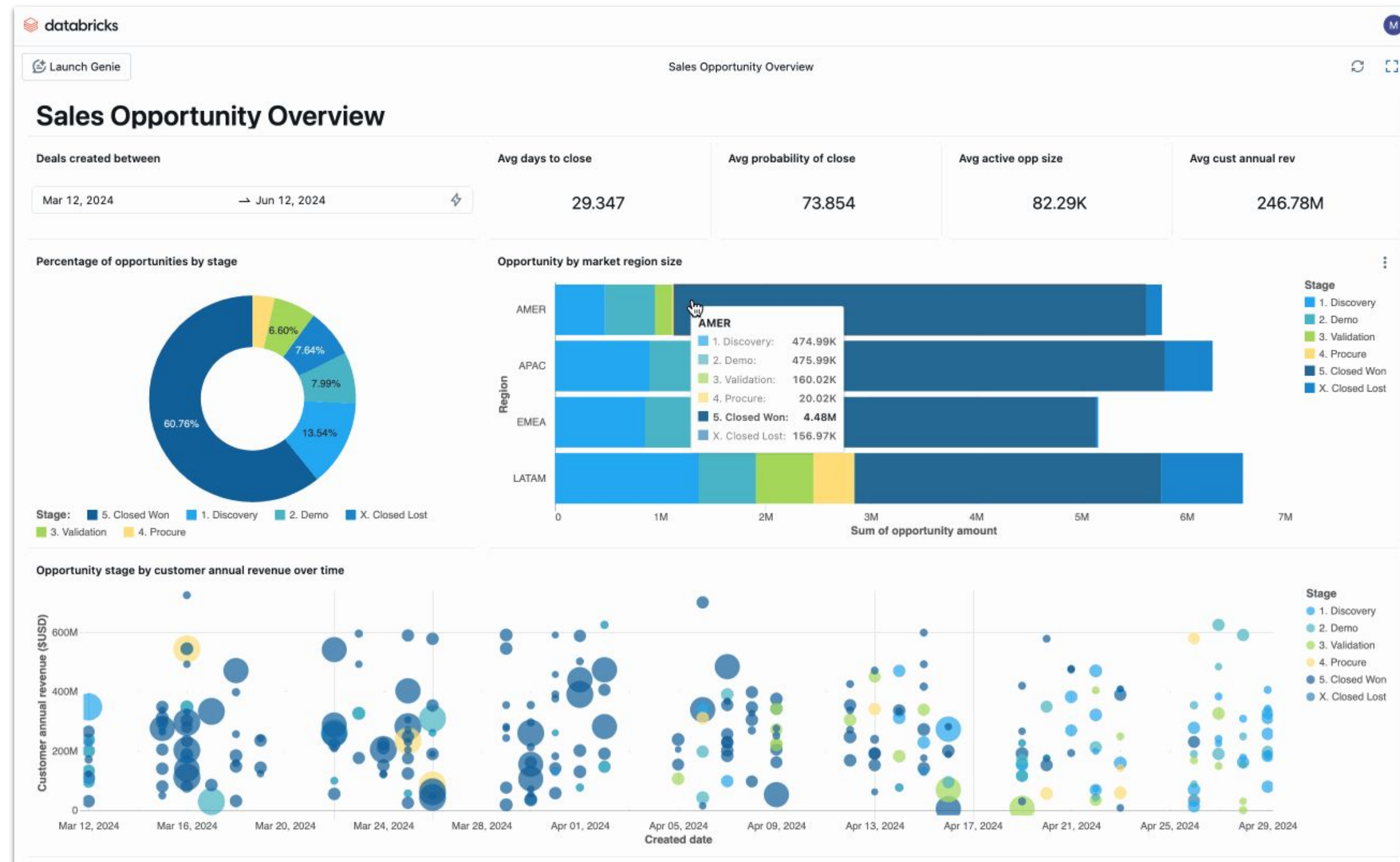
- Presentation environment for visualizations, titles, filters, and other features traditionally found in presentation software.



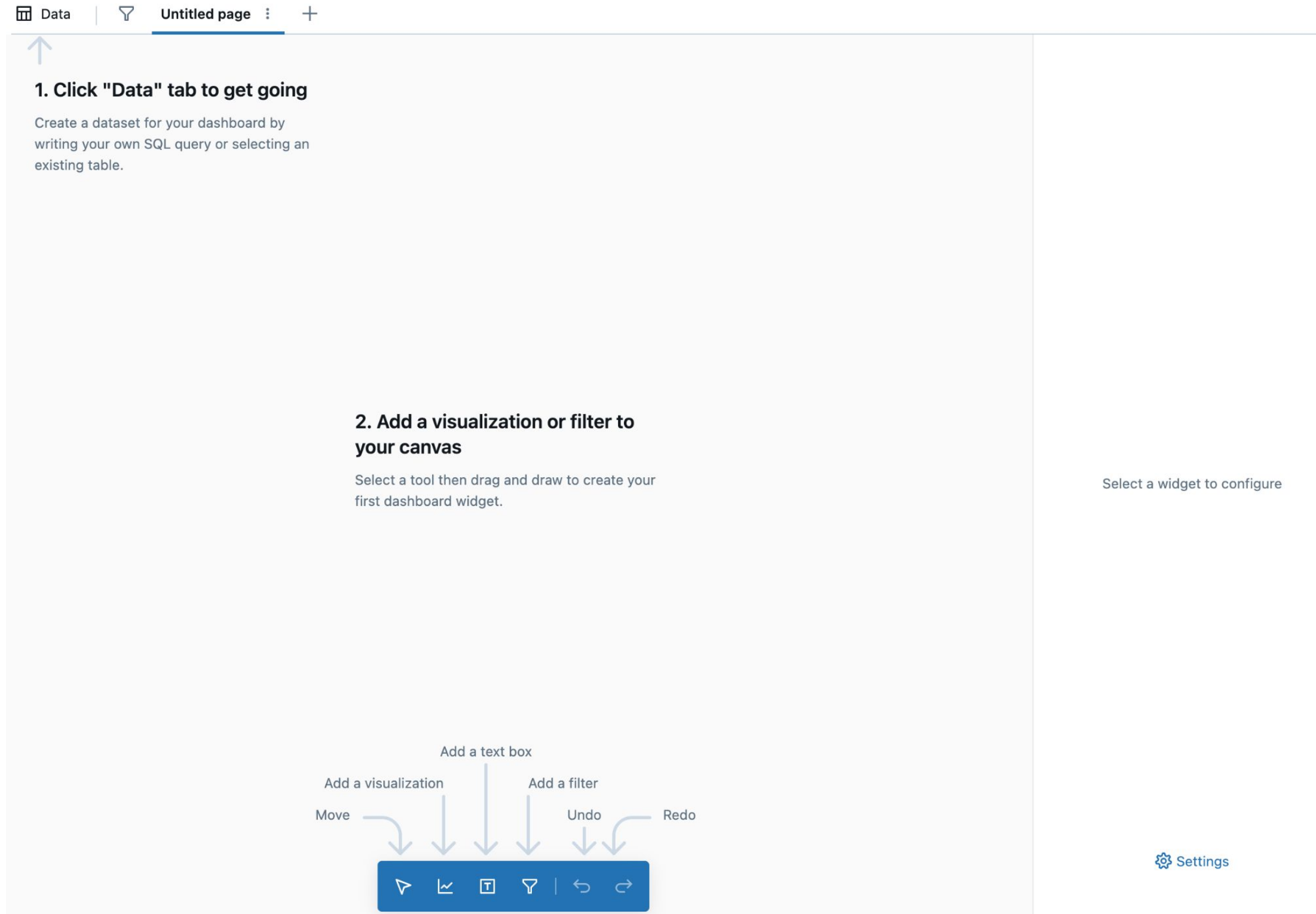
AI/BI Dashboards

All-in-one visualization and presentation environment

- Add datasets, write queries, and design visualizations in one environment.
- Easily share with others in your organization for presentation and review.
- Use natural language to support visualization creation.



AI/BI Dashboards



- Support for technical and non-technical users.
- Simple, easy to use interface.
- All-in-one sharing manages the permissions for the data, queries, and dashboard all from one interface.

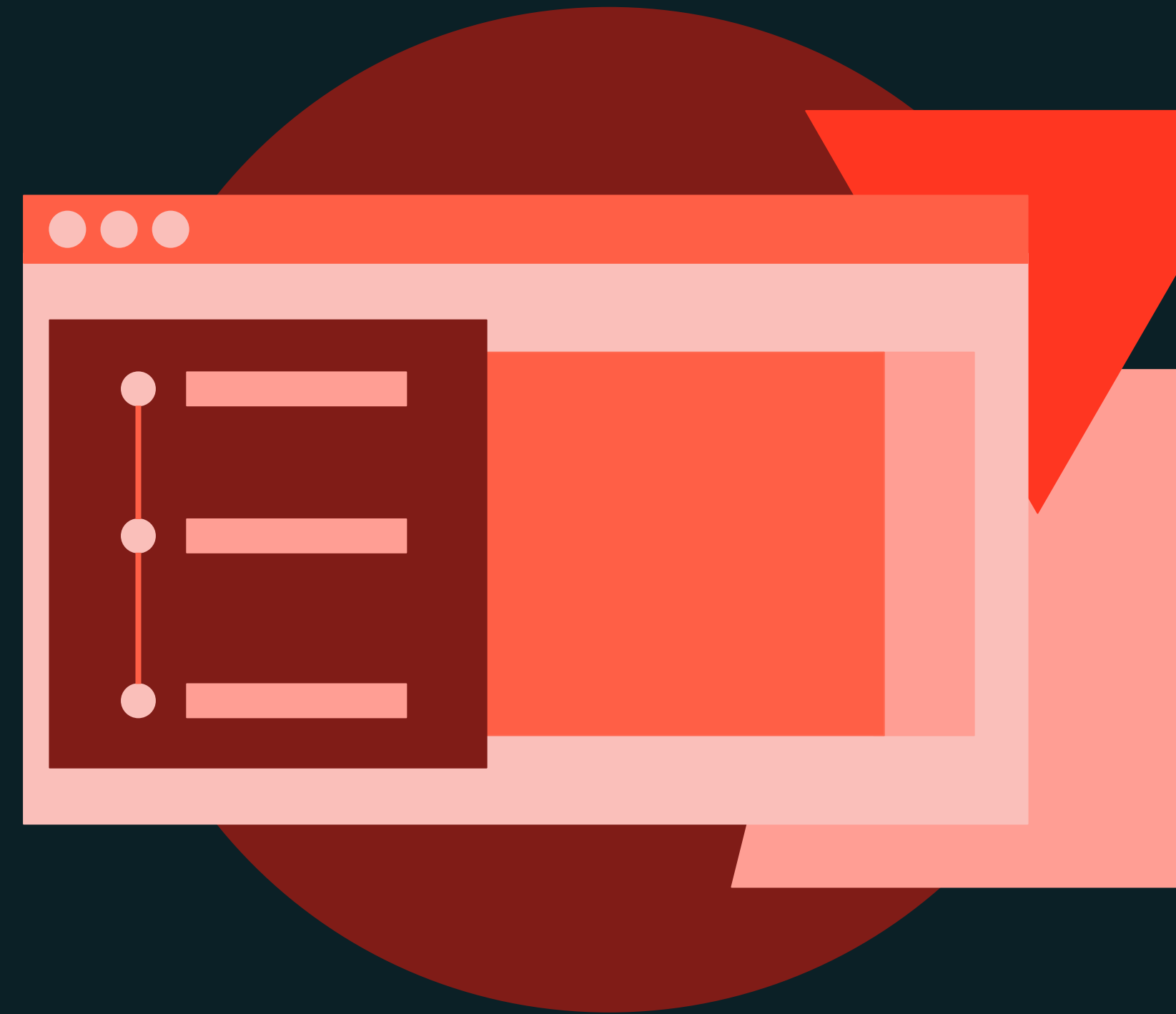




Databricks AI/BI

DEMONSTRATION

Creating a Dashboard in Databricks



Follow-along Instructions

Estimated Time: 10 minutes

For this demonstration, the instructor will guide you through the Dashboards interface, including adding data sets both with the UI and through a query, creating visualizations, and adding features like filters to the dashboard. This demo closes with a brief introduction to sharing your dashboards with others in your business. If you have access to the Vocareum lab environment, feel free to follow-along to explore the workspace and get oriented with where future labs will be completed.





Data Warehousing with Databricks

LAB EXERCISE

Creating a Dashboard Lab



Lab Exercise Instructions

Estimated Time: 15 minutes

In this lab, you will go through the process of creating and populating an AI/BI Dashboard.

To begin:

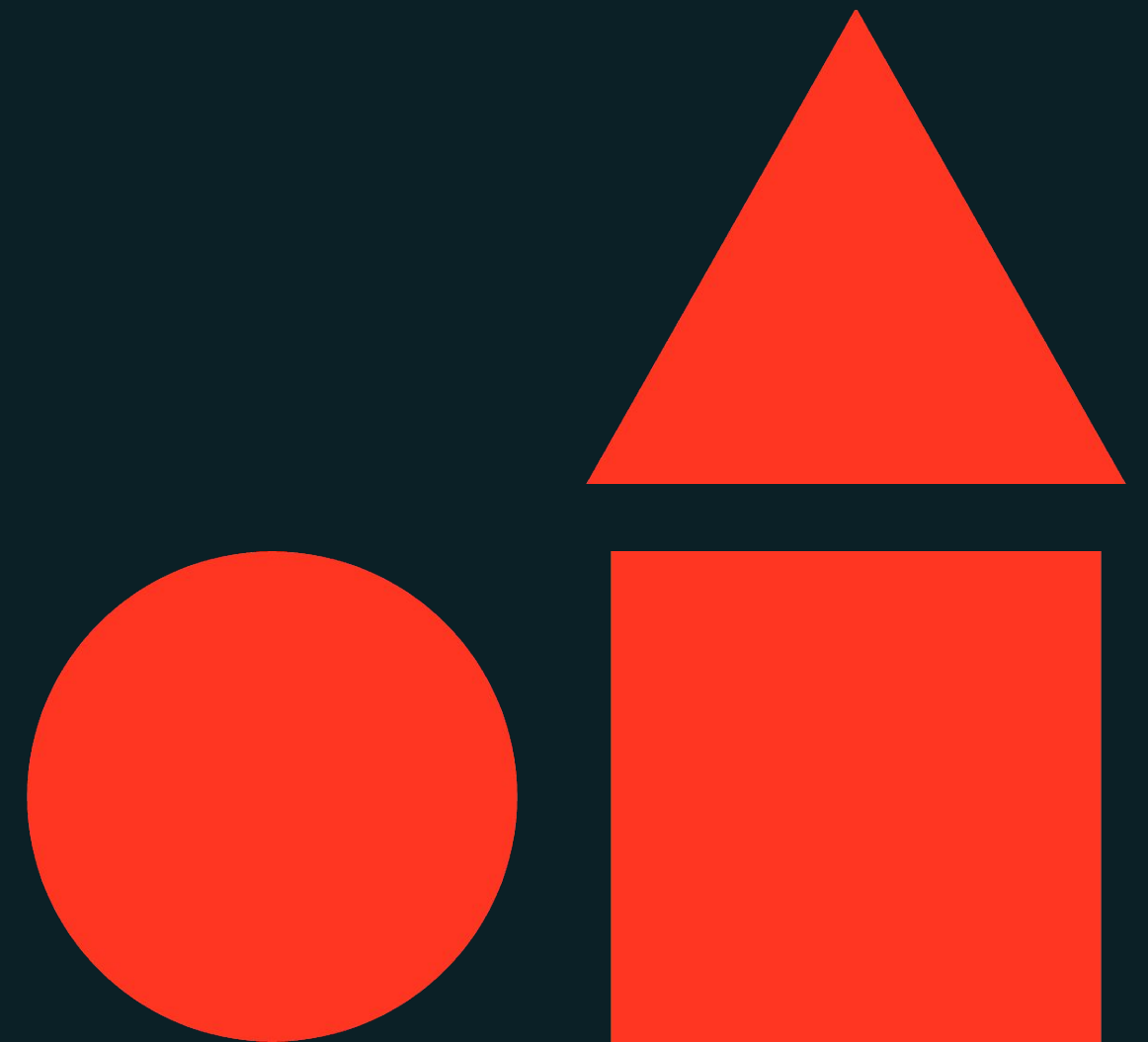
- Open up Vocareum through Databricks Academy.
- Open the notebook **05L Creating a Dashboard Lab** and follow the instructions in the notebook.





Genie

Data Warehousing with Databricks



Objectives

- Create Genie Spaces that leverage natural language to garner insights from your data





Genie

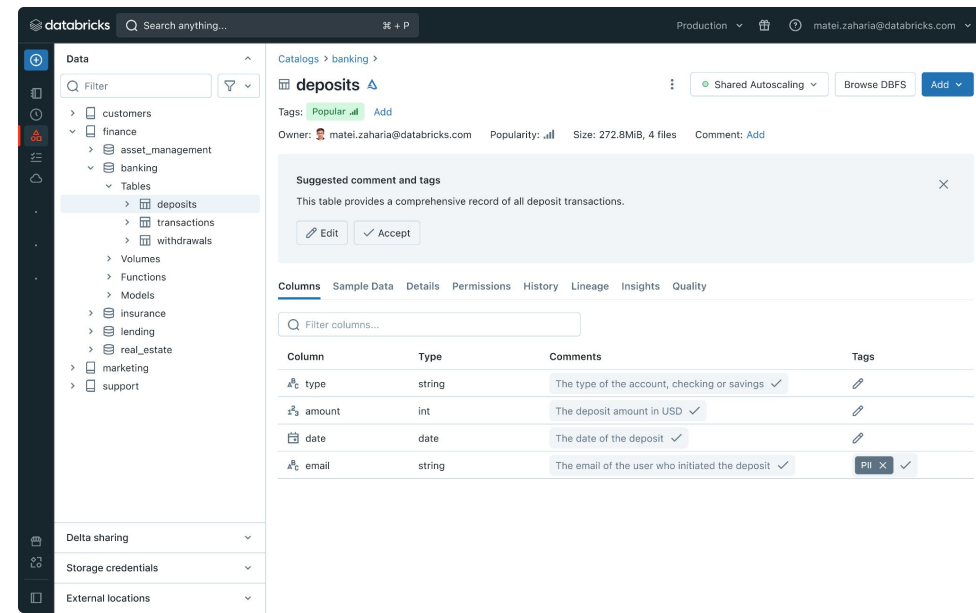
LECTURE

Introduction to Genie

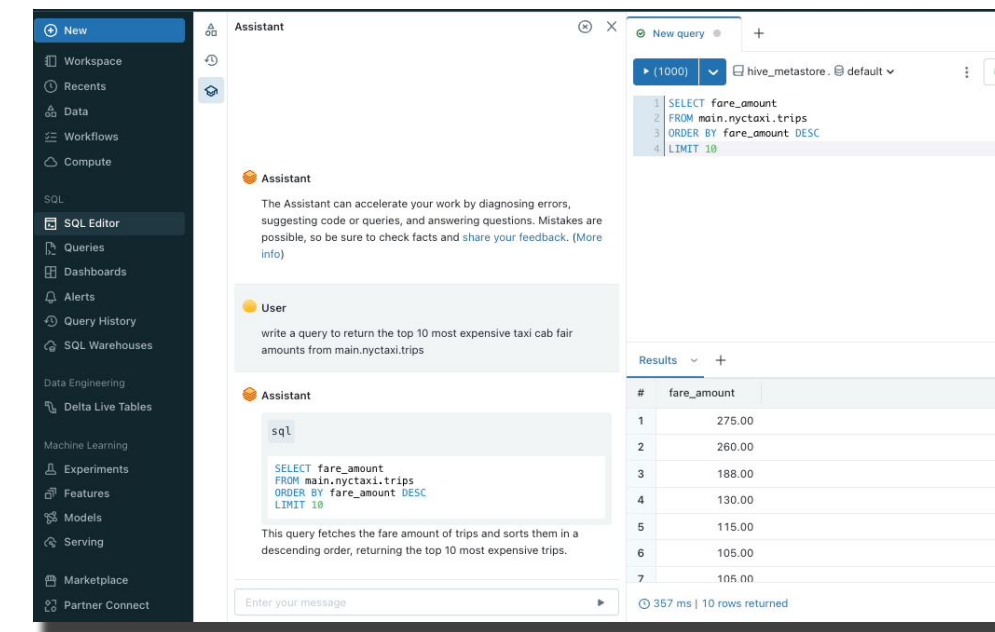


Infusing AI in our user experience

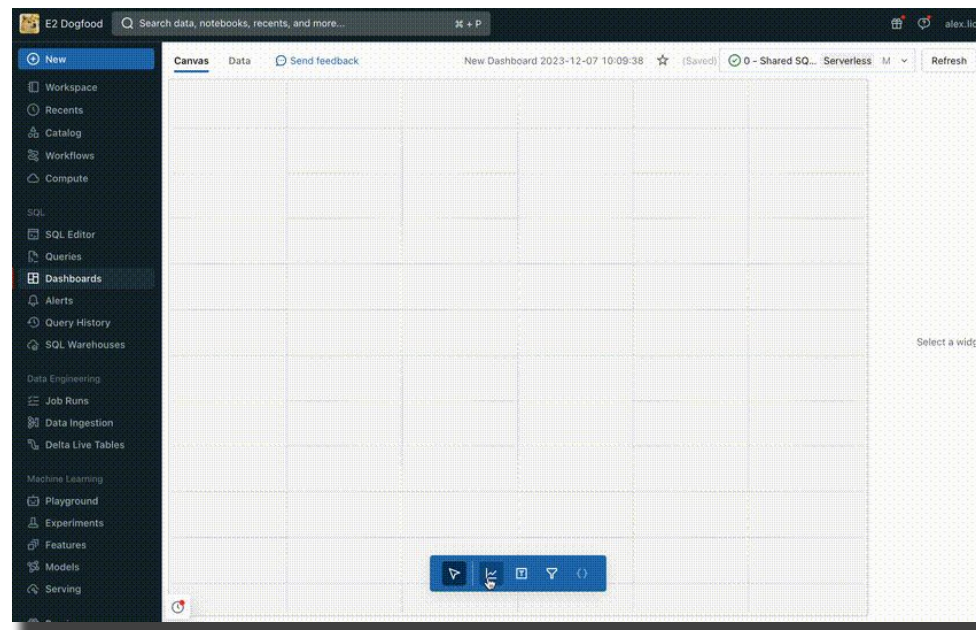
AI-generated docs for easier search and discoverability



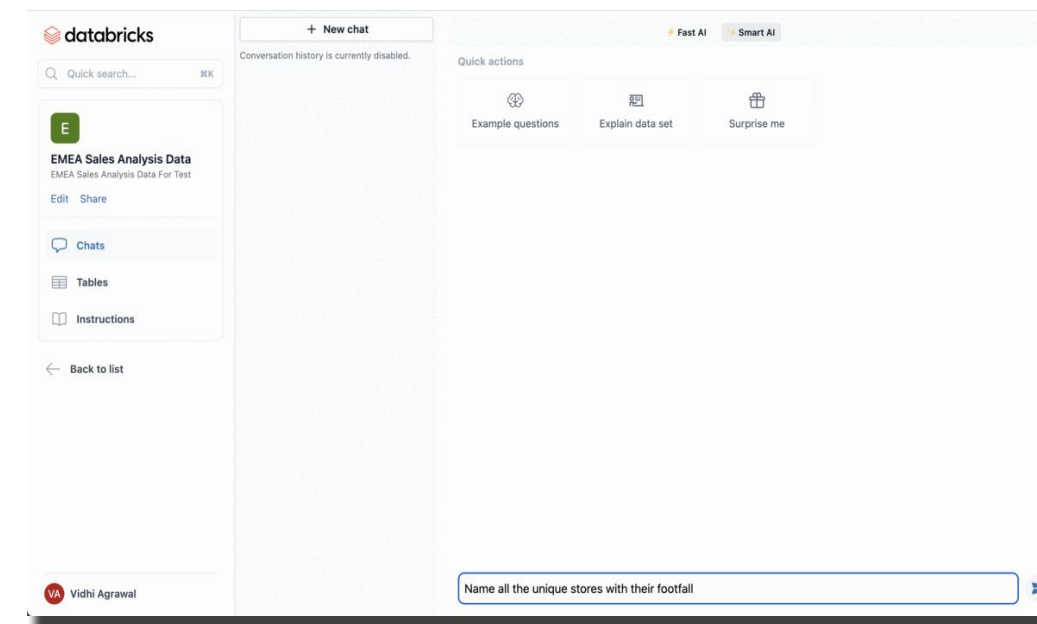
AI Assistant for quick code fixes, queries and descriptions



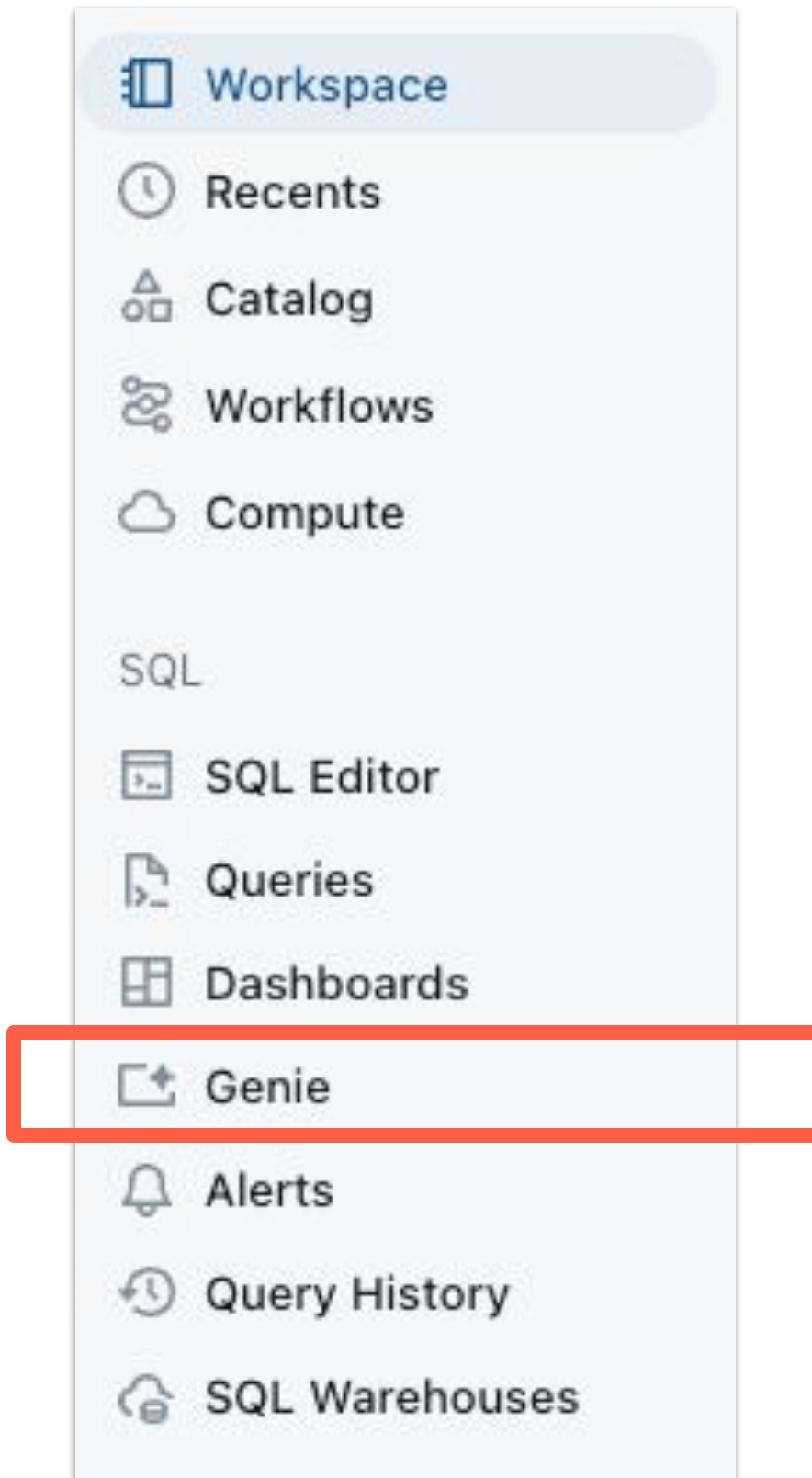
AI Assistant for quick creation of visualizations



AI-powered Genie Spaces to quickly understand your data using natural language



Databrick's AI/BI Offerings



Located under the SQL header in the navigation:

AI/BI Genie

- Unique space associated to a specific curated data set for asking questions using natural language to gain further insight or alternative visualizations without needing a data analyst to oversee the development.



AI/BI Genie Spaces

Ask questions of your data in natural language using your terms

TPC-H Samples ☆

What tables are there and how are they connected? Give me a short summary.

There are eight tables in the database:

1. **orders**: Contains order details such as order key, customer key, order status, total price, order date, and other related information.
2. **customer**: Contains customer details including customer key, name, address, nation key, phone, account balance, market segment, and comments.
3. **lineitem**: Contains line item details for orders, including order key, part key, supplier key, line number, quantity, extended price, discount, tax, return flag, line status, ship date, commit date, receipt date, shipping instructions, shipping mode, and comments.
4. **nation**: Contains nation details such as nation key, name, region key, and comments.
5. **part**: Contains part details including part key, name, manufacturer, brand, type, size, container, retail price, and comments.
6. **partsupp**: Contains part-supplier details including part key, supplier key, available quantity, supply cost, and comments.
7. **region**: Contains region details such as region key, name, and comments.
8. **supplier**: Contains supplier details including supplier key, name, address, nation key, phone, account balance, and comments.

Connections between tables:

- **orders** is connected to **customer** via `o_custkey = c_custkey`.
- **lineitem** is connected to **orders** via `l_orderkey = o_orderkey`.
- **lineitem** is connected to **part** via `l_partkey = p_partkey`.
- **lineitem** is connected to **supplier** via `l_suppkey = s_suppkey`.
- **customer** and **supplier** are connected to **nation** via `c_nationkey = n_nationkey` and `s_nationkey = n_nationkey`, respectively.
- **nation** is connected to **region** via `n_regionkey = r_regionkey`.
- **partsupp** is connected to **part** via `ps_partkey = p_partkey` and to **supplier** via `ps_suppkey = s_suppkey`.

Is this correct?

Yes

Fix it

What is the total number of orders in the database?

How many customers are t

→

Ask your question... ➤

+ New chat

History

Configure

Monitoring

Share

Data

Instructions

Settings

×

Data

Link existing or upload new data to start discovering insights.

+ Add

Name	Type
orders samples.tpch	Table
customer samples.tpch	Table
lineitem samples.tpch	Table
nation samples.tpch	Table
part samples.tpch	Table
partsupp samples.tpch	Table
region samples.tpch	Table
supplier samples.tpch	Table

- Supports non technical users with a no-code solution to gaining data insights using natural language.
- Provides a self-service access point to the data for follow-up questions without needing to involve a data analytics specialist.

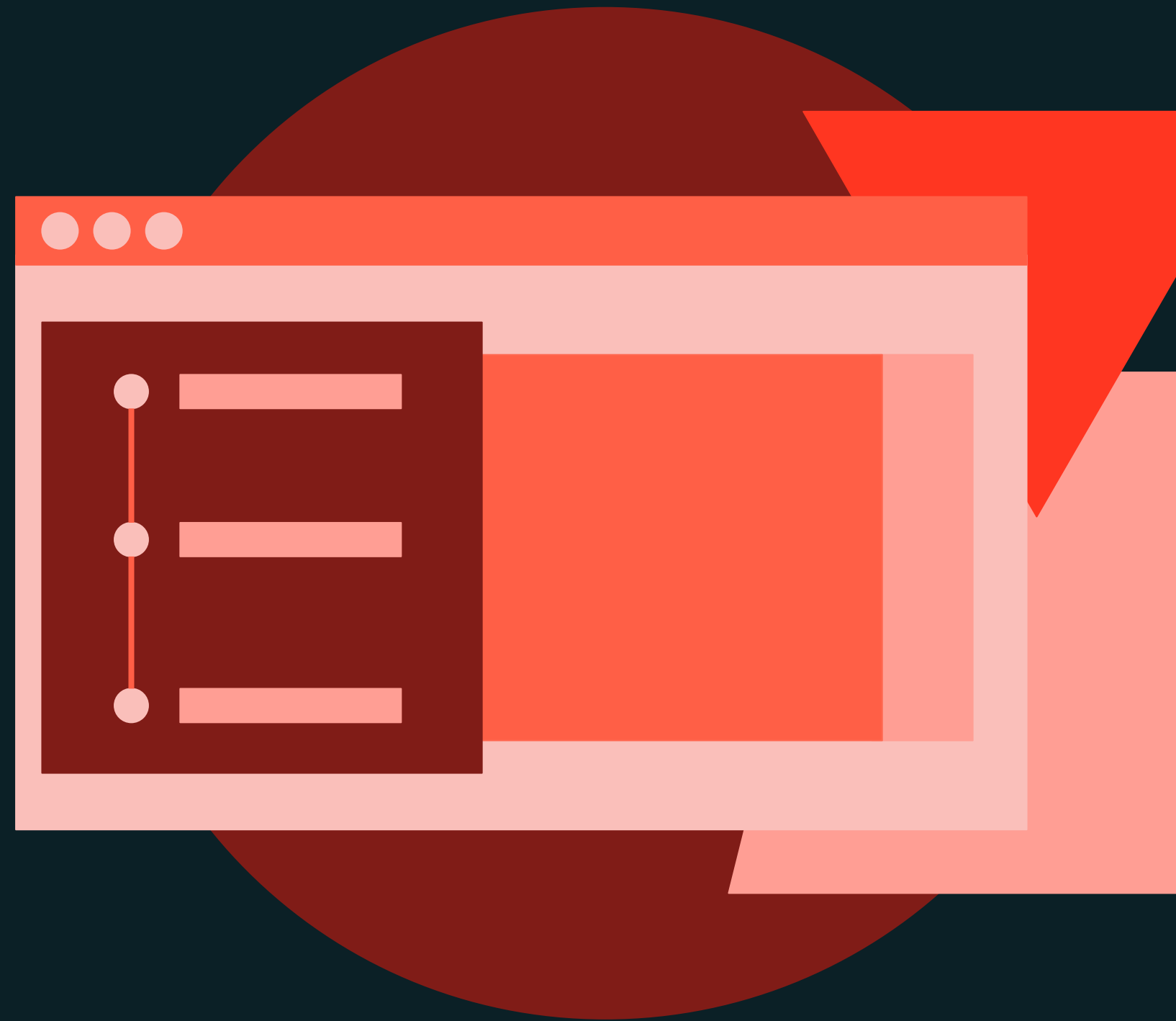




Genie

DEMONSTRATION

Creating Genie Spaces



Create a Genie Space

Simple to use interface

- Name
- Description
- Default Warehouse
- Tables
 - Catalog
 - Schema
 - Table
- Sample Questions

Connect your data



Genie spaces empower you to uncover meaningful insights from your data. Just upload your datasets, provide instructions, and simply ask your data questions.



For you

All

All catalogs > samples > tpch

- ✓  customer
- ✓  lineitem
- ✓  nation
- ✓  orders
- ✓  part
- ✓  partsupp
- ✓  region
- ✓  supplier

Selected: customer × lineitem × nation × orders × part × partsupp × region ×
supplier ×

Cancel

Create



Interacting with Genie

TPCH Samples ☆

What is the monthly total price of orders placed?

The user wants to see the total price of all orders placed each month, organized by year and month.  orders

80 rows  [Add as instruction](#) [Show code](#) 

	1 ² ₃ year	1 ² ₃ month	1.2 total_price
1	1992	1	14,632,465,461.03
2	1992	2	13,702,881,250.25
3	1992	3	14,471,982,842.46
4	1992	4	14,154,839,483.68
5	1992	5	14,488,605,743.39
6	1992	6	14,141,698,293.18
7	1992	7	14,623,719,531.75
8	1992	8	14,504,784,195.86
9	1992	9	14,155,852,017.99
10	1992	10	14,561,557,263.21
11	1992	11	14,133,978,950.19



↪ What is the total price of orders placed in the last month?

↪ How does the total p →

Ask your question...



Options in Genie

Instructions

DataInstructionsSettings

✕

TextJoinsSQL Queries

General Instructions

Add general instructions on how you want Genie to behave.

Examples:

- * MCA stands for My Company Abbreviation
- * Countries in the country_code column are stored with two characters (e.g. US, IT)
- * The rating column is for parental guidance and not popularity
- * When a user asks for performance they are interested in a products revenue

(You can use markdown text)

Save



Options in Genie

Instructions

Data

DataInstructionsSettings

Data

+ Add

Link existing or upload new data to start discovering insights.

Name	Type
customer samples.tpch	Table
lineitem samples.tpch	Table
nation samples.tpch	Table
orders samples.tpch	Table
part samples.tpch	Table
partsupp samples.tpch	Table
region samples.tpch	Table
supplier	Table



Options in Genie

Instructions

Data

Settings

History

DataInstructionsSettings

Title

TPCH Samples

Default warehouse

TPCDI_2X-Small

Description

Describe the available data, provide example questions, and add links to external documents so that users understand how to use your space.

Space ID

01f088dbfe8617d39b06e88feea4e8ab

Sample questions

Sample questions will be presented in new chat windows for users to ask the Space.

Save

Options in Genie

Instructions

Data

Settings

History

Past chats



Today

What is the monthly total price of orders placed?



Demonstration Details

Estimated Time: 10 minutes

For this demonstration, the instructor will walk you through creating Genie Spaces both from a dashboard but also from scratch and discuss the capabilities of this tool.





Genie

LAB EXERCISE

Genie Spaces Lab



Lab Exercise Instructions

Estimated Time: 15 minutes

In this lab, you will go through the process of creating a Genie space and formulate prompts to achieve the requested results.

To begin:

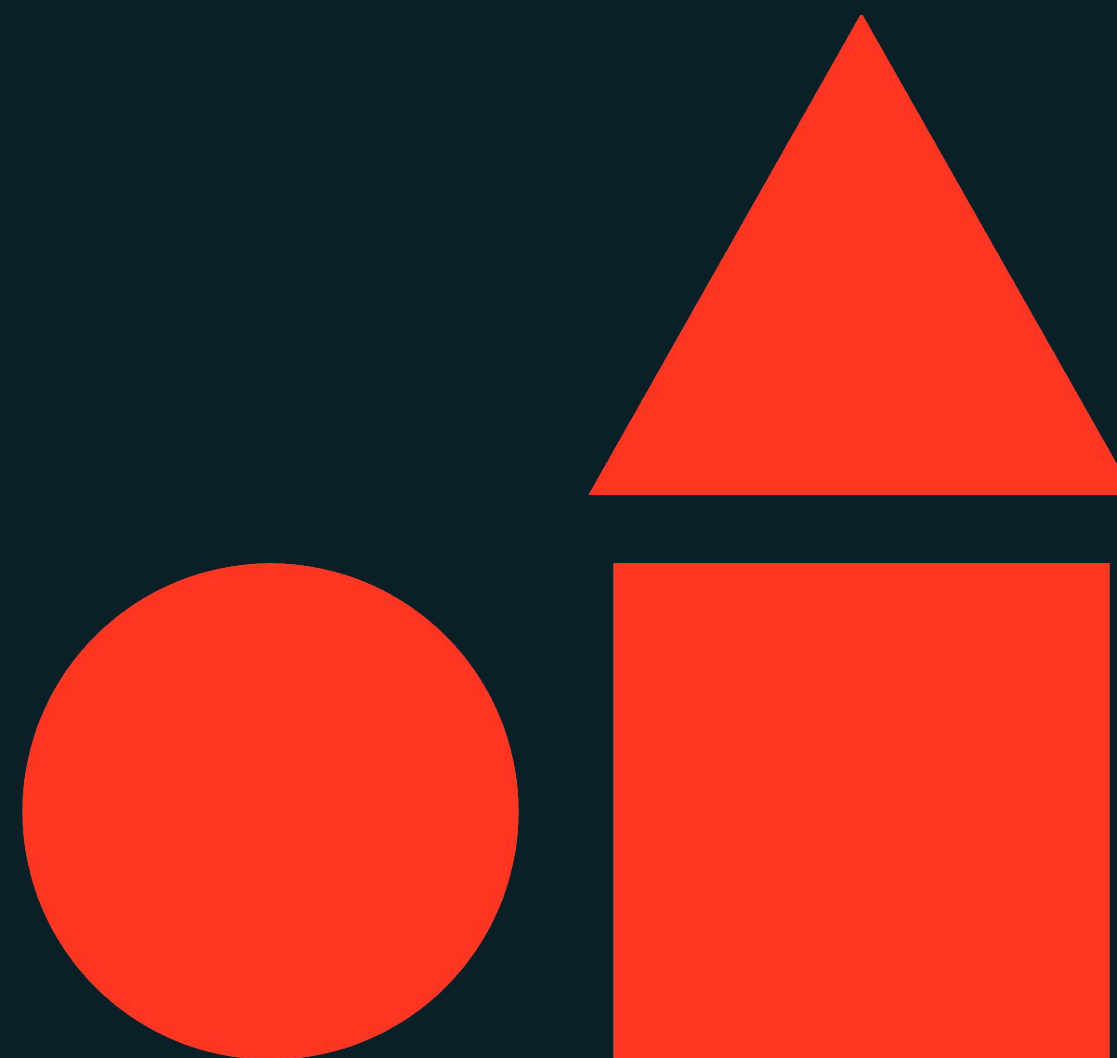
- Open up Vocareum through Databricks Academy.
- Open the notebook **06L Genie Lab** and follow the instructions in the notebook.





Power BI Integration

Data Warehousing with Databricks



Objectives

- Understand how to publish your data from Databricks the Power BI workspace
- Understand the star schema data modelling technique and why it's beneficial





Power BI Integration

LECTURE

Introduction to Databricks and Power BI



Power BI Integration

Overview

Create Power BI datasets directly from Databricks

Making it easy to share and visualize data in Power BI

Suitable for all business users

The screenshot shows the Databricks Catalog Explorer interface for the 'tpcdi.gold_market' schema. The 'Publish to' dropdown menu is open, showing options to 'Open in Tableau Desktop', 'Open in Power BI Desktop', and 'Publish to Power BI workspace'. The 'Publish to Power BI workspace' option is highlighted with a red box. Below the menu, there are tabs for 'Tables 4', 'Volumes 0', 'Models 0', and 'Functions 0'. A search bar labeled 'Filter tables' is present. A table lists the schema's contents:

Name	Owner	Created at	Popularity
dim_company	julio.granados@data...	2024-07-03 16:54:...	
dim_date	julio.granados@data...	2024-07-03 17:21:...	
dim_security	julio.granados@data...	2024-07-03 17:01:...	
fact_markethistory	julio.granados@data...	2024-07-03 17:52:...	



Power BI Integration

Requirements

Unity Catalog schemas only

Premium Power BI license (Desktop version 2.98.683.0 or above)

Private link workspaces requires additional credential configuration

Use star schema data model

For more info:

<https://learn.microsoft.com/en-us/azure/databricks/partners/bi/power-bi#requirements>





Power BI Integration

LECTURE

Star Schema Overview



Star Schema

Overview

Common data modelling technique

Consists of two types of tables:

- Fact tables: quantitative data with foreign keys
- Dimension tables: descriptive attributes providing additional context



Star Schema

Example

Common data modelling technique

Consists of two types of tables:

- Fact tables: quantitative data with foreign keys
- Dimension tables: descriptive attributes providing additional context

Sales (fact table)

SaleID	ProductID	SaleDate	Quantity	TotalAmount



Star Schema

Example

Common data modelling technique

Consists of two types of tables:

- Fact tables: quantitative data with foreign keys
- Dimension tables: descriptive attributes providing additional context

Sales (fact table)

SaleID	ProductID	SaleDate	Quantity	TotalAmount
Primary key				



Star Schema

Example

Common data modelling technique

Consists of two types of tables:

- Fact tables: quantitative data with foreign keys
- Dimension tables: descriptive attributes providing additional context

Sales (fact table)

SaleID	ProductID	SaleDate	Quantity	TotalAmount
Primary key				

Products (dimension table)

ProductID	Name	Category



Star Schema

Example

Common data modelling technique

Consists of two types of tables:

- Fact tables: quantitative data with foreign keys
- Dimension tables: descriptive attributes providing additional context

Sales (fact table)

SaleID	ProductID	SaleDate	Quantity	TotalAmount
Primary key				

Products (dimension table)

ProductID	Name	Category
Primary key		



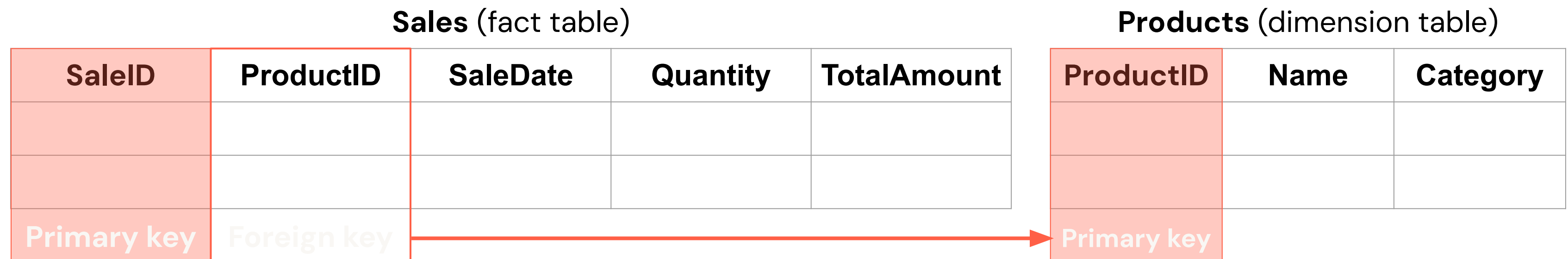
Star Schema

Example

Common data modelling technique

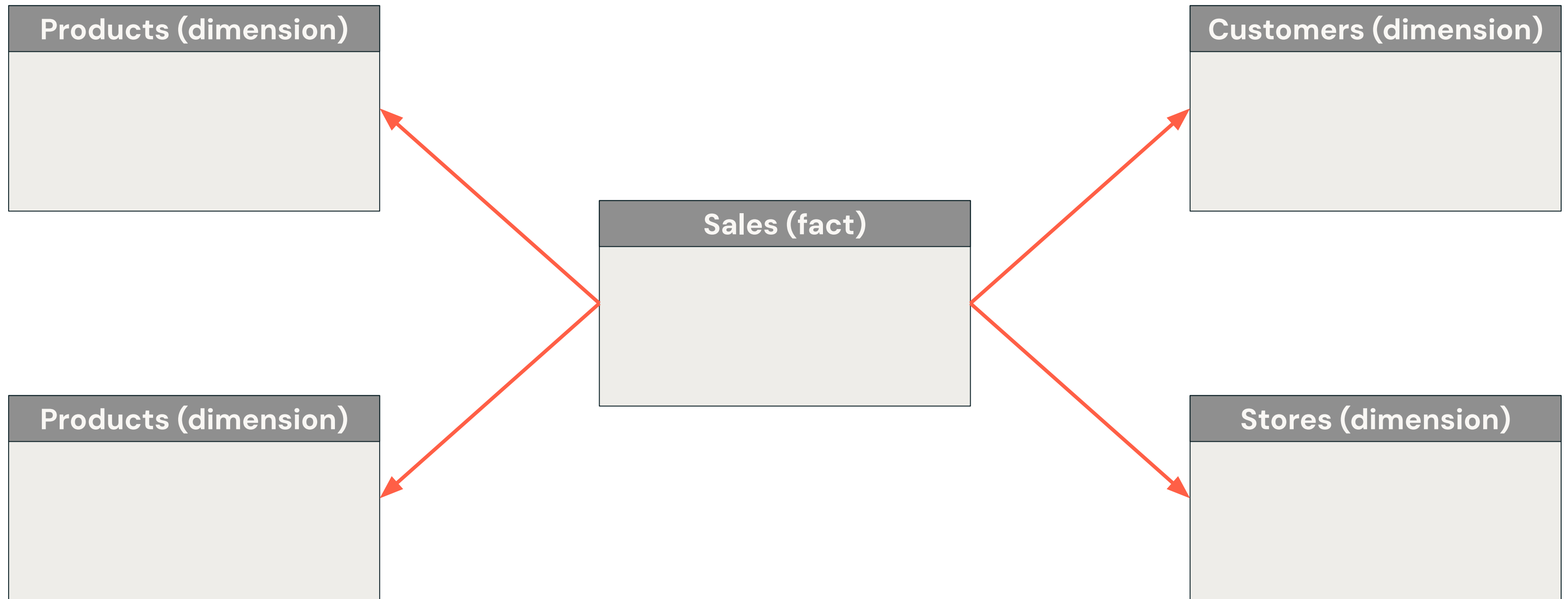
Consists of two types of tables:

- Fact tables: quantitative data with foreign keys
- Dimension tables: descriptive attributes providing additional context



Star Schema

Example



Star Schema

Rationale

Improved query performance

User-friendly design

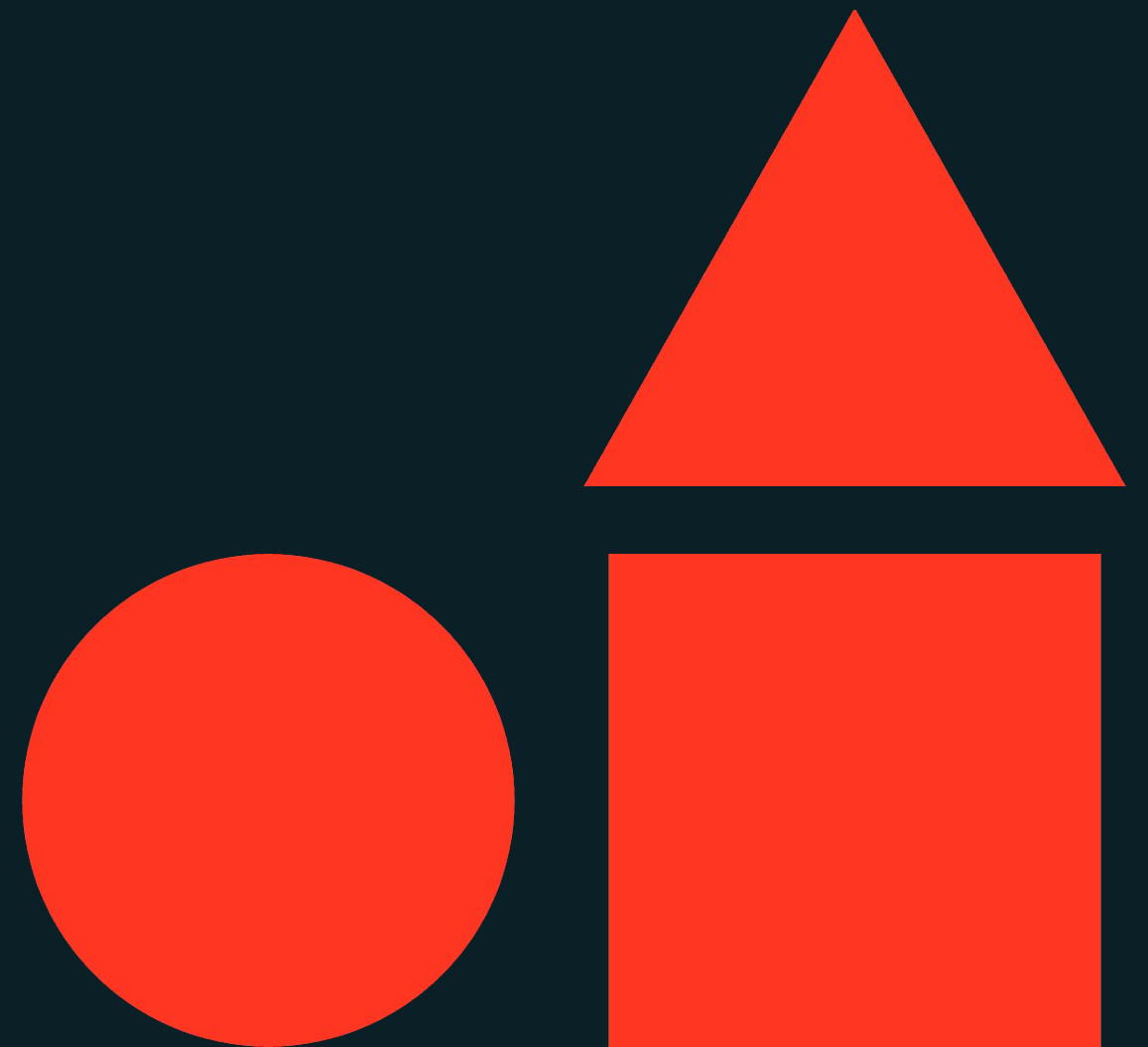
Scalability

Optimized for Power BI Features

Consistent with Industry Standards



Summary and Next Steps



Course Learning Objective Recap

- Describe how data warehousing works on Databricks
- Describe the elements of Unity Catalog that support its role as the unified governance solution in Databricks
- Leverage Databricks capabilities to ingest, transform, and explore your data
- Use Lakeflow jobs to automate data ingestion and transformation
- Apply best practices for controlling cost and optimizing your data warehouse
- Create meaningful visualizations of your data using AI/BI dashboards
- Create Genie Spaces that leverage natural language to garner insights from your data



Next Steps

Additional resources for continuing the learning journey

Data Analysis with Databricks

- Continue your learning through [self-paced](#) or [instructor-led](#) offerings
- The courses offer hands-on instruction in:
 - Databricks SQL
 - Lakehouse Architecture
 - Delta Lake
 - Data Security
 - Data visualizations and dashboards

Data Analysis Associate Certification

- Validate your data and AI skills on Databricks by earning a Databricks credential
- [Exam information](#) and [exam guide](#)
- The exam covers:
 - Databricks SQL – 22%
 - Data Management – 20%
 - SQL – 29%
 - Data Visualization and Dashboards – 18%
 - Analytics Applications – 11%



